Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()



    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```
    nan_ratio = df[cols_to_check].isna().mean()
    valid_features = nan_ratio[nan_ratio <= 0.15].index
    valid_features_list = list(valid_features)

    #df = df[valid_features]
    # Removing of last np.nan
    mask = np.isfinite(df[valid_features]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(valid_features_list + #target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    new_feats_X = [x for x in features_X if x in valid_features_list]
    new_feats_add = [x for x in feats_eng if x in valid_features_list]

    data["df"]=df
    data["features_X"] = new_feats_X
    data["feats_added"] = new_feats_add

    column_removed = [x for x in cols_to_check if x not in valid_features_list]
    print(f'Column removed because np.isnan > 15%: {column_removed}')

    return data
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
```

```
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r")
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r")as
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)




    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
```

```
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Column removed because np.isnan > 15%: ['r_NH3_fft_2h_fft_f2_cph', 'r_NH3_fft_2h_fft_period2_h', 'r_NH3_fft_2h_fft_power2', 'r_NH3_fft_2h_fft_f3_cph', 'r_NH3_fft_2h_fft_period3_h', 'r_NH3_fft_2h_fft_power3', 'r_temperature_fft_2h_fft_f2_cph', 'r_temperature_fft_2h_fft_period2_h', 'r_temperature_fft_2h_fft_power2', 'r_temperature_fft_2h_fft_f3_cph', 'r_temperature_fft_2h_fft_period3_h', 'r_temperature_fft_2h_fft_power3', 'r_umidity_fft_2h_fft_f2_cph', 'r_umidity_fft_2h_fft_period2_h', 'r_umidity_fft_2h_fft_power2', 'r_umidity_fft_2h_fft_f3_cph', 'r_umidity_fft_2h_fft_period3_h', 'r_umidity_fft_2h_fft_power3']

1.2 Metadata

Nombre de lignes df	8092
Nombre de lignes df clean	8092
Nombre de features	14
% de NaN	0.0
Taille dataset entraînement	5664
Taille dataset test	2427
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_2h_fft_f1_cph, r_NH3_fft_2h_fft_period1_h, r_NH3_fft_2h_fft_power1, r_temperature_fft_2h_fft_f1_cph, r_temperature_fft_2h_fft_period1_h, r_temperature_fft_2h_fft_power1, r_umidity_fft_2h_fft_f1_cph, r_umidity_fft_2h_fft_period1_h, r_umidity_fft_2h_fft_power1

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

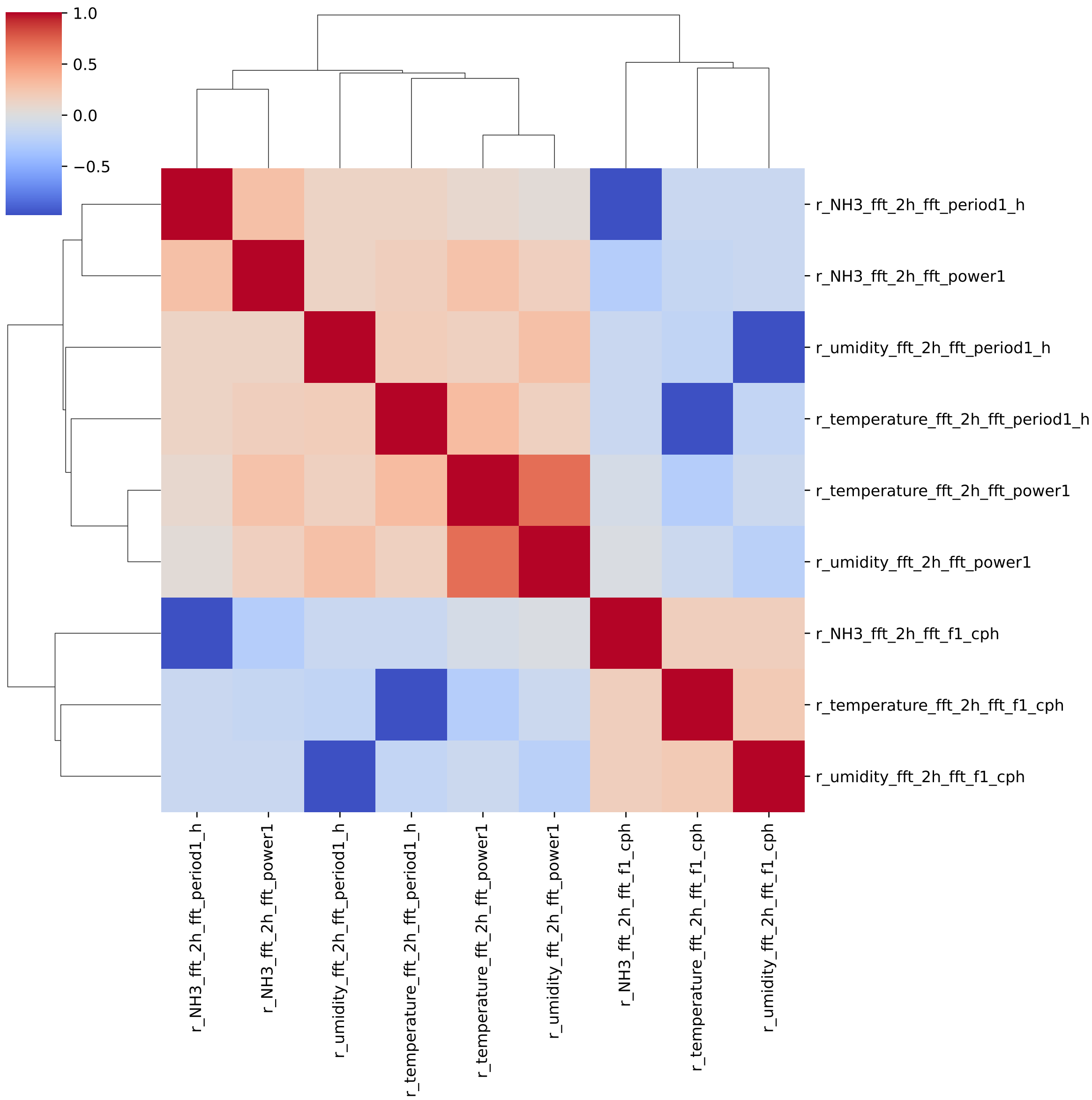
The following section has been done with a threshold choosen of : 0.8

cluster 1: r, *, N, H, 3 ;* ***cluster 2:*** *r,* , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, *, u, m, i, d, i, t, y ;* ***cluster 4:*** *r,* , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_fft_2h_fft_period1_h, r_NH3_fft_2h_fft_f1_cph ; **cluster 7:** r_NH3_fft_2h_fft_power1 ; **cluster 8:** r_temperature_fft_2h_fft_period1_h, r_temperature_fft_2h_fft_f1_cph ; **cluster 9:** r_temperature_fft_2h_fft_power1 ; **cluster 10:** r_umidity_fft_2h_fft_period1_h, r_umidity_fft_2h_fft_f1_cph ; **cluster 11:** r_umidity_fft_2h_fft_power1 ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_2h_fft_period1_h, r_NH3_fft_2h_fft_power1, r_temperature_fft_2h_fft_period1_h, r_temperature_fft_2h_fft_power1, r_umidity_fft_2h_fft_period1_h, r_umidity_fft_2h_fft_power1



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_2h_fft_period1_h, r_NH3_fft_2h_fft_power1, r_temperature_fft_2h_fft_period1_h, r_temperature_fft_2h_fft_power1, r_umidity_fft_2h_fft_period1_h, r_umidity_fft_2h_fft_power1

Len(Features_X): 11

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model': None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

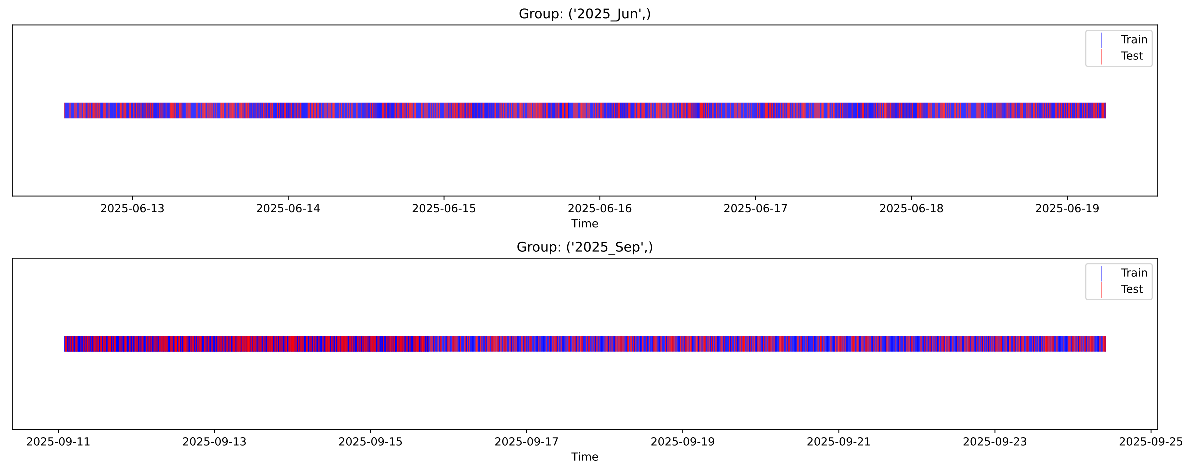
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

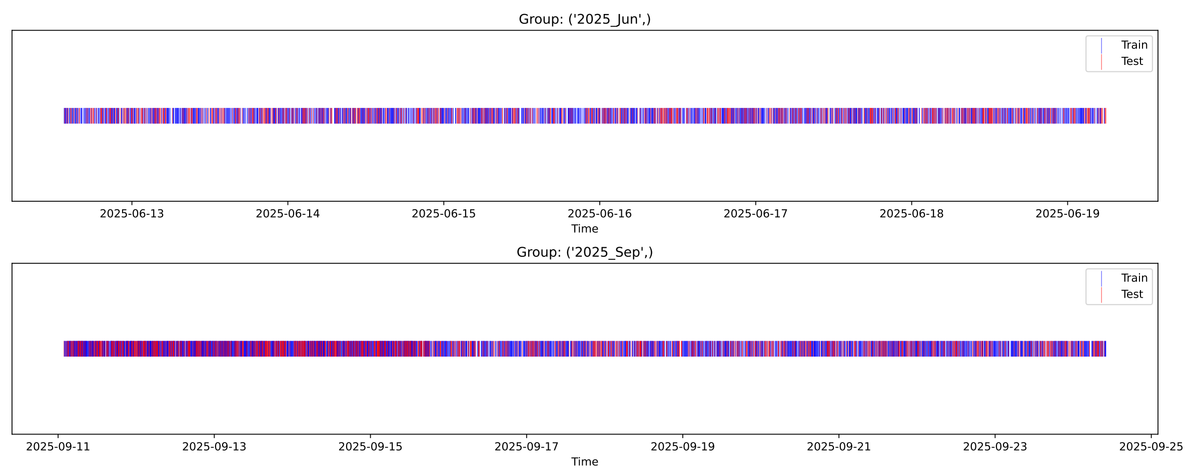
External Split: Train: 5664 samples Test: 2428 samples



3.3.2 Internal CV Folds

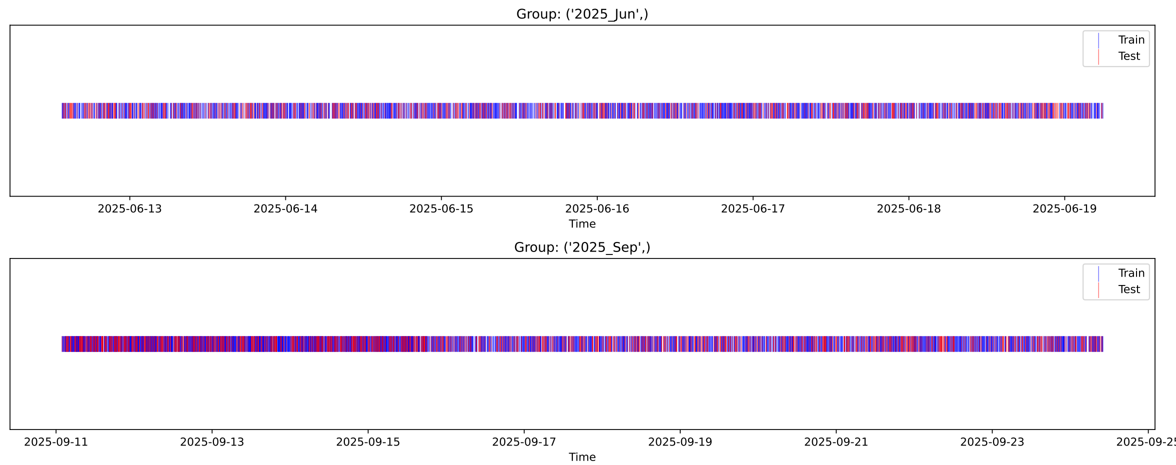
Fold 1/5

Train: 3964 samples Test: 1700 samples Excluded: 0 samples Window exclusion: 0h



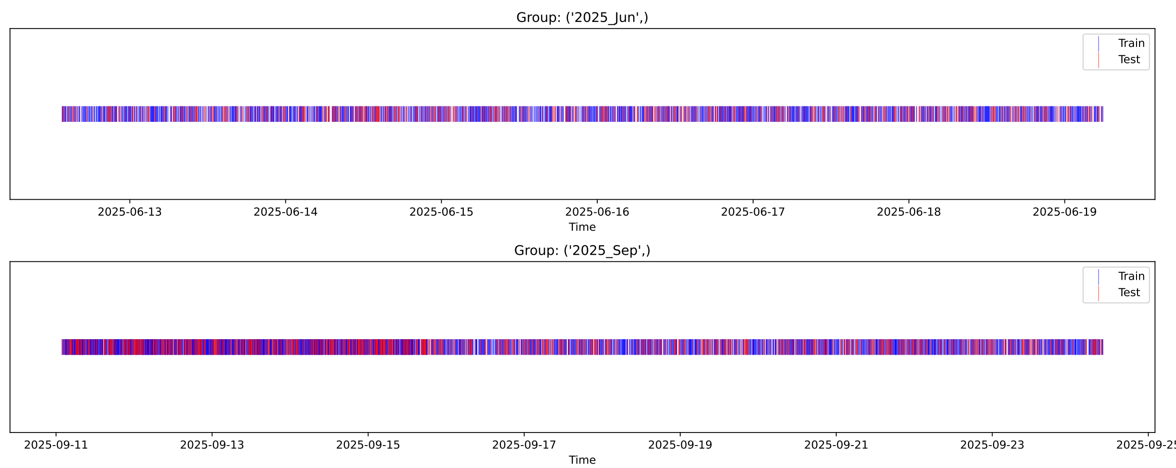
Fold 2/5

Train: 3964 samples Test: 1700 samples Excluded: 0 samples Window exclusion: 0h



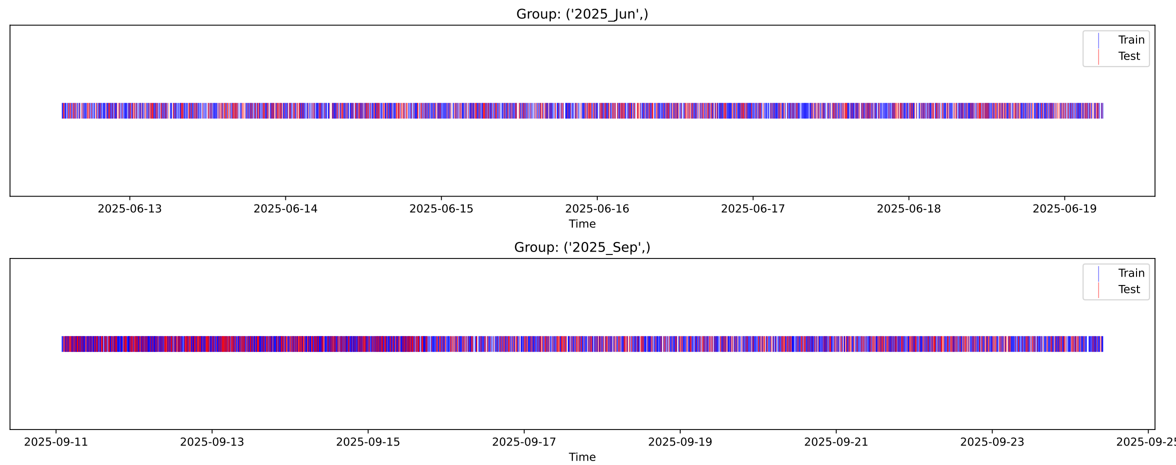
Fold 3/5

Train: 3964 samples Test: 1700 samples Excluded: 0 samples Window exclusion: 0h



Fold 4/5

Train: 3964 samples Test: 1700 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3964 samples Test: 1700 samples Excluded: 0 samples Window exclusion: 0h

3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.351, std=0.804, min=0.959, max=13.845 y_test: mean=2.352, std=0.780, min=1.041, max=8.190

Fold 0 — y_test: mean=2.36, std=0.84, y_train: mean=2.35, std=0.79

Fold 1 — y_test: mean=2.35, std=0.81, y_train: mean=2.35, std=0.80

Fold 2 — y_test: mean=2.37, std=0.83, y_train: mean=2.34, std=0.79

Fold 3 — y_test: mean=2.36, std=0.78, y_train: mean=2.35, std=0.81

Fold 4 — y_test: mean=2.35, std=0.79, y_train: mean=2.35, std=0.81

train scores CV: [0.51827096 0.50354676 0.52141346 0.49415974 0.48516088]

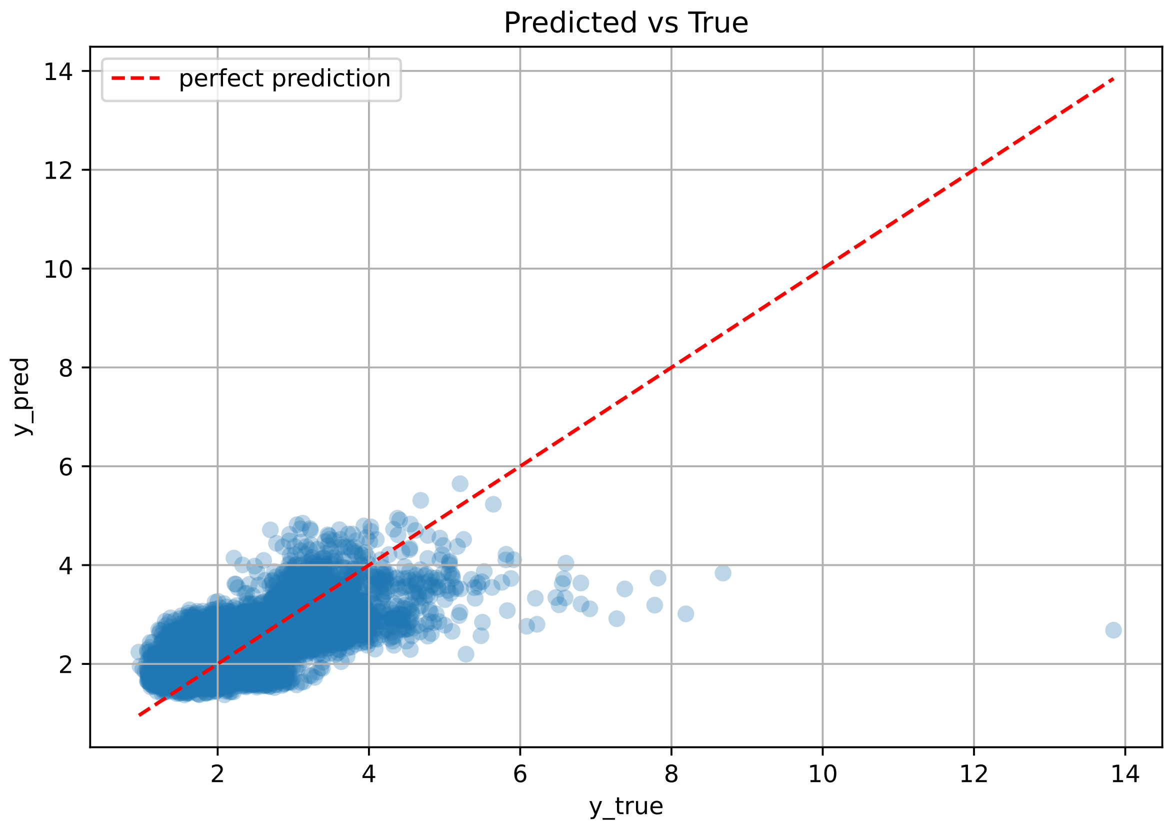
test scores CV: [0.46869327 0.4977403 0.46066416 0.52226285 0.54485527]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5045	0.4988	0.0317	0.4783	0.4791	-0.0198	0.9603	1.0114

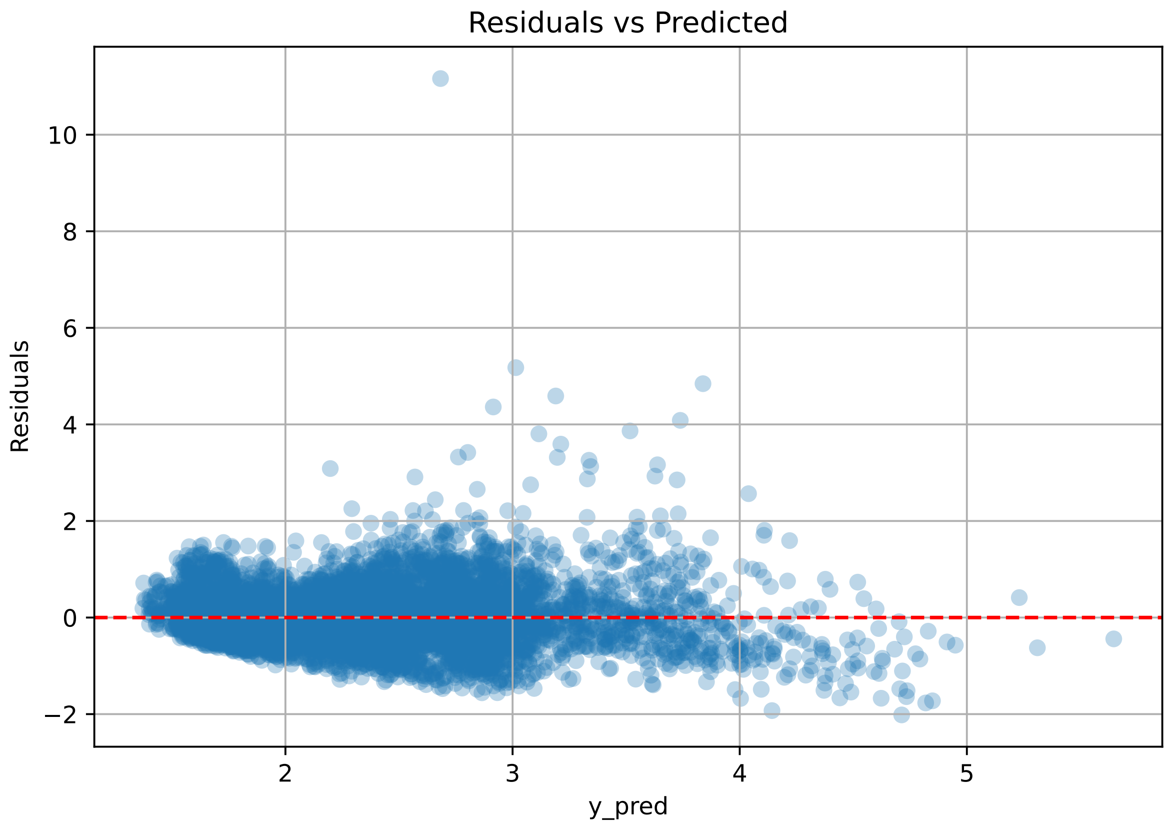
Coefficients:

coef_r_CO2 = -0.0462, coef_r_NH3 = -0.0754, coef_r_NH3_fft_2h_fft_period1_h = -0.0147, coef_r_NH3_fft_2h_fft_power1 = 0.0390, coef_r_THI = -0.0042, coef_r_temperature = 0.0550, coef_r_temperature_fft_2h_fft_period1_h = -0.0531, coef_r_temperature_fft_2h_fft_power1 = 0.3038, coef_r_umidity = -0.3545, coef_r_umidity_fft_2h_fft_period1_h = -0.0085, coef_r_umidity_fft_2h_fft_power1 = 0.0304, intercept = 2.3509,

3.4.2 Scatter


Predicted vs True
perfect prediction
14
12
10
8
6
4
2
y_pred
2
4
6
8
10
12
14
y_true

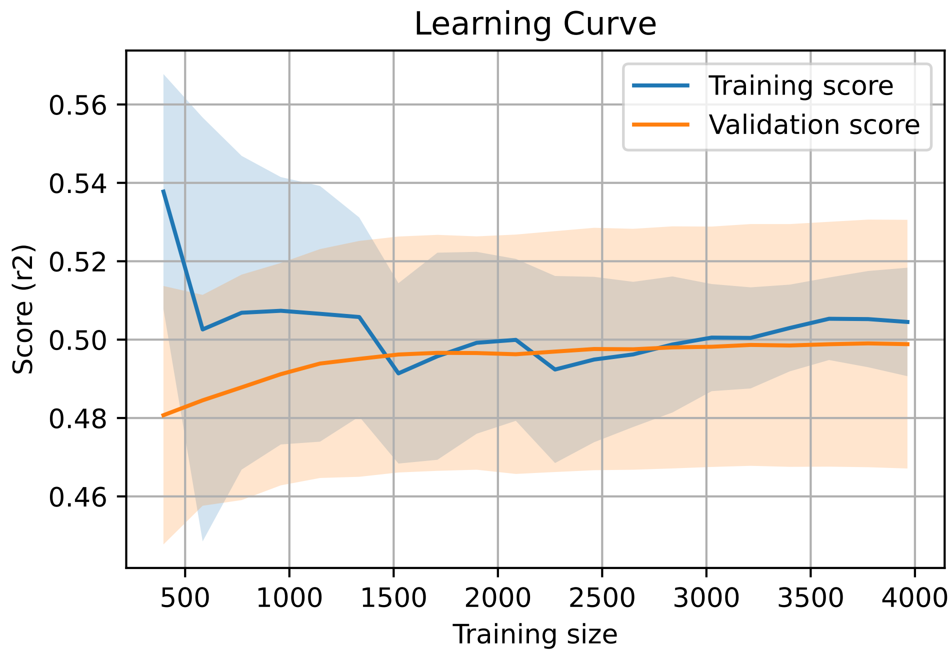
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 8092, **Len(training)** : 5665,

Len(training_real) : 3966, **Len(test_of_training)** : 1699, **Len(test)** : 2427

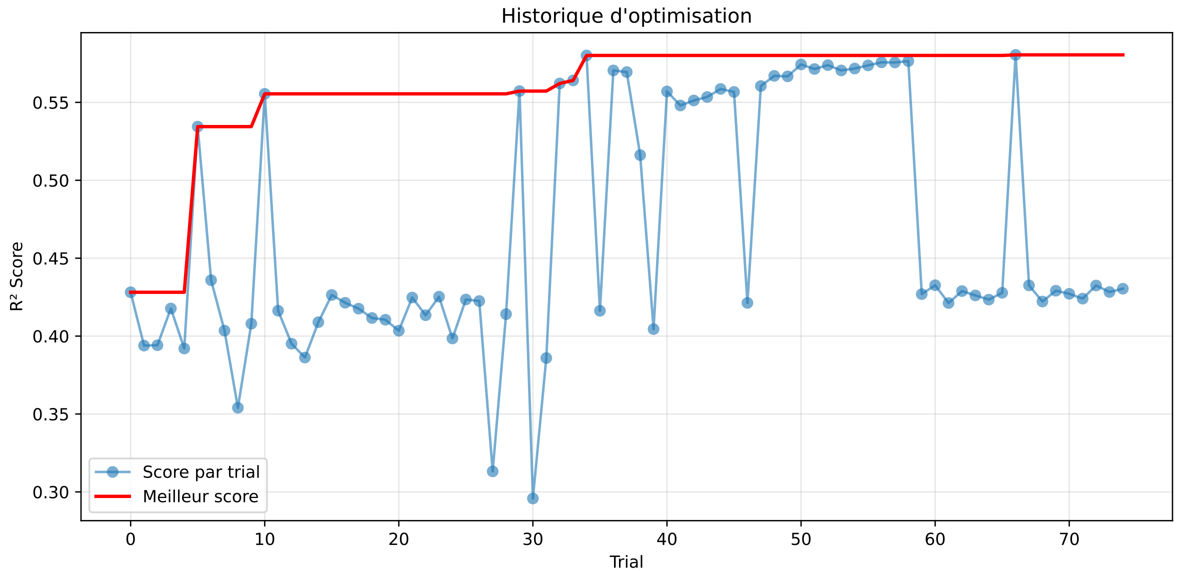


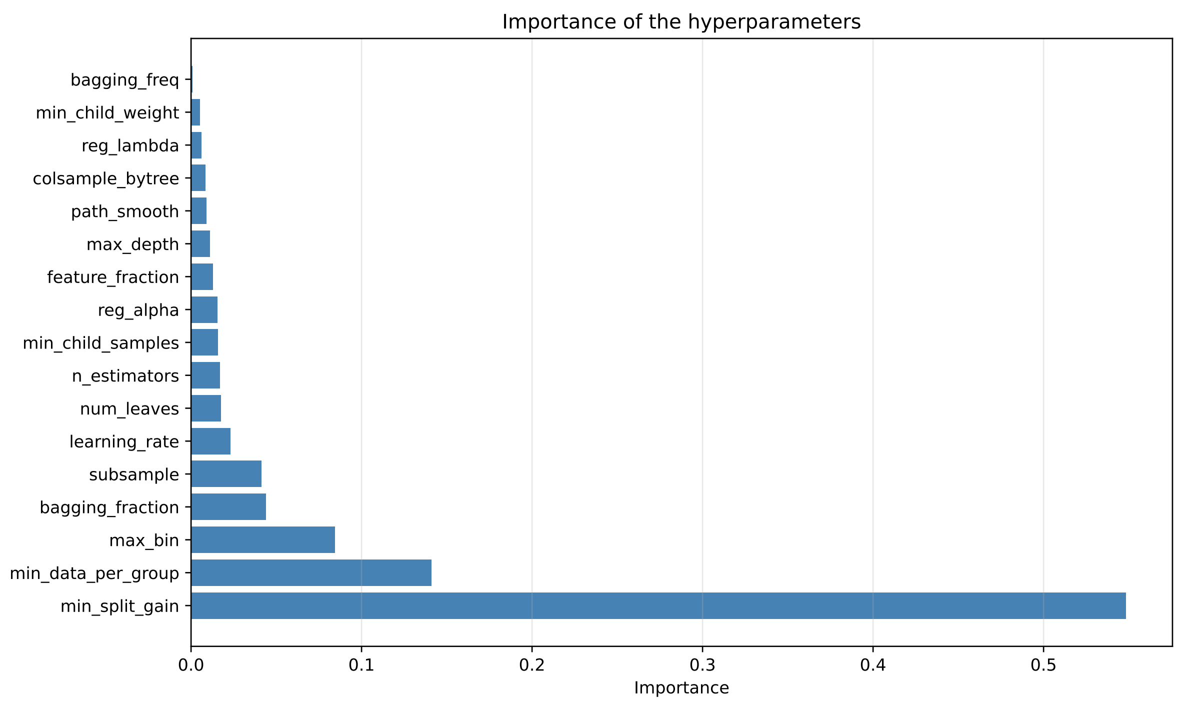
3.5 Model : LGBM

3.5.1 Gridsearch

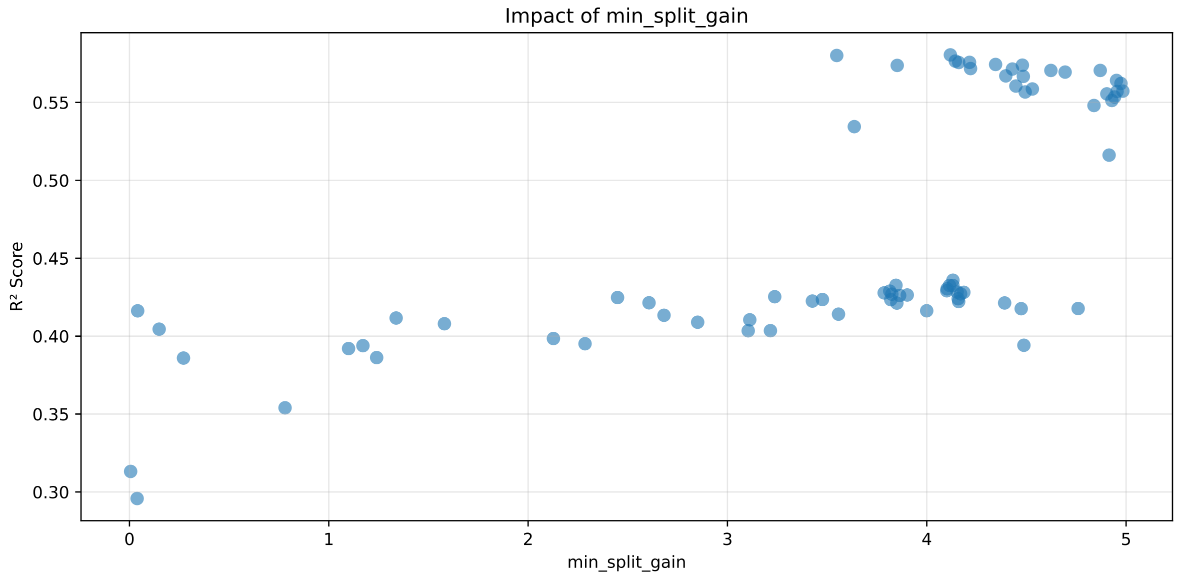
Distribution y_train vs y_test: y_train: mean=2.351, std=0.804, min=0.959, max=13.845 y_test: mean=2.352, std=0.780, min=1.041, max=8.190

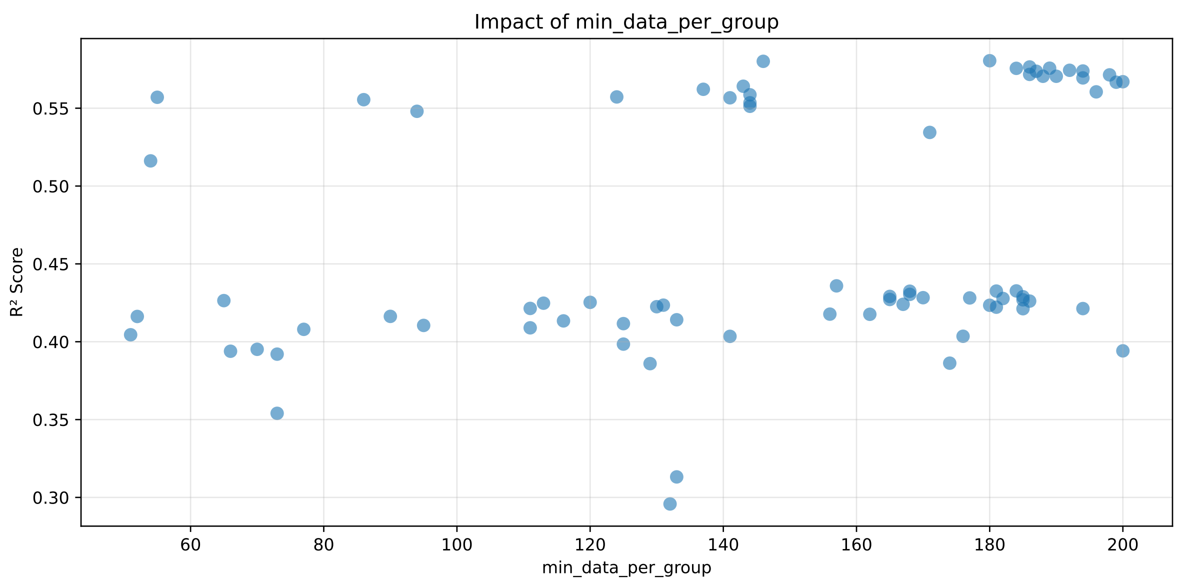
==================== TOP Importance FEATURES ==================== feature importance r_CO2 153 r_temperature_fft_2h_fft_power1 149 r_umidity 133 r_temperature_fft_2h_fft_period1_h 108 r_NH3 101 r_umidity_fft_2h_fft_period1_h 62 r_temperature 56 r_NH3_fft_2h_fft_power1 44 r_umidity_fft_2h_fft_power1 44 r_THI 41 r_NH3_fft_2h_fft_period1_h 12

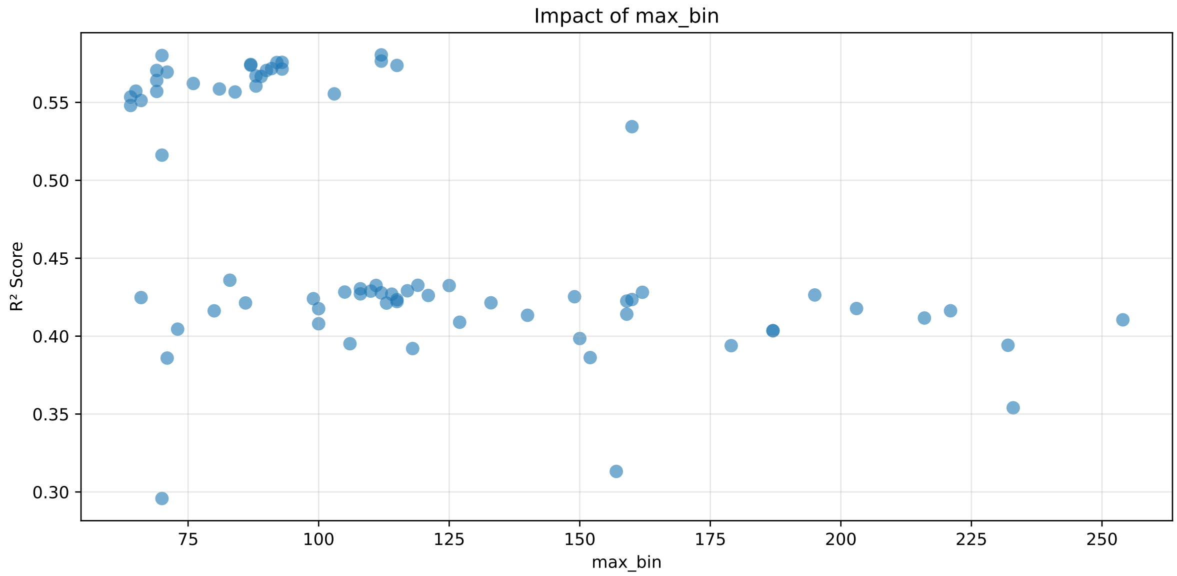




Paramètres avec >5% d'importance : ['min_split_gain', 'min_data_per_group', 'max_bin']

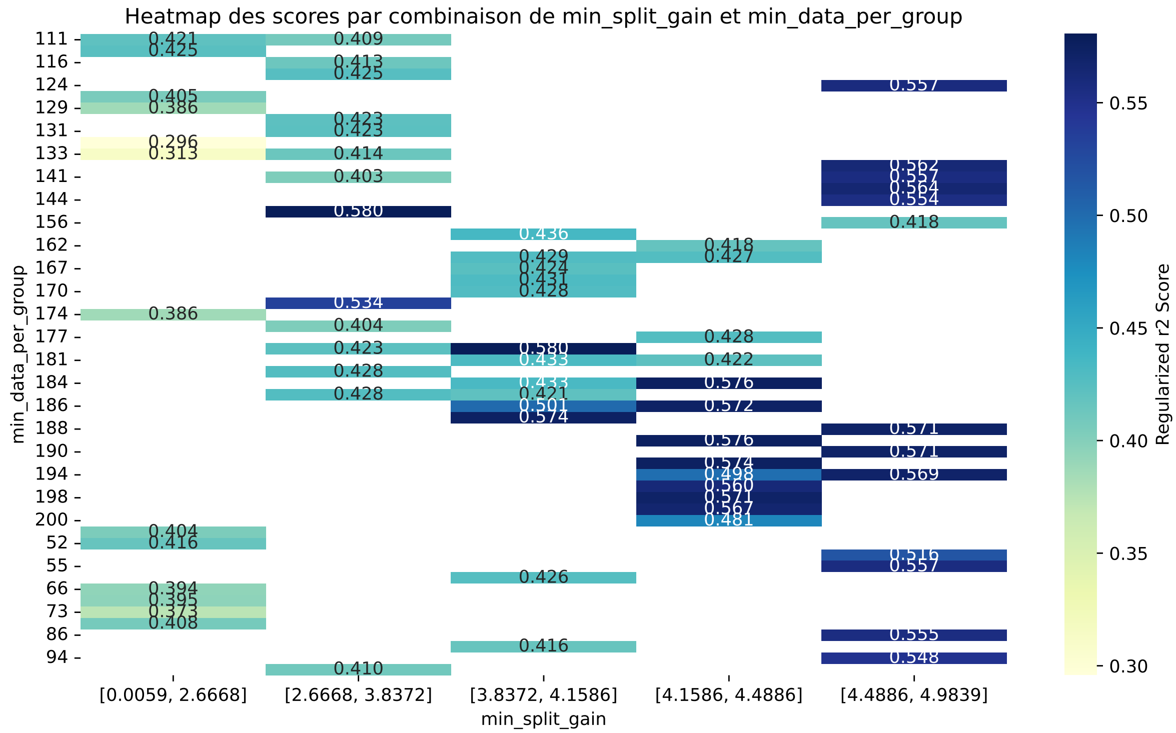

Impact of min_split_gain
R² Score
0.30
0.35
0.40
0.45
0.50
0.55
0
1
2
3
4
5
min_split_gain


Impact of min_data_per_group
R² Score
0.30
0.35
0.40
0.45
0.50
0.55
60
80
100
120
140
160
180
200
min_data_per_group




```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6094	0.5805	0.5613	0.0288	0.5805	0.5797	-0.0008	0.9987	1.0498
2	0.6081	0.5801	0.561	0.0277	0.5801	0.5761	-0.004	0.9931	1.0483
3	0.6045	0.5765	0.557	0.0304	0.5765	0.574	-0.0025	0.9956	1.0485
4	0.6043	0.5757	0.5567	0.0316	0.5757	0.5719	-0.0038	0.9934	1.0497
5	0.6039	0.5756	0.5571	0.0317	0.5756	0.5741	-0.0015	0.9973	1.0491
6	0.6021	0.5743	0.556	0.03	0.5743	0.5762	0.0019	1.0033	1.0484
7	0.602	0.5739	0.5543	0.0308	0.5739	0.5701	-0.0038	0.9935	1.049
8	0.6022	0.5737	0.5554	0.0302	0.5737	0.5744	0.0006	1.0011	1.0495
9	0.5983	0.5717	0.5519	0.0311	0.5717	0.5653	-0.0065	0.9887	1.0466
10	0.5989	0.5714	0.5523	0.0291	0.5714	0.5697	-0.0017	0.9971	1.0482
---	---	---	---	---	---	---	---	---	---
75	0.748	0.6727	0.6432	0.0257	0.2958	0.6591	-0.0136	0.9798	1.1121

Hyperparameters:

Rank 1: {'reg___n_estimators': 500, 'reg___max_depth': 4, 'reg___learning_rate': 0.02910877350796739, 'reg___num_leaves': 22, 'reg___subsample': 0.5667093435602438, 'reg___colsample_bytree': 0.5290030867901523, 'reg___min_child_samples': 34, 'reg___reg_alpha': 1.5311176039331071, 'reg___reg_lambda': 40.07641990117104, 'reg___min_split_gain': 4.118382485699875, 'reg___path_smooth': 4.038721047868482, 'reg___min_child_weight': 0.06984256146717256, 'reg___feature_fraction': 0.558476714190737, 'reg___bagging_fraction': 0.7647901845925154, 'reg___bagging_freq': 2, 'reg___max_bin': 112, 'reg___min_data_per_group': 180}

Rank 2: {'reg___n_estimators': 700, 'reg___max_depth': 4, 'reg___learning_rate': 0.016948169332819653, 'reg___num_leaves': 43, 'reg___subsample': 0.6087647491787104, 'reg___colsample_bytree': 0.6637190384515469, 'reg___min_child_samples': 42, 'reg___reg_alpha': 2.82962431988083, 'reg___reg_lambda': 30.288457744154517, 'reg___min_split_gain': 3.548390280708008, 'reg___path_smooth': 3.6084786193679963, 'reg___min_child_weight': 0.2000174757641451, 'reg___feature_fraction': 0.601466017058022, 'reg___bagging_fraction': 0.6660269952592672, 'reg___bagging_freq': 4, 'reg___max_bin': 70, 'reg___min_data_per_group': 146}

Rank 3: {'reg___n_estimators': 600, 'reg___max_depth': 4, 'reg___learning_rate': 0.022903769040647545, 'reg___num_leaves': 32, 'reg___subsample': 0.5325989168666353, 'reg___colsample_bytree': 0.5765792629530088, 'reg___min_child_samples': 39, 'reg___reg_alpha': 2.1068207292468086, 'reg___reg_lambda': 37.145887161210986, 'reg___min_split_gain': 4.143247647196447, 'reg___path_smooth': 2.7050796319413872, 'reg___min_child_weight': 0.1515296640469024, 'reg___feature_fraction': 0.5688328207316222, 'reg___bagging_fraction': 0.719939916165148, 'reg___bagging_freq': 2, 'reg___max_bin': 112, 'reg___min_data_per_group': 186}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.011373557063604567, 'reg__num_leaves': 32, 'reg__subsample': 0.5304060567251662, 'reg__colsample_bytree': 0.5960252965910035, 'reg__min_child_samples': 38, 'reg__reg_alpha': 2.1950113905655138, 'reg__reg_lambda': 27.279300332973776, 'reg__min_split_gain': 4.214593333033391, 'reg__path_smooth': 4.183512942449053, 'reg__min_child_weight': 0.10785372644341334, 'reg__feature_fraction': 0.5715703977742999, 'reg__bagging_fraction': 0.7127886277735117, 'reg__bagging_freq': 2, 'reg__max_bin': 93, 'reg__min_data_per_group': 189}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.011432761725748187, 'reg__num_leaves': 31, 'reg__subsample': 0.5348943286879461, 'reg__colsample_bytree': 0.5921251348929397, 'reg__min_child_samples': 38, 'reg__reg_alpha': 2.0642232825181326, 'reg__reg_lambda': 36.32381661494525, 'reg__min_split_gain': 4.160458577958341, 'reg__path_smooth': 4.197586964056782, 'reg__min_child_weight': 0.1299764335562686, 'reg__feature_fraction': 0.5728830341529819, 'reg__bagging_fraction': 0.7197255853088215, 'reg__bagging_freq': 2, 'reg__max_bin': 92, 'reg__min_data_per_group': 184}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.02571059614972063, 'reg__num_leaves': 32, 'reg__subsample': 0.5015773695873713, 'reg__colsample_bytree': 0.5878501917430865, 'reg__min_child_samples': 46, 'reg__reg_alpha': 1.0850506676579252, 'reg__reg_lambda': 27.4675783731971, 'reg__min_split_gain': 4.345245748958681, 'reg__path_smooth': 3.271098334297397, 'reg__min_child_weight': 0.10160609741412735, 'reg__feature_fraction': 0.5713962042698453, 'reg__bagging_fraction': 0.70465414197852, 'reg__bagging_freq': 3, 'reg__max_bin': 87, 'reg__min_data_per_group': 192}

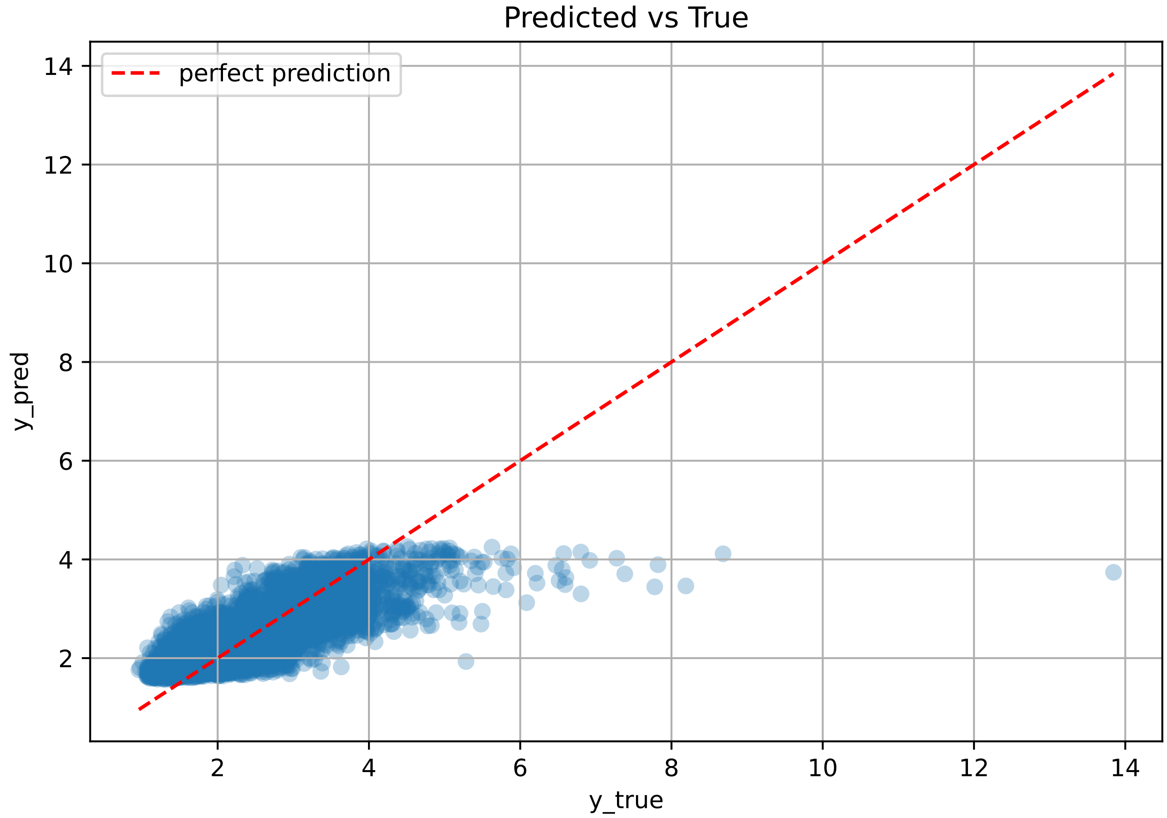
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.013603735620226633, 'reg__num_leaves': 31, 'reg__subsample': 0.5357041062941476, 'reg__colsample_bytree': 0.590747498467615, 'reg__min_child_samples': 38, 'reg__reg_alpha': 2.3432732610525324, 'reg__reg_lambda': 28.248769812378832, 'reg__min_split_gain': 4.479693285231169, 'reg__path_smooth': 4.162563366509822, 'reg__min_child_weight': 0.10451347513816021, 'reg__feature_fraction': 0.5735686139980629, 'reg__bagging_fraction': 0.7163597415912542, 'reg__bagging_freq': 3, 'reg__max_bin': 87, 'reg__min_data_per_group': 194}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.01145708835787607, 'reg__num_leaves': 33, 'reg__subsample': 0.5352472530628856, 'reg__colsample_bytree': 0.5854360929094033, 'reg__min_child_samples': 39, 'reg__reg_alpha': 2.127838006953959, 'reg__reg_lambda': 36.3448122713688, 'reg__min_split_gain': 3.8518084009071987, 'reg__path_smooth': 4.137655034192063, 'reg__min_child_weight': 0.10668216411814088, 'reg__feature_fraction': 0.5805224094842423, 'reg__bagging_fraction': 0.6562640192874258, 'reg__bagging_freq': 4, 'reg__max_bin': 115, 'reg__min_data_per_group': 187}

Rank 9: {'reg___n_estimators': 600, 'reg___max_depth': 4, 'reg___learning_rate': 0.011533191869015495, 'reg___num_leaves': 32, 'reg___subsample': 0.5308089104024999, 'reg___colsample_bytree': 0.598881384055844, 'reg___min_child_samples': 39, 'reg___reg_alpha': 2.3151990749243434, 'reg___reg_lambda': 36.13365571458496, 'reg___min_split_gain': 4.21968414493001, 'reg___path_smooth': 4.163161016453244, 'reg___min_child_weight': 0.1149021222276661, 'reg___feature_fraction': 0.5785270999495782, 'reg___bagging_fraction': 0.6628192661874022, 'reg___bagging_freq': 4, 'reg___max_bin': 91, 'reg___min_data_per_group': 186}

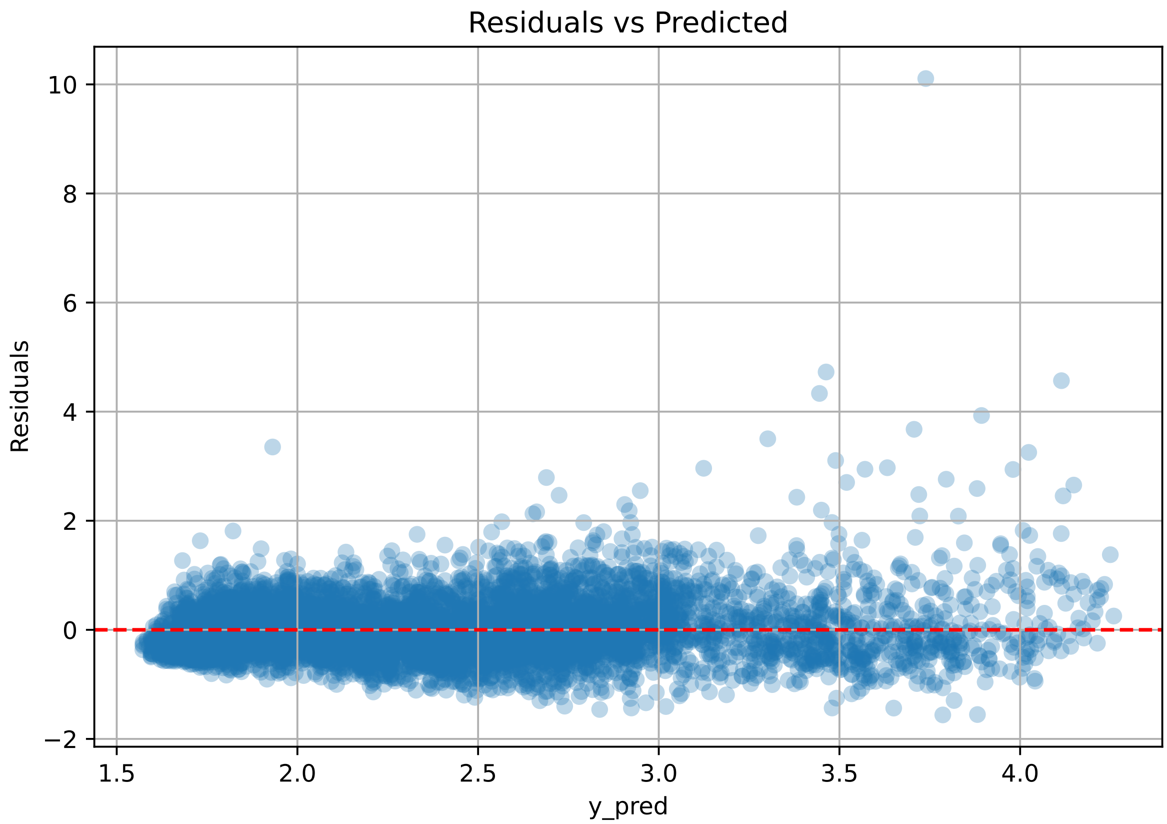
Rank 10: {'reg___n_estimators': 600, 'reg___max_depth': 4, 'reg___learning_rate': 0.013441823297093066, 'reg___num_leaves': 32, 'reg___subsample': 0.5293739013476546, 'reg___colsample_bytree': 0.5960634386709518, 'reg___min_child_samples': 38, 'reg___reg_alpha': 3.3454225757734535, 'reg___reg_lambda': 27.213731889182014, 'reg___min_split_gain': 4.4296480460415655, 'reg___path_smooth': 4.248494512169012, 'reg___min_child_weight': 0.1320022315665814, 'reg___feature_fraction': 0.5731617001002249, 'reg___bagging_fraction': 0.7119730835367742, 'reg___bagging_freq': 3, 'reg___max_bin': 93, 'reg___min_data_per_group': 198}

3.5.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

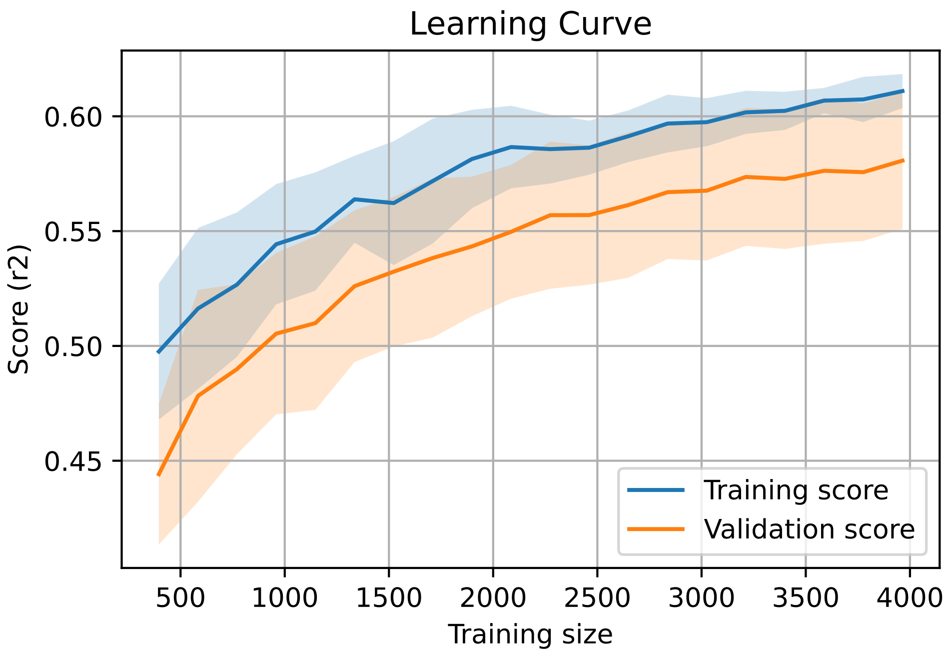
3.5.3 Scatter residuals



3.5.4 Learning curve

$\mathbf{Len(group)}$: 8092, $\mathbf{Len(training)}$: 5665,

$\mathbf{Len(training\_real)}$: 3966, $\mathbf{Len(test\_of\_training)}$: 1699, $\mathbf{Len(test)}$: 2427

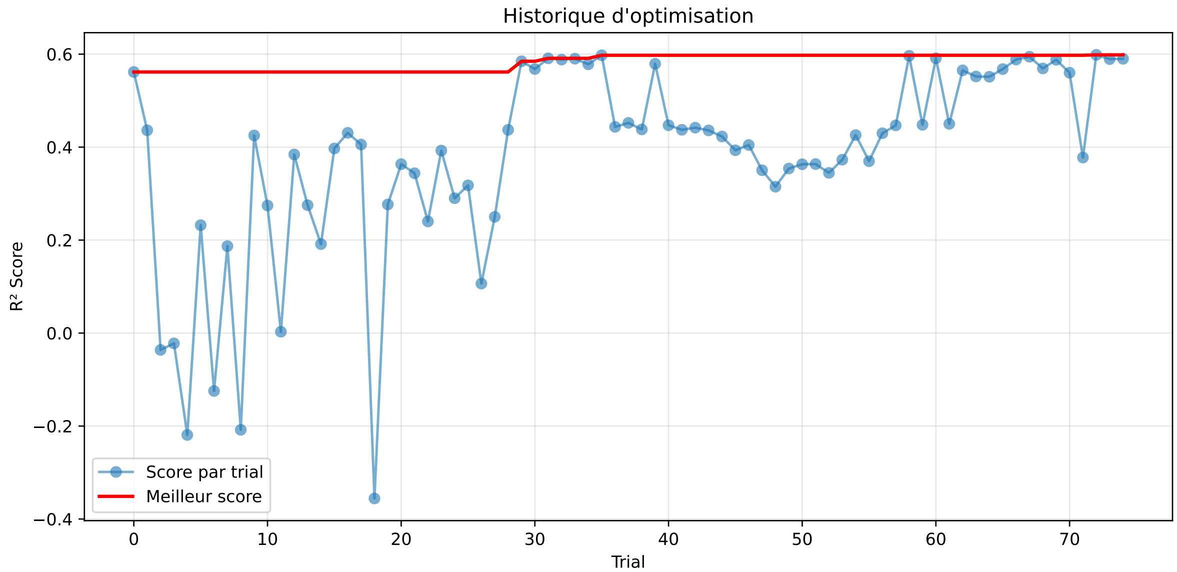


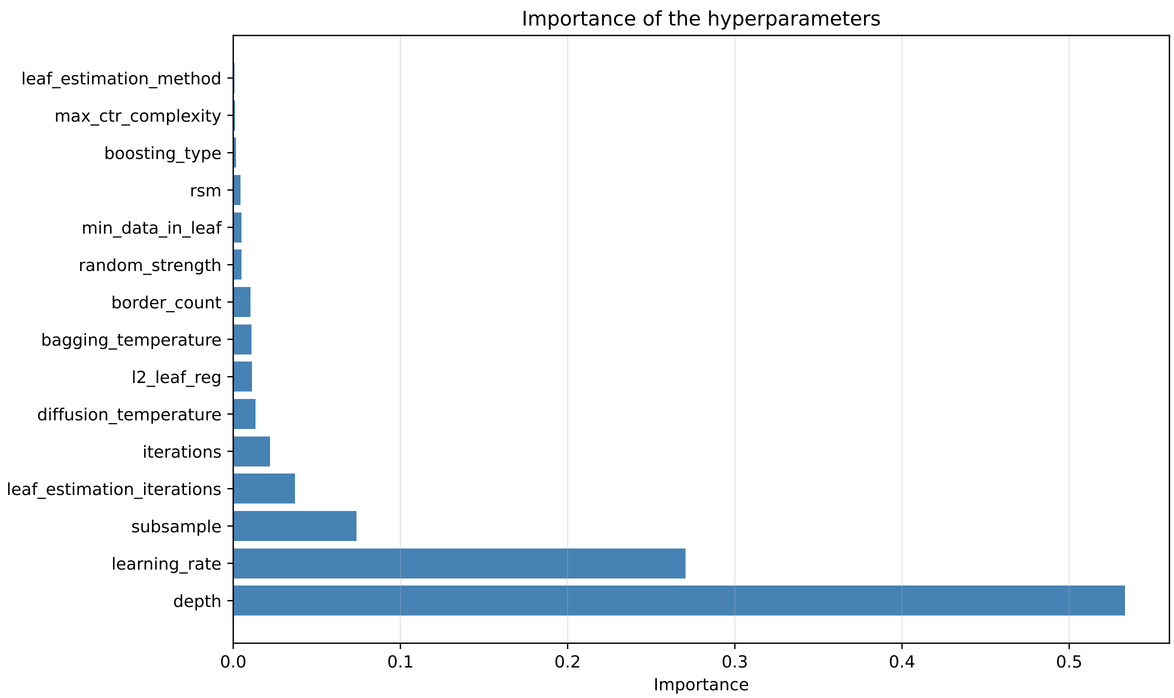
3.6 Model : CatBoost

3.6.1 Gridsearch

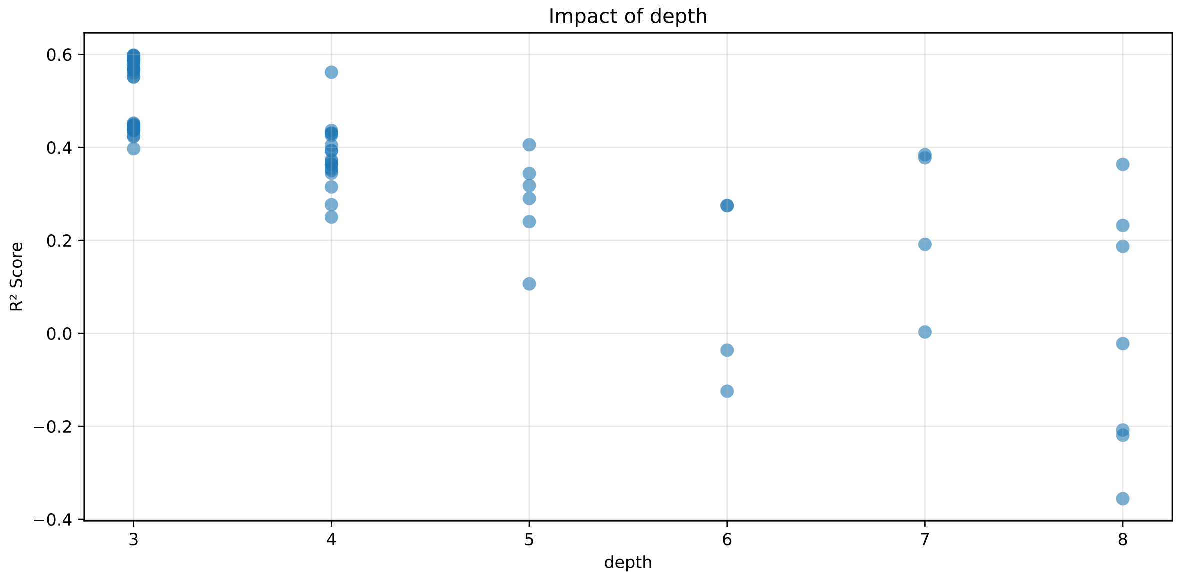
Distribution y_train vs y_test: y_train: mean=2.351, std=0.804, min=0.959, max=13.845 y_test: mean=2.352, std=0.780, min=1.041, max=8.190

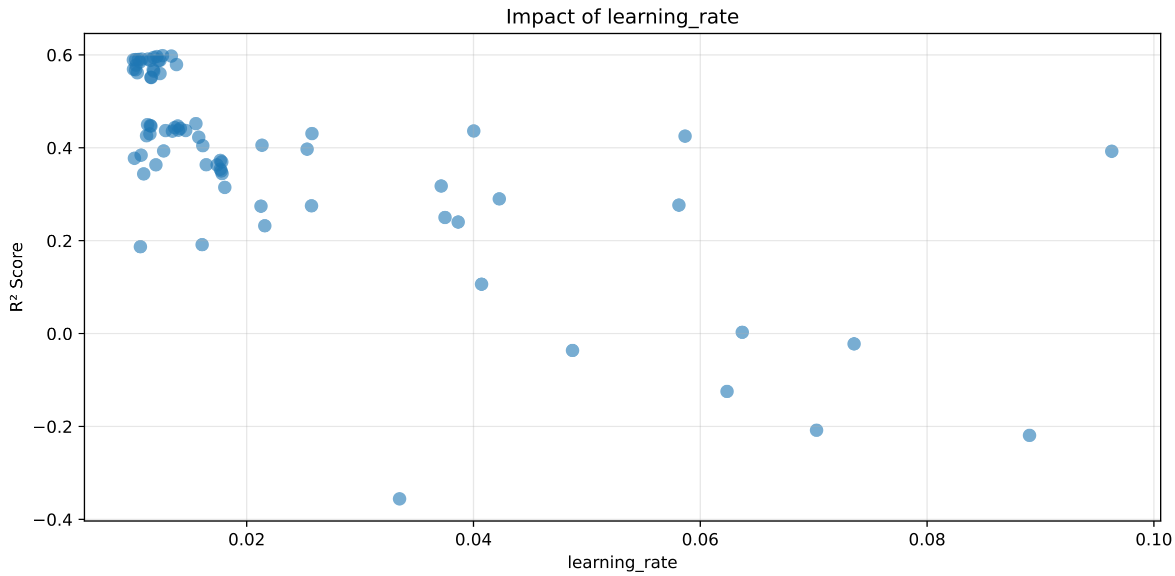
==================== TOP Importance FEATURES ==================== feature importance r_umidity 32.062821 r_temperature_fft_2h_fft_power1 25.718057 r_temperature 7.777311 r_CO2 7.730120 r_NH3 6.813152 r_temperature_fft_2h_fft_period1_h 6.092044 r_umidity_fft_2h_fft_power1 4.097816 r_NH3_fft_2h_fft_power1 2.839355 r_THI 2.702331 r_umidity_fft_2h_fft_period1_h 2.492992 r_NH3_fft_2h_fft_period1_h 1.674003

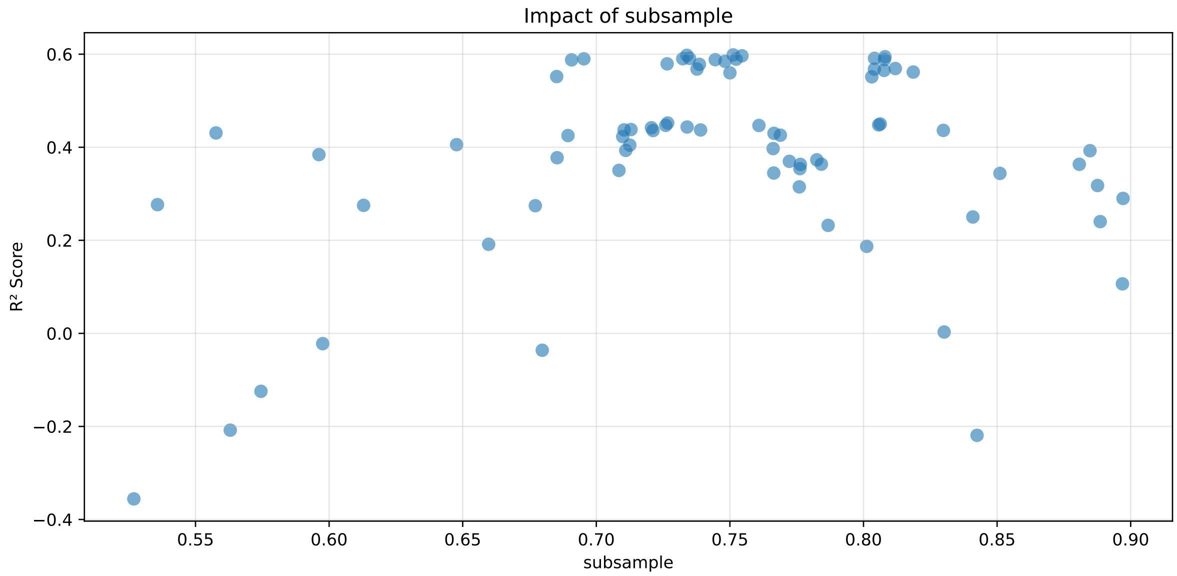




Paramètres avec >5% d'importance : ['depth', 'learning_rate', 'subsample']

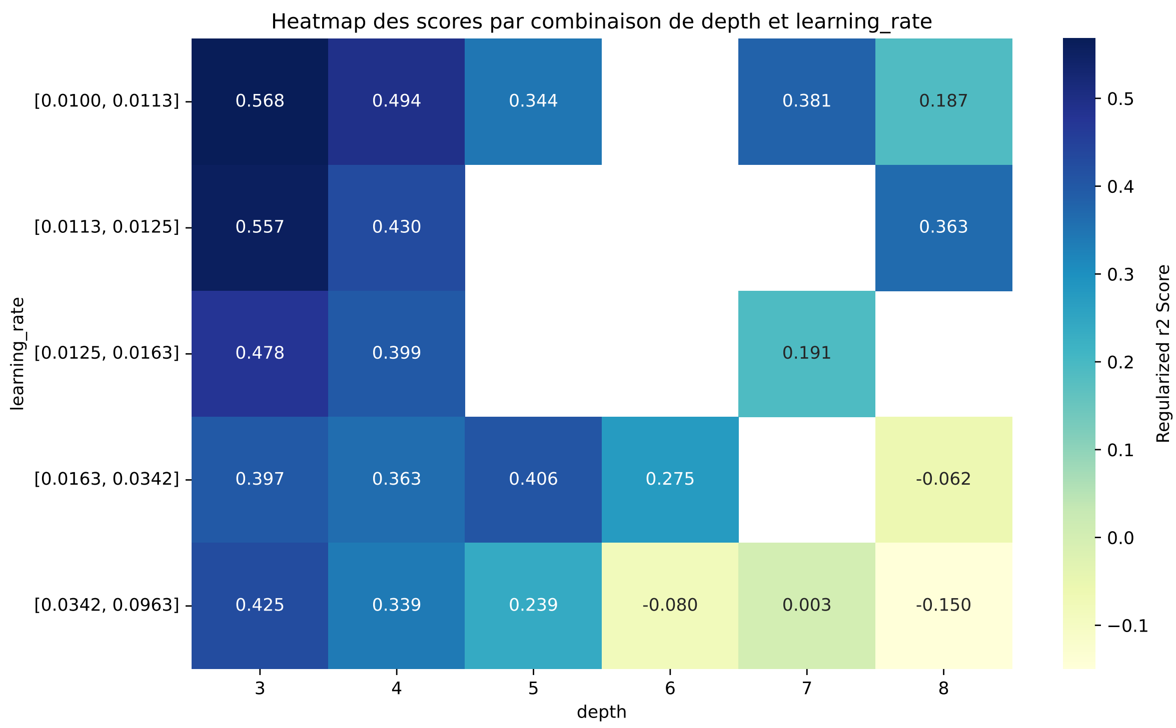

Impact of depth
R² Score
depth
0.6
0.4
0.2
0.0
−0.2
−0.4
3
4
5
6
7
8


Impact of learning_rate
R² Score
learning_rate
0.6
0.4
0.2
0.0
−0.2
−0.4
0.02
0.04
0.06
0.08
0.10




```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6263	0.5984	0.5758	0.0269	0.5984	0.5816	-0.0169	0.9718	1.0465
2	0.6273	0.5975	0.5784	0.0281	0.5975	0.584	-0.0135	0.9774	1.0498
3	0.6243	0.5965	0.5764	0.0281	0.5965	0.5801	-0.0164	0.9725	1.0465
4	0.6208	0.5945	0.5732	0.0277	0.5945	0.5797	-0.0148	0.9751	1.0444
5	0.6191	0.5911	0.5738	0.0279	0.5911	0.5797	-0.0114	0.9808	1.0474
6	0.6173	0.591	0.5701	0.0274	0.591	0.5745	-0.0165	0.9721	1.0446
7	0.6172	0.5902	0.5714	0.0283	0.5902	0.5784	-0.0118	0.9799	1.0458
8	0.6168	0.59	0.5719	0.0292	0.59	0.5794	-0.0106	0.9821	1.0453
9	0.6157	0.5893	0.5704	0.0291	0.5893	0.5763	-0.013	0.9779	1.0448
10	0.6137	0.5883	0.5689	0.0277	0.5883	0.5739	-0.0144	0.9755	1.0432
---	---	---	---	---	---	---	---	---	---
75	0.9327	0.7179	0.6959	0.0235	-0.3558	0.7135	-0.0044	0.9939	1.2991

Hyperparameters:

Rank 1: {’reg__iterations’: 900, ’reg__depth’: 3, ’reg__learning_rate’: 0.012575865788336117, ’reg__l2_leaf_reg’: 19.067692137625876, ’reg__bagging_temperature’: 0.3378447024855285, ’reg__random_strength’: 3.1474452517723077, ’reg__subsample’: 0.7512246119072429, ’reg__min_data_in_leaf’: 44, ’reg__leaf_estimation_iterations’: 1, ’reg__rsm’: 0.7534586512741687, ’reg__border_count’: 39, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 2, ’reg__diffusion_temperature’: 2378.215090092179}

Rank 2: {’reg__iterations’: 700, ’reg__depth’: 3, ’reg__learning_rate’: 0.013356938956044666, ’reg__l2_leaf_reg’: 1.662449592108692, ’reg__bagging_temperature’: 0.7968339194431586, ’reg__random_strength’: 2.74604823466592, ’reg__subsample’: 0.7338983744054953, ’reg__min_data_in_leaf’: 17, ’reg__leaf_estimation_iterations’: 2, ’reg__rsm’: 0.674976787947056, ’reg__border_count’: 63, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 3, ’reg__diffusion_temperature’: 2057.2651064878096}

Rank 3: {’reg__iterations’: 900, ’reg__depth’: 3, ’reg__learning_rate’: 0.01208023703077072, ’reg__l2_leaf_reg’: 14.035054513747053, ’reg__bagging_temperature’: 0.5801266747686633, ’reg__random_strength’: 3.4860122428719325, ’reg__subsample’: 0.7545226455013189, ’reg__min_data_in_leaf’: 15, ’reg__leaf_estimation_iterations’: 1, ’reg__rsm’: 0.7235837364687119, ’reg__border_count’: 33, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Plain’, ’reg__max_ctr_complexity’: 2, ’reg__diffusion_temperature’: 1140.5866232755657}

Rank 4: {’reg__iterations’: 900, ’reg__depth’: 3, ’reg__learning_rate’: 0.011799452900833977, ’reg__l2_leaf_reg’: 18.236124518550263, ’reg__bagging_temperature’: 0.30977786289296305, ’reg__random_strength’: 3.209480988243406, ’reg__subsample’: 0.8080482847341189, ’reg__min_data_in_leaf’: 20, ’reg__leaf_estimation_iterations’: 1, ’reg__rsm’:

0.7759055853730286, 'reg___border_count': 57, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 634.9422471082032}

Rank 5: {'reg___iterations': 900, 'reg___depth': 3, 'reg___learning_rate': 0.011298947926229331, 'reg___l2_leaf_reg': 14.613100510090089, 'reg___bagging_temperature': 1.2007097180993027, 'reg___random_strength': 3.517921222644533, 'reg___subsample': 0.8041260747130191, 'reg___min_data_in_leaf': 29, 'reg___leaf_estimation_iterations': 1, 'reg___rsm': 0.7776417116909238, 'reg___border_count': 75, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 967.0728725098851}

Rank 6: {'reg___iterations': 800, 'reg___depth': 3, 'reg___learning_rate': 0.01074262966012559, 'reg___l2_leaf_reg': 24.365971002650472, 'reg___bagging_temperature': 1.4555384726996539, 'reg___random_strength': 2.703334413647759, 'reg___subsample': 0.7348932395962637, 'reg___min_data_in_leaf': 38, 'reg___leaf_estimation_iterations': 4, 'reg___rsm': 0.6850121922569306, 'reg___border_count': 64, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 1932.6037196180123}

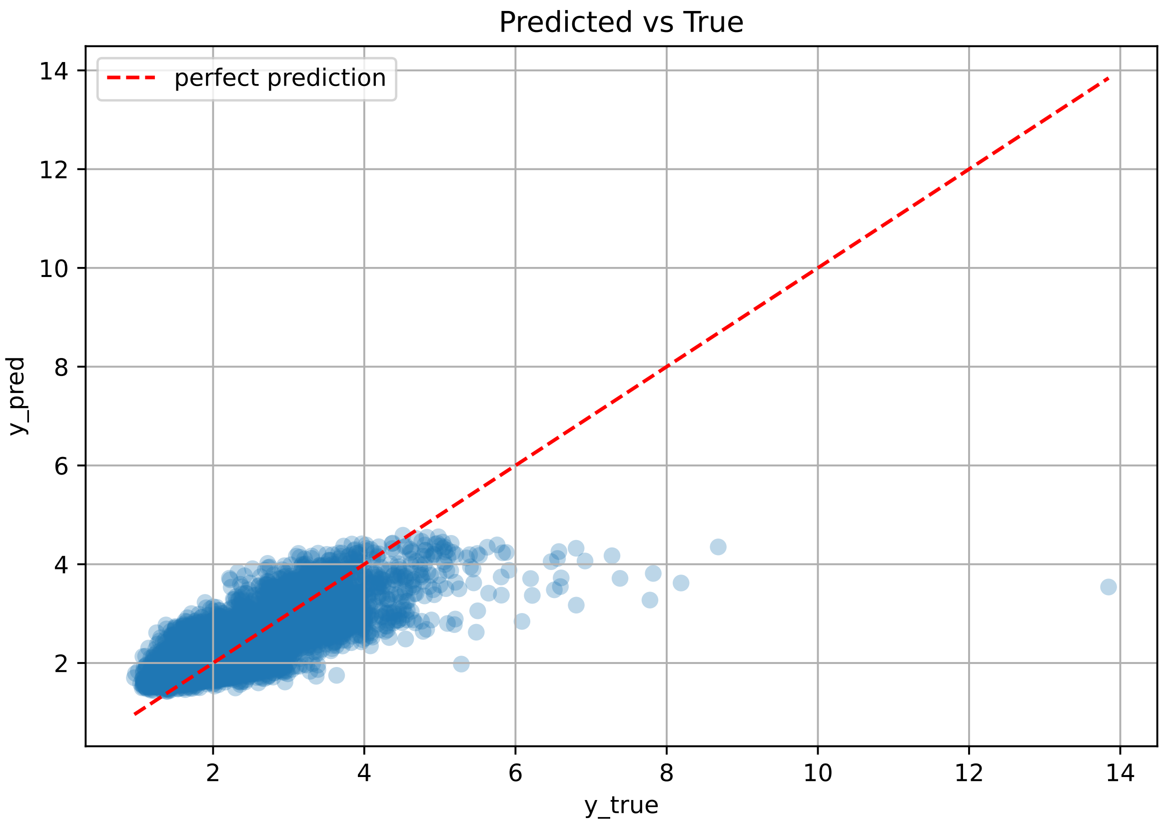
Rank 7: {'reg___iterations': 800, 'reg___depth': 3, 'reg___learning_rate': 0.010453780789191795, 'reg___l2_leaf_reg': 2.220843791685534, 'reg___bagging_temperature': 1.2562615117038254, 'reg___random_strength': 2.5863729756047498, 'reg___subsample': 0.7323575691411689, 'reg___min_data_in_leaf': 18, 'reg___leaf_estimation_iterations': 2, 'reg___rsm': 0.6615911424914236, 'reg___border_count': 65, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 1905.983170314546}

Rank 8: {'reg___iterations': 900, 'reg___depth': 3, 'reg___learning_rate': 0.010205994510705979, 'reg___l2_leaf_reg': 18.658202869745125, 'reg___bagging_temperature': 0.23008673982490802, 'reg___random_strength': 3.1792039383265096, 'reg___subsample': 0.6953450403358963, 'reg___min_data_in_leaf': 20, 'reg___leaf_estimation_iterations': 3, 'reg___rsm': 0.7790375002649138, 'reg___border_count': 91, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 2369.7052326986527}

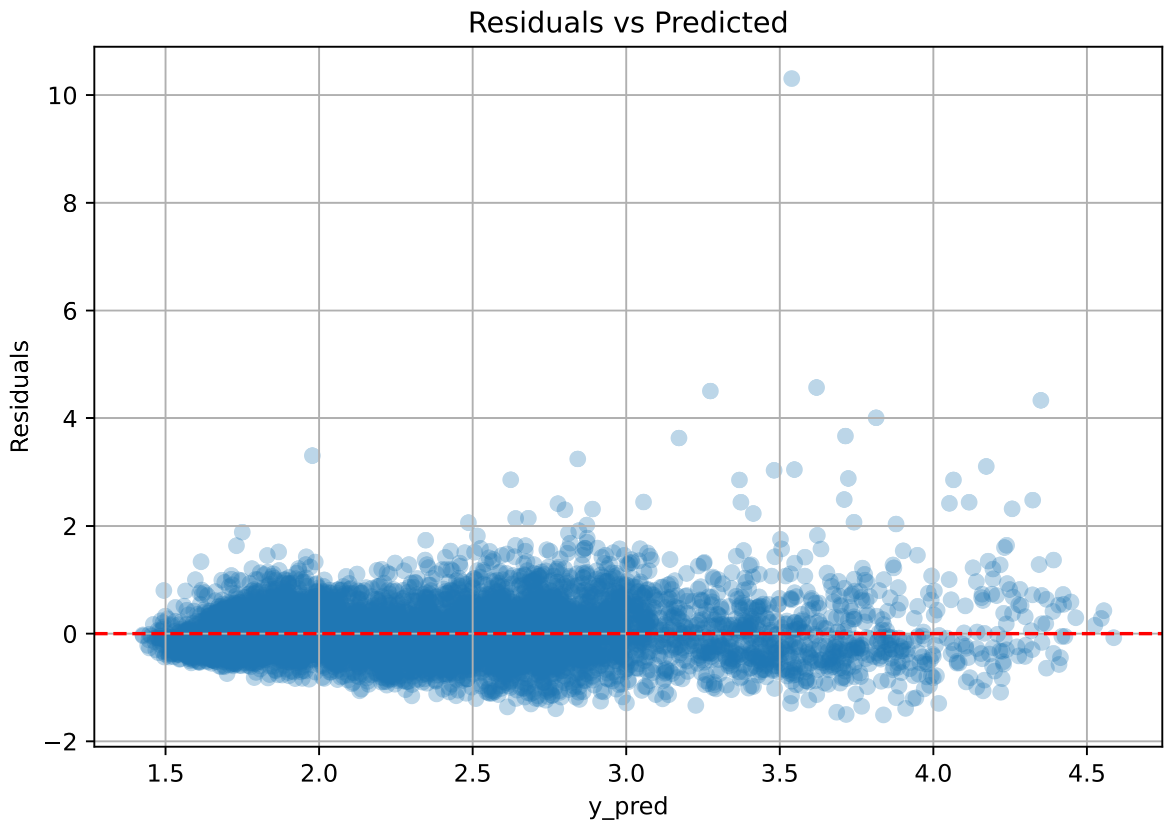
Rank 9: {'reg___iterations': 900, 'reg___depth': 3, 'reg___learning_rate': 0.010006997051422882, 'reg___l2_leaf_reg': 10.225875002227065, 'reg___bagging_temperature': 0.274479231491775, 'reg___random_strength': 3.1825900540918424, 'reg___subsample': 0.7523336796653131, 'reg___min_data_in_leaf': 20, 'reg___leaf_estimation_iterations': 3, 'reg___rsm': 0.7526237764599517, 'reg___border_count': 67, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 2434.8971791454956}

Rank 10: {'reg___iterations': 900, 'reg___depth': 3, 'reg___learning_rate': 0.012353051164004897, 'reg___l2_leaf_reg': 31.55279323851432, 'reg___bagging_temperature': 0.33938042420296477, 'reg___random_strength': 4.370843367410838, 'reg___subsample': 0.8078363704252927, 'reg___min_data_in_leaf': 26, 'reg___leaf_estimation_iterations': 1, 'reg___rsm': 0.6054461938590596, 'reg___border_count': 57, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 1399.502831026168}

3.6.2 Scatter



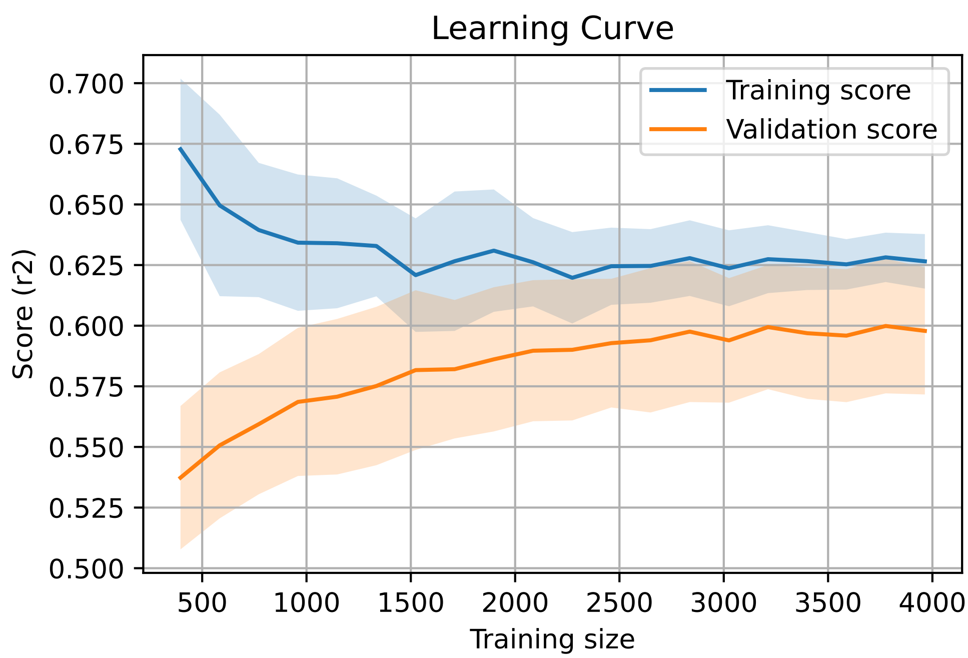
3.6.3 Scatter residuals



3.6.4 Learning curve

Len(group) : 8092, **Len(training)** : 5665,

Len(training_real) : 3966, **Len(test_of_training)** : 1699, **Len(test)** : 2427



3.7 Sumup-table

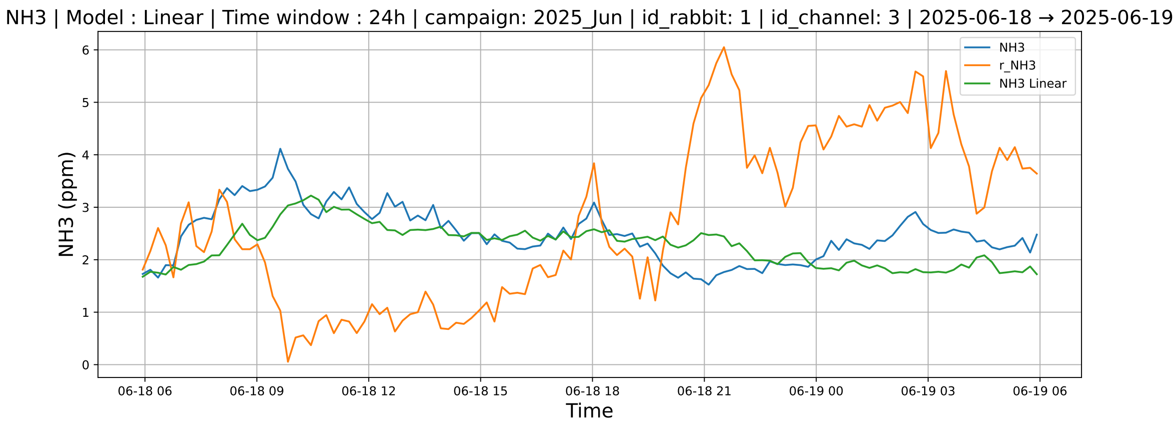
model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.504	0.499	0.032	0.478	0.479	-0.02	0.96	1.011	nan
LGBM	0.609	0.58	0.029	0.561	0.58	-0.001	0.999	1.05	0.5805
CatBoost	0.626	0.598	0.027	0.576	0.582	-0.017	0.972	1.046	0.5984

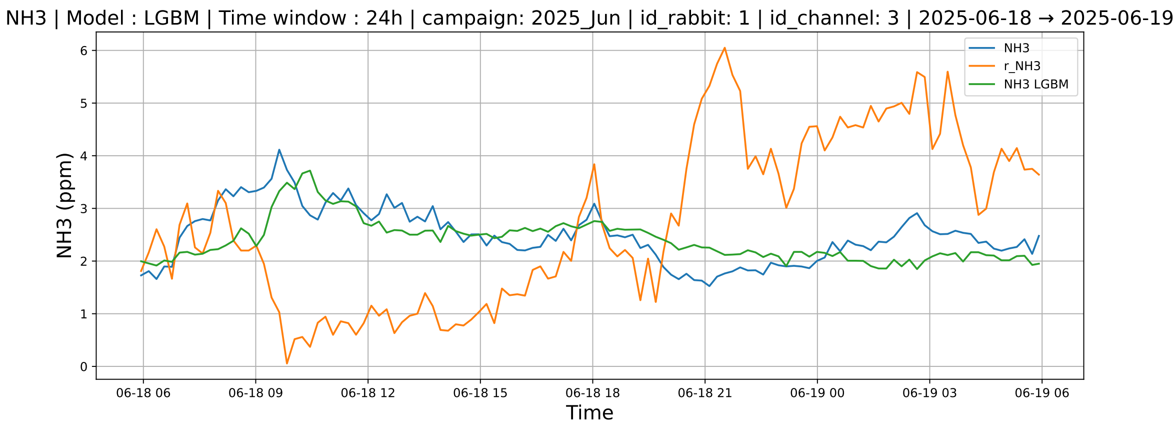
4 Plot

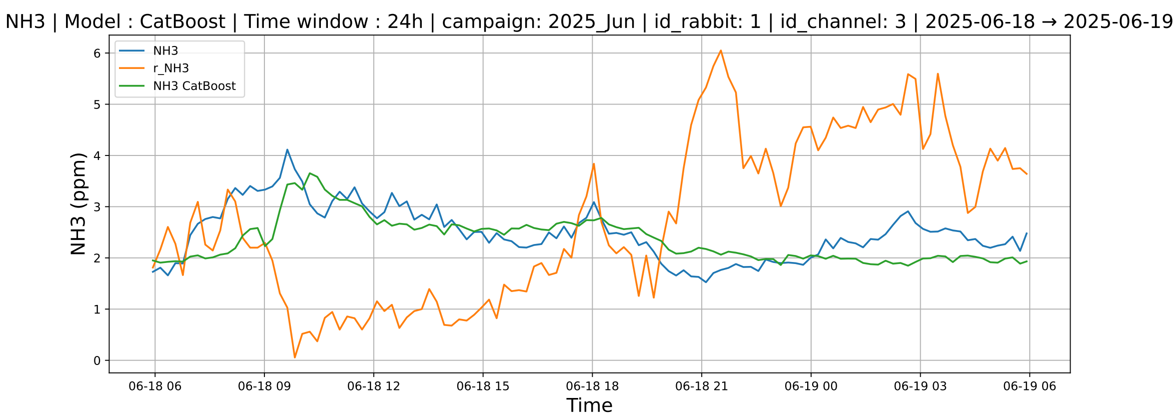
4.1 Standalone plots

4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

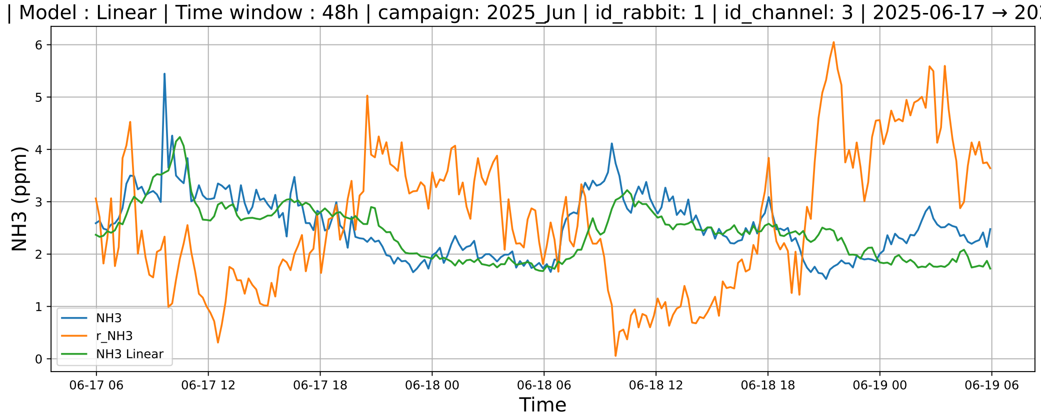




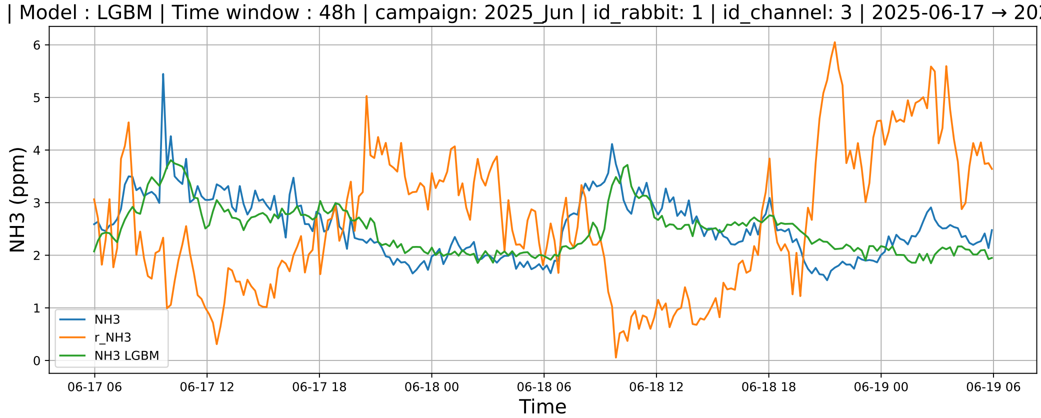


4.1.1.2 Time window : 48

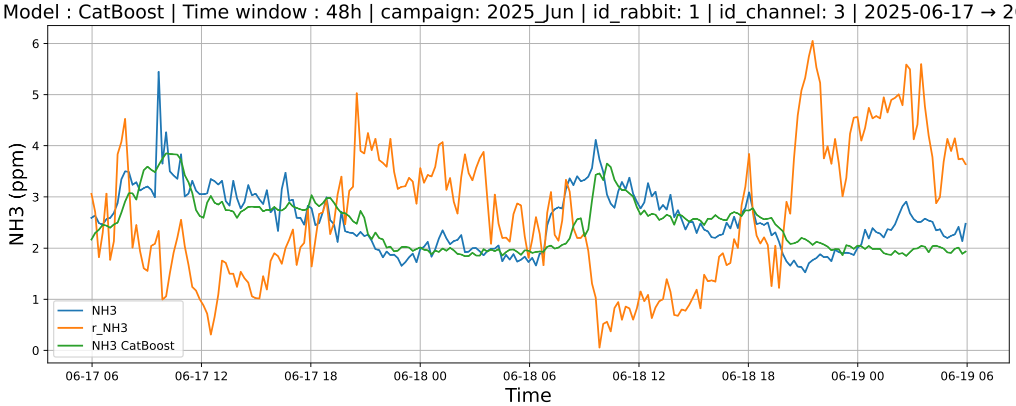
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



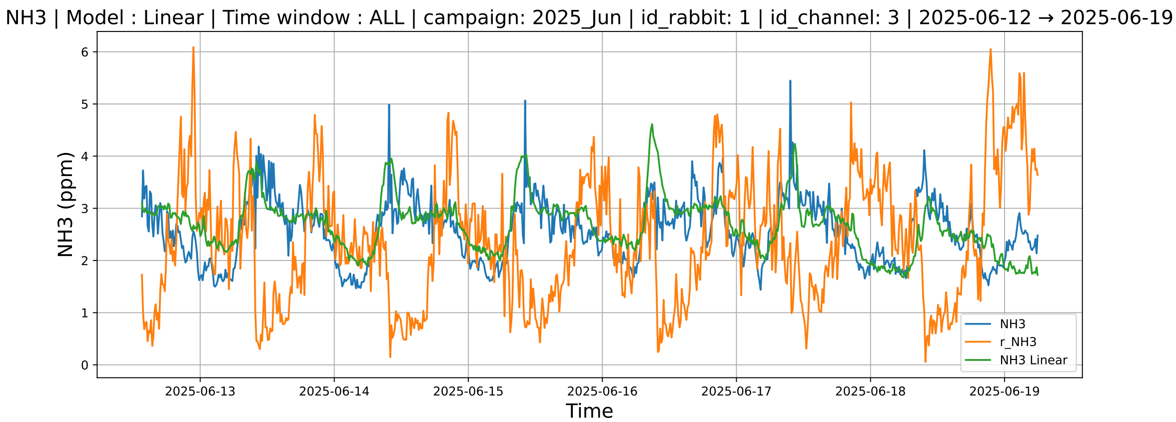
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

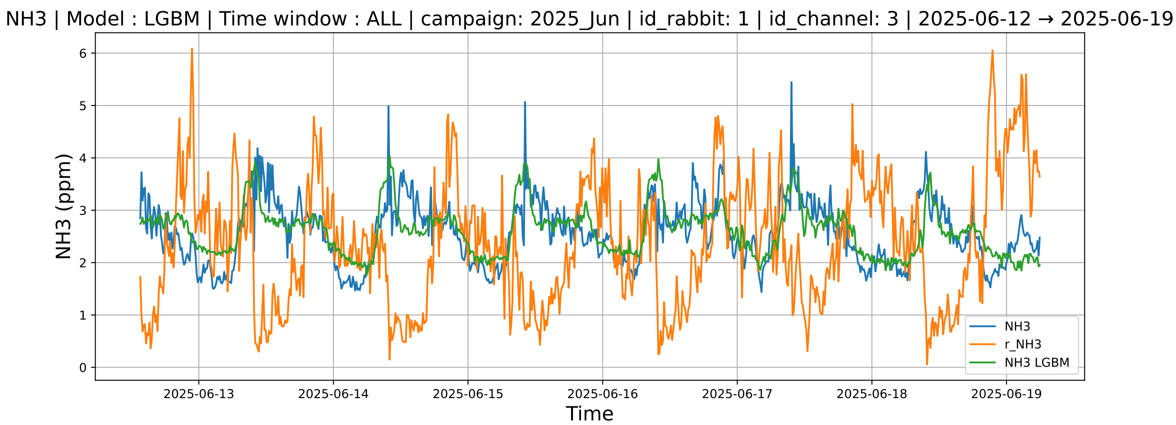


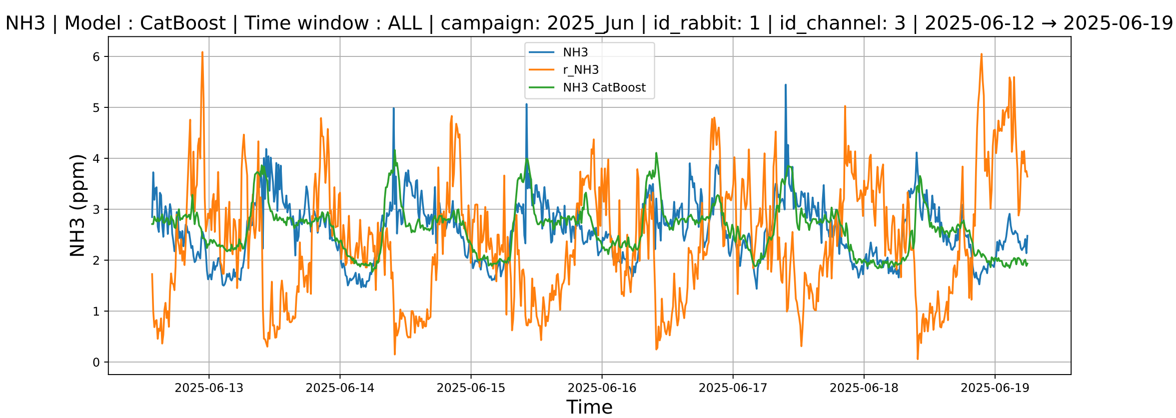
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.1.1.3 Time window : ALL

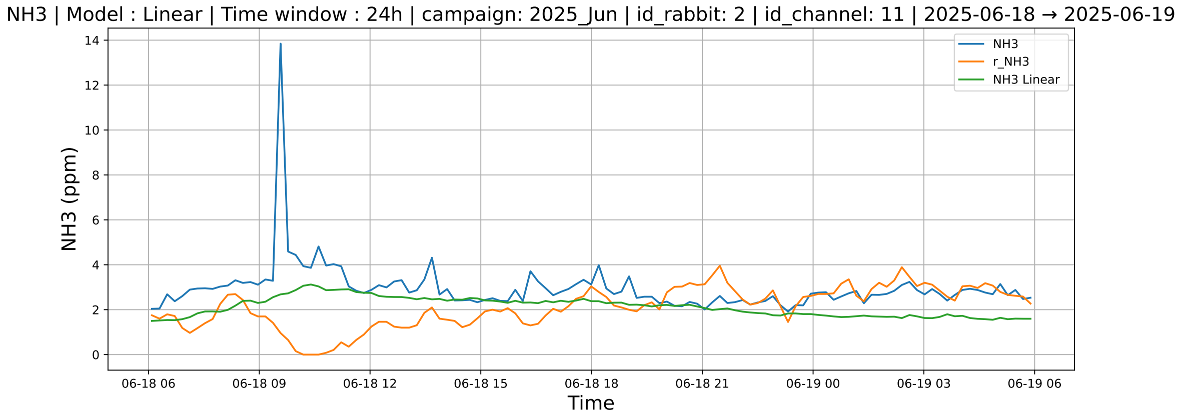


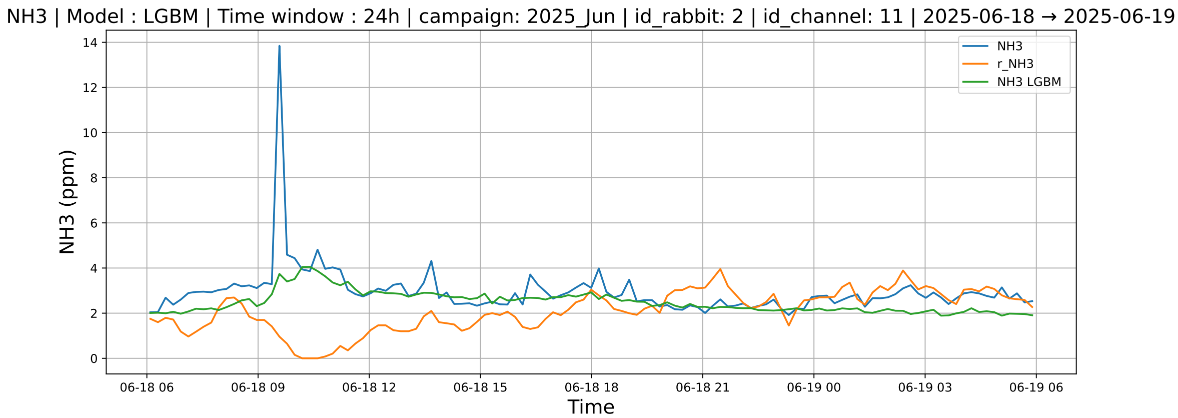


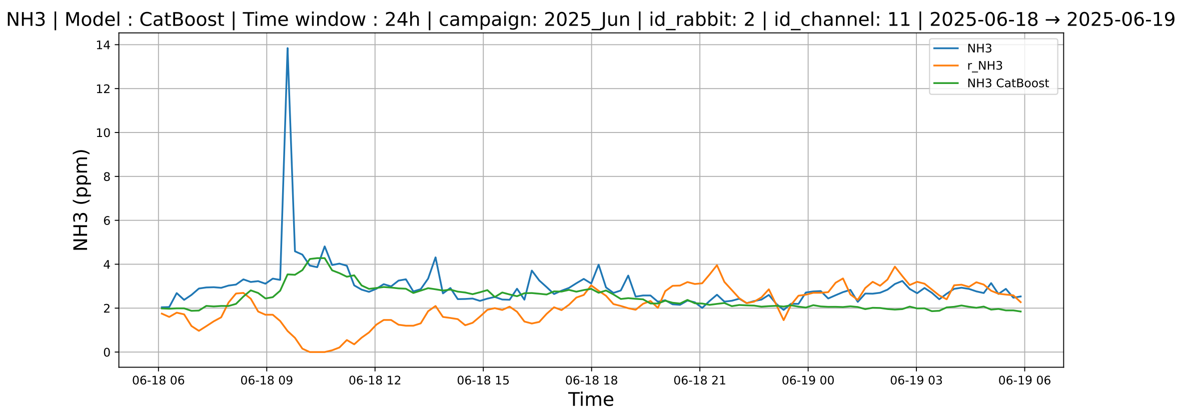


4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

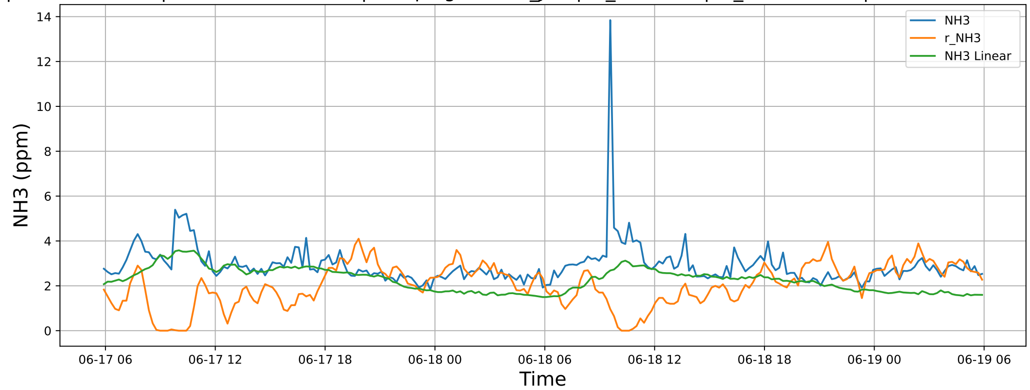




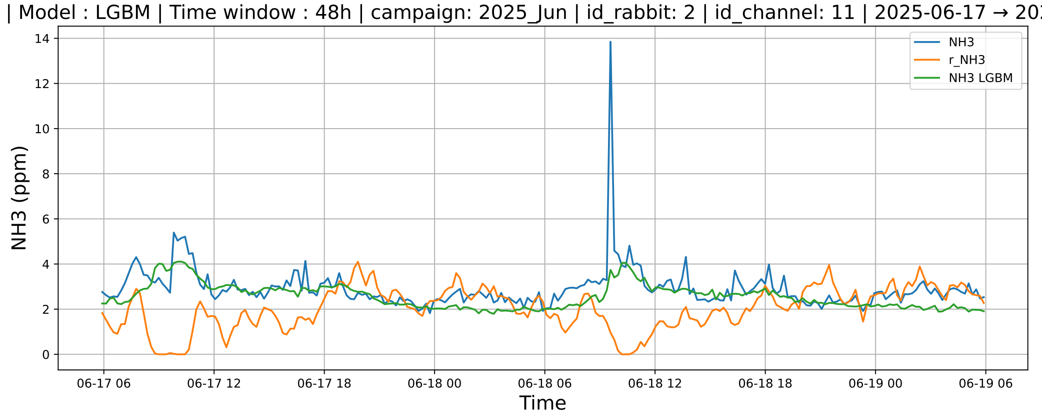


4.1.2.2 Time window : 48

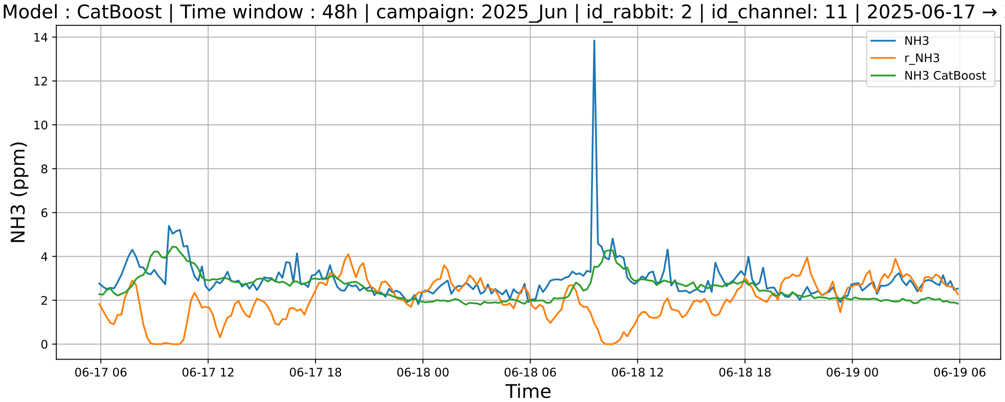
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



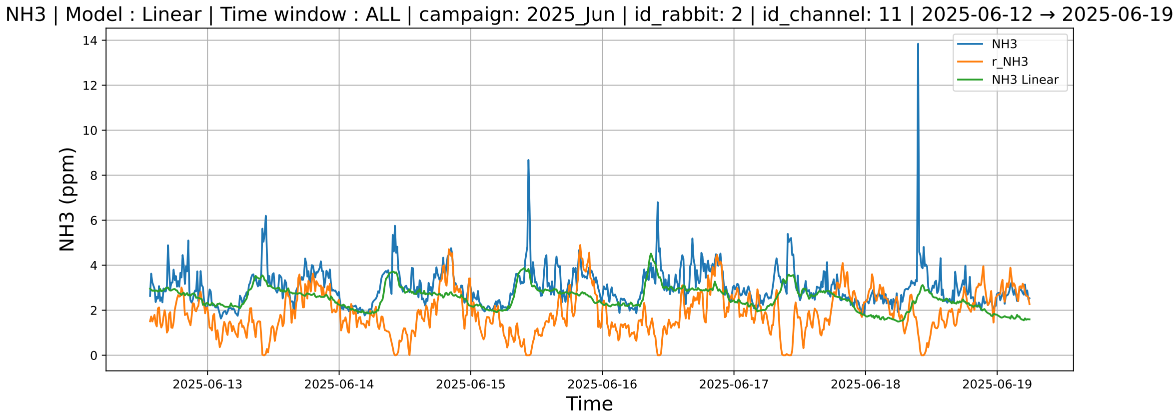
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

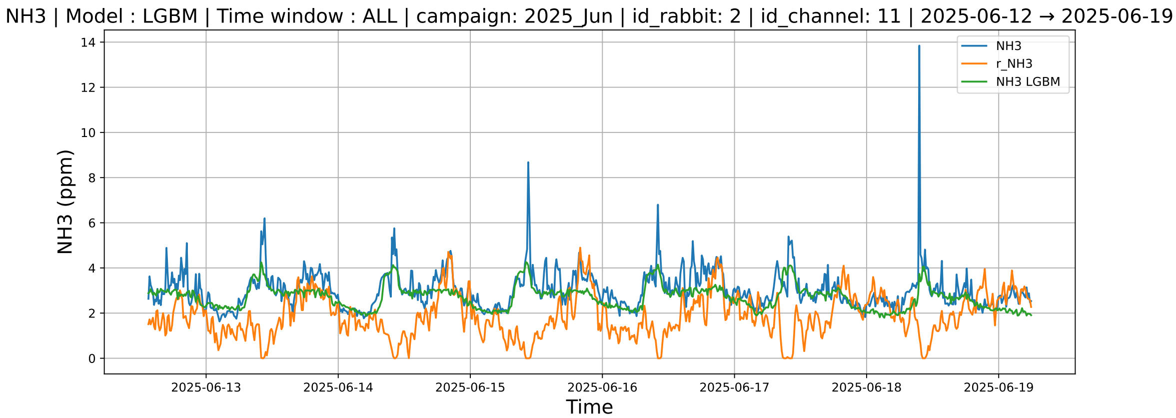


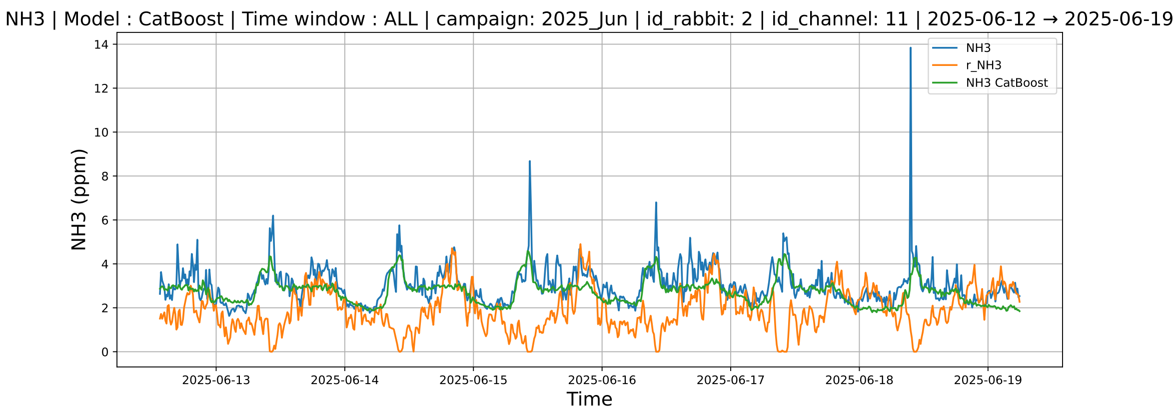
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.1.2.3 Time window : ALL

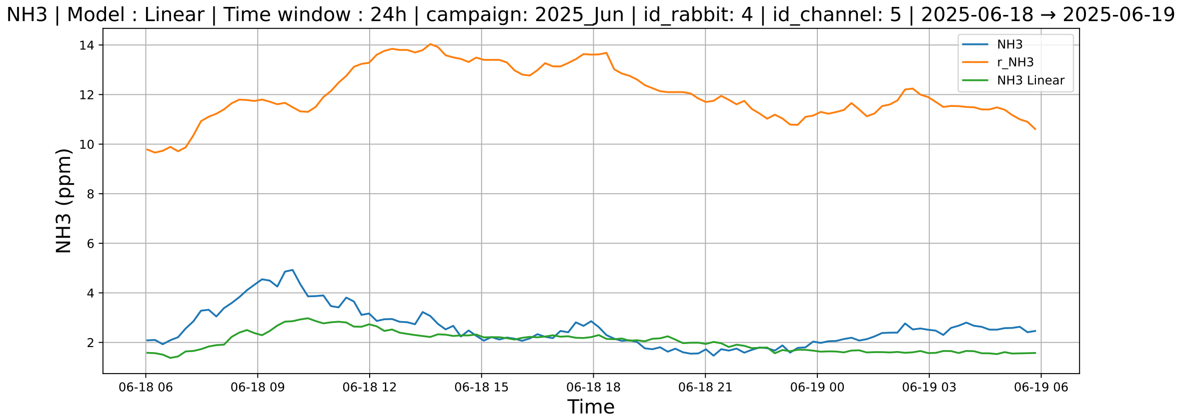


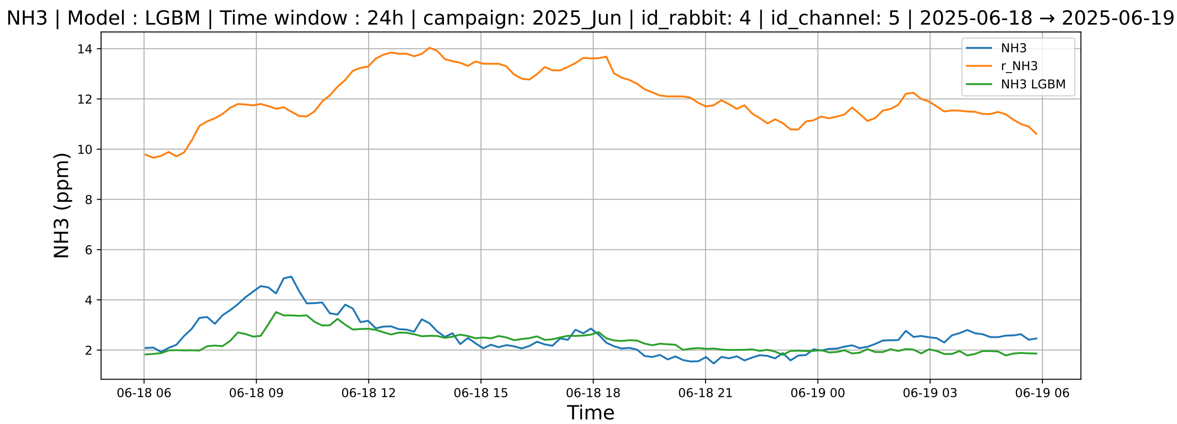


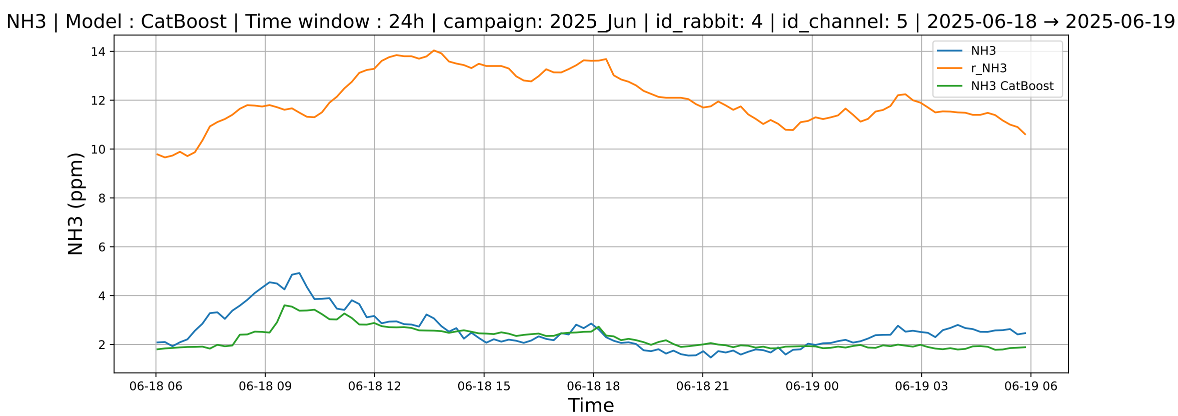


4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

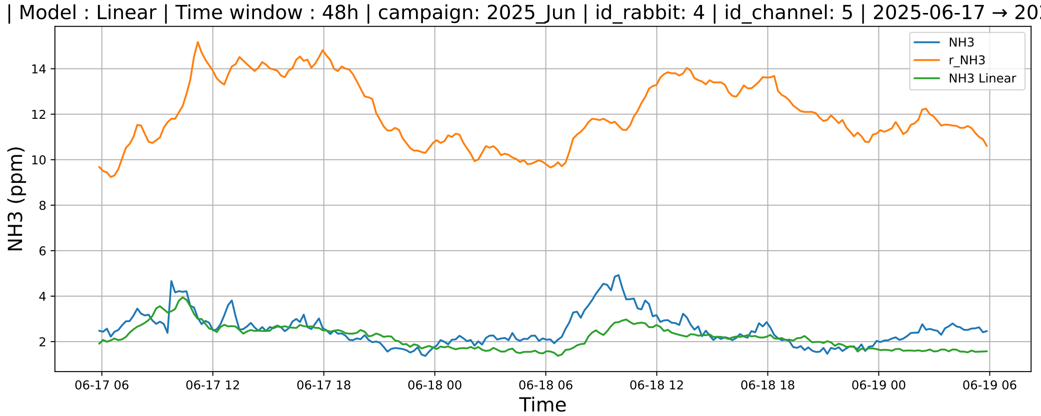




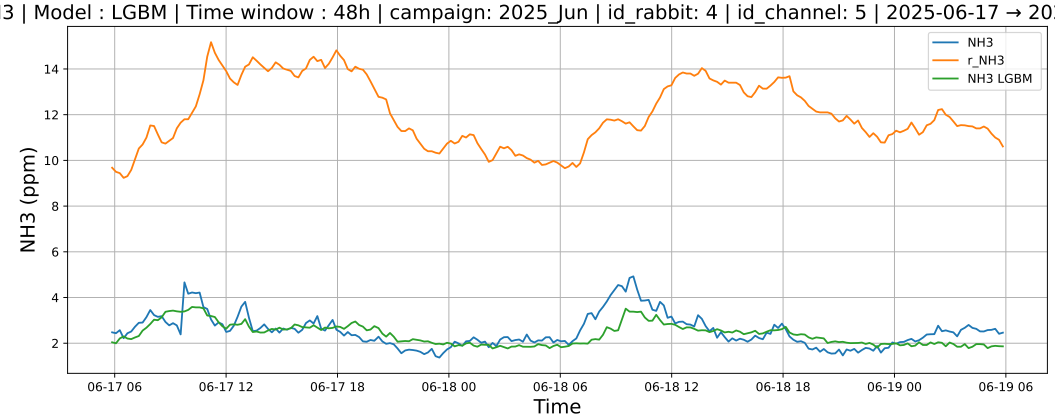


4.1.3.2 Time window : 48

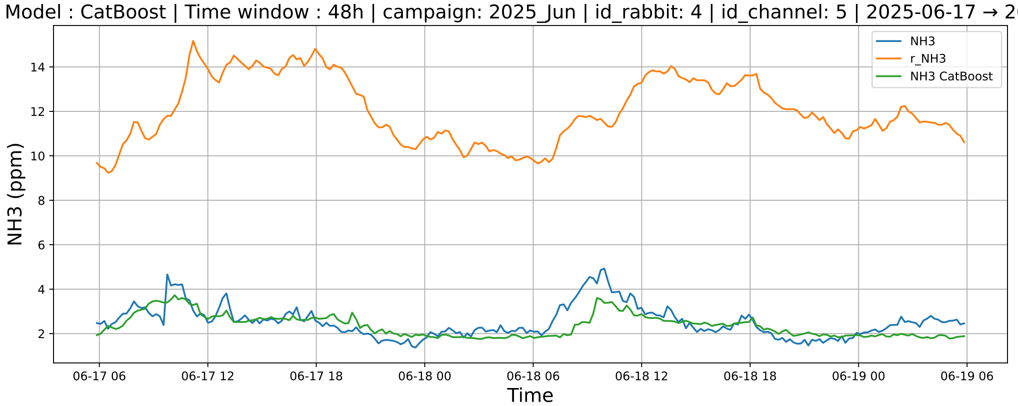
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



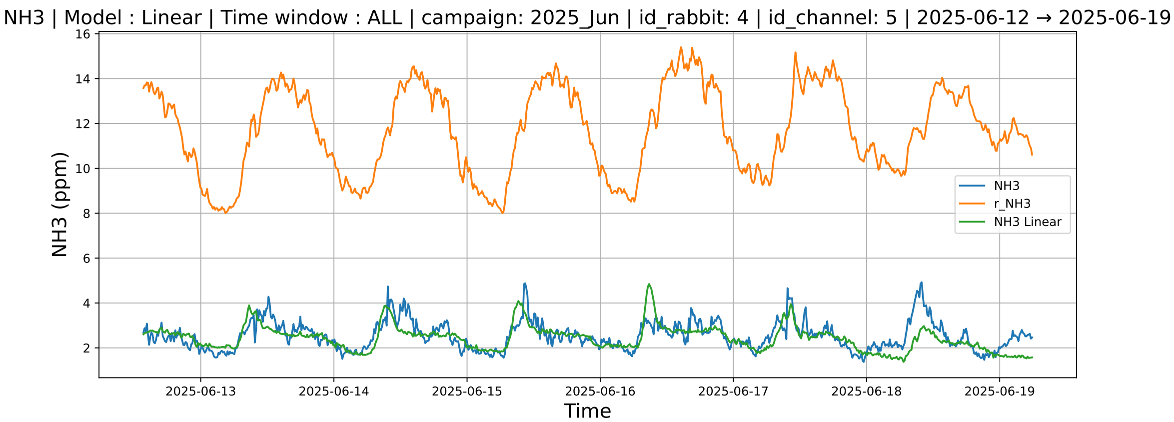
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

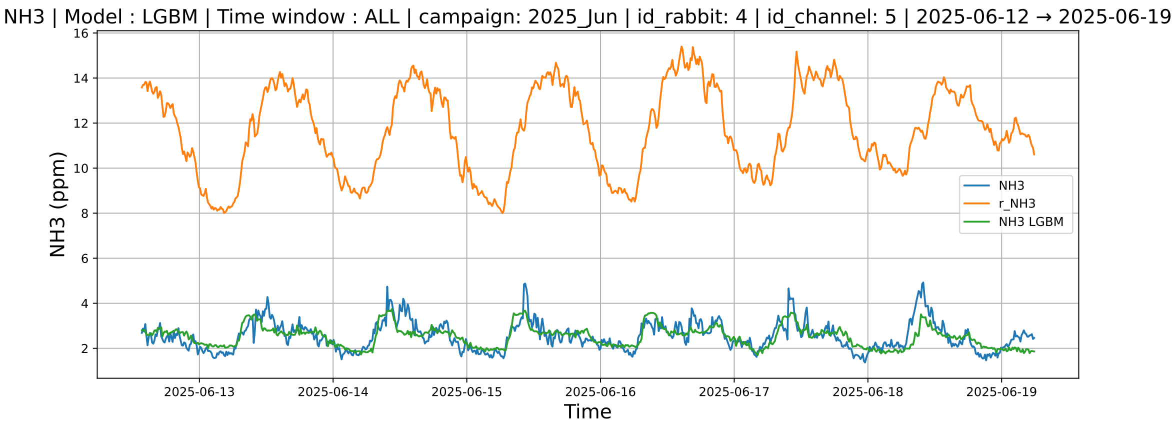


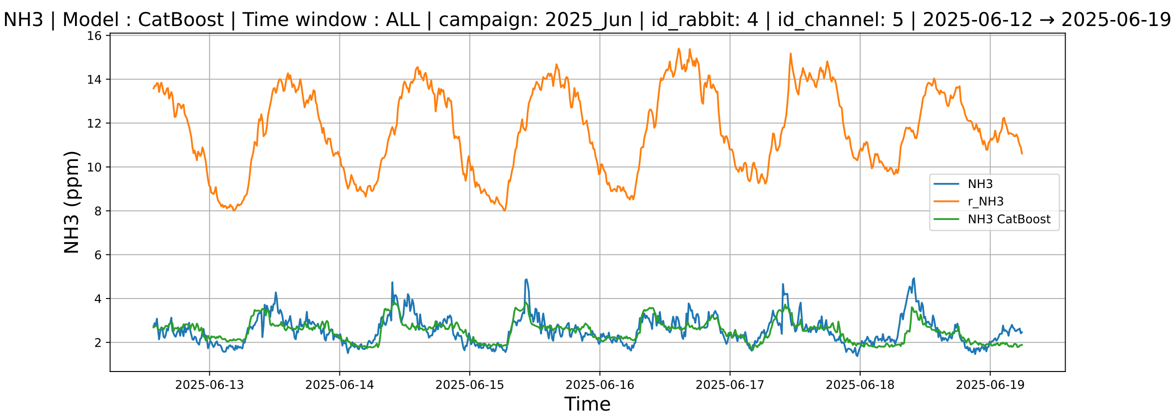
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.1.3.3 Time window : ALL

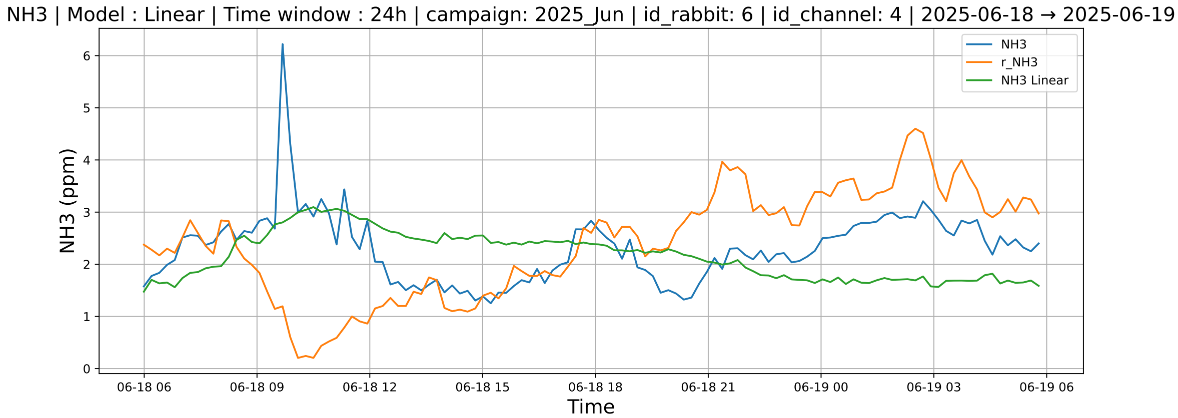


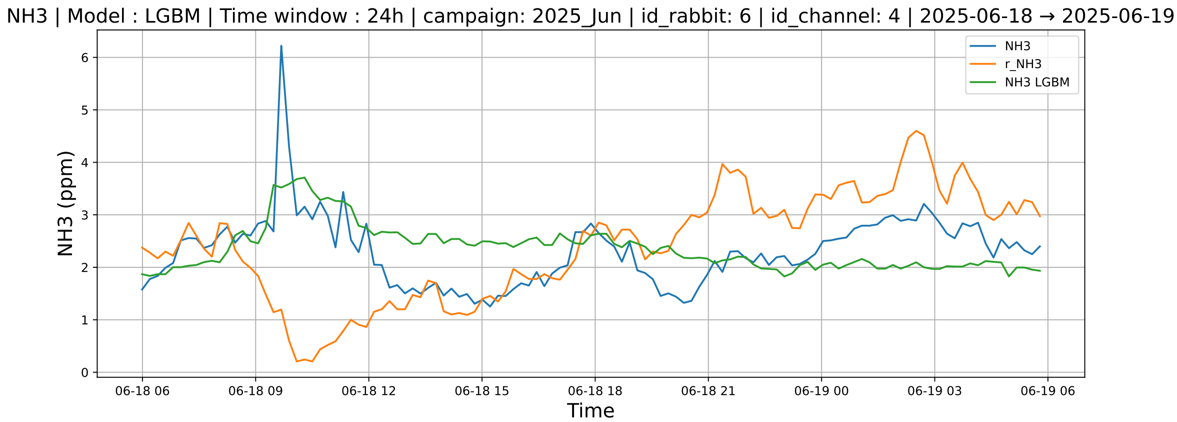


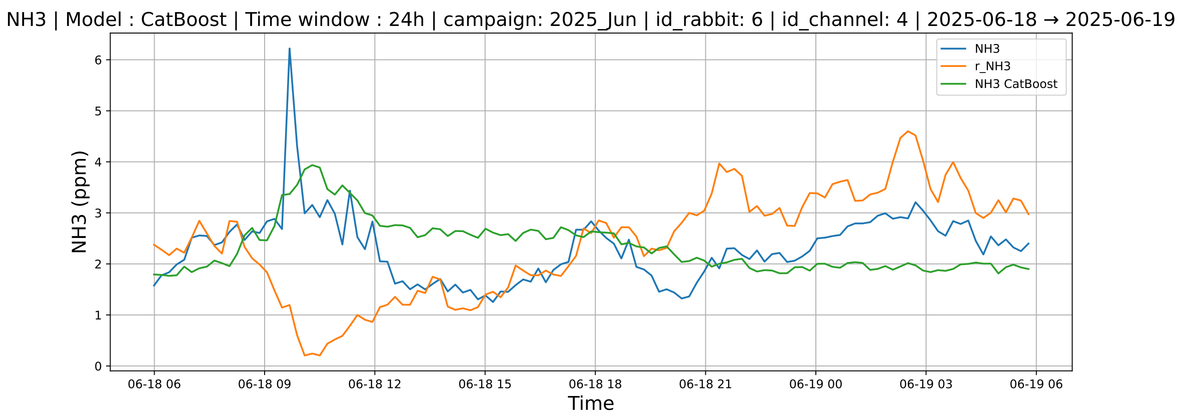


4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

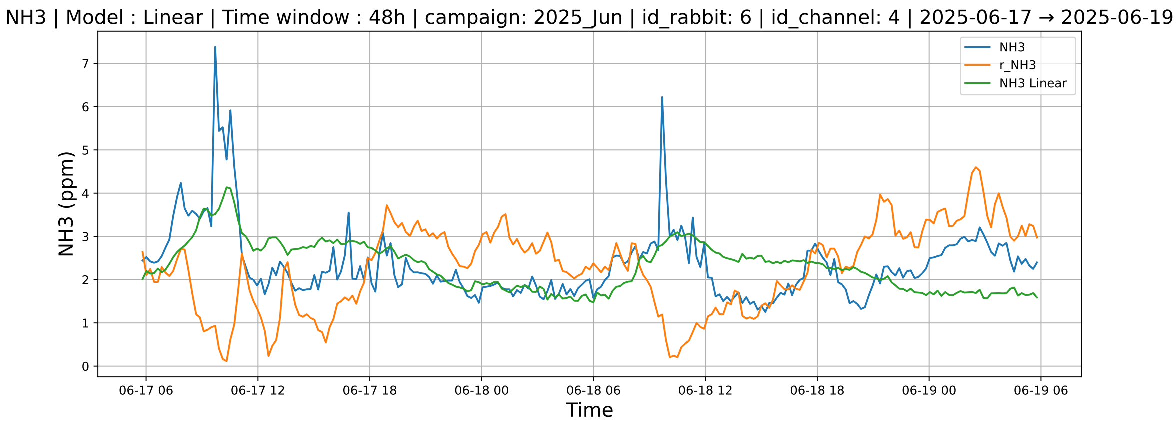
4.1.4.1 Time window : 24

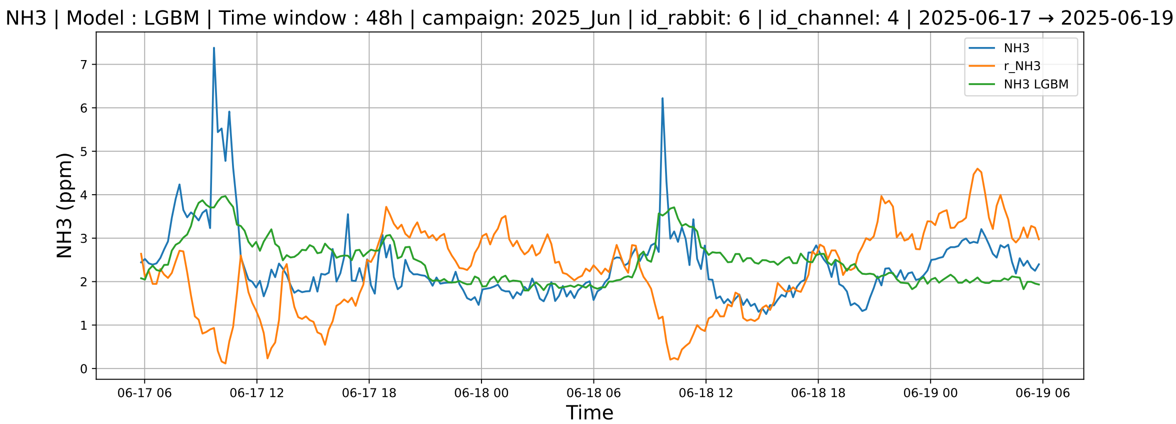


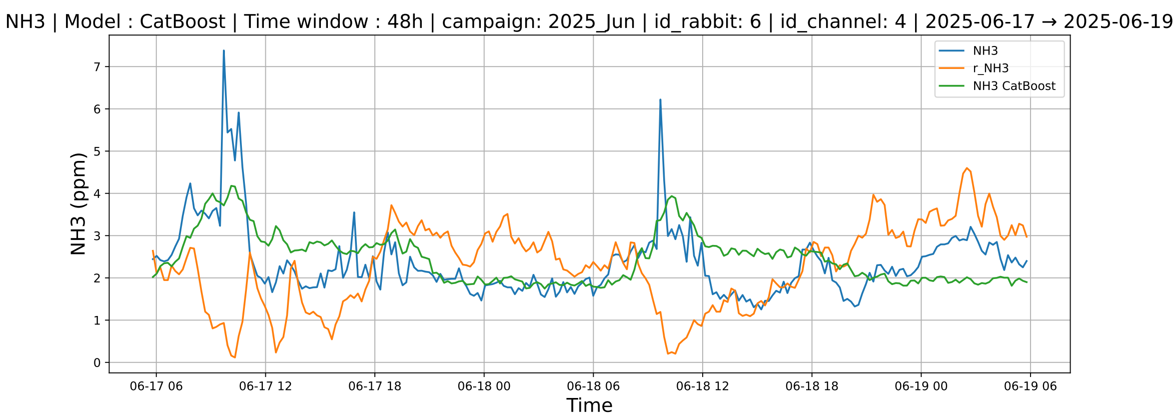




4.1.4.2 Time window : 48

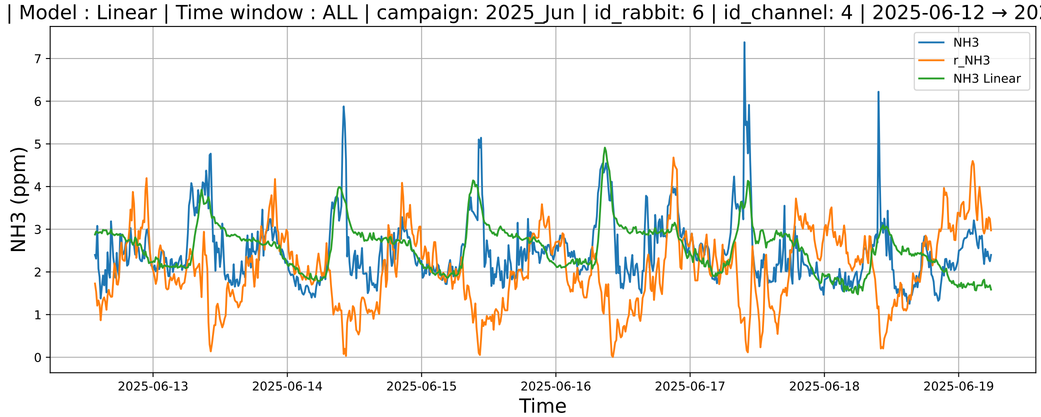




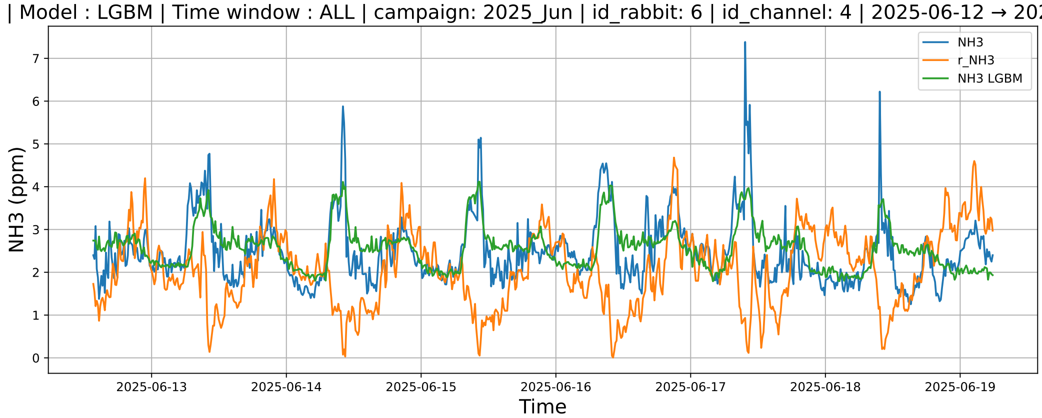


4.1.4.3 Time window : ALL

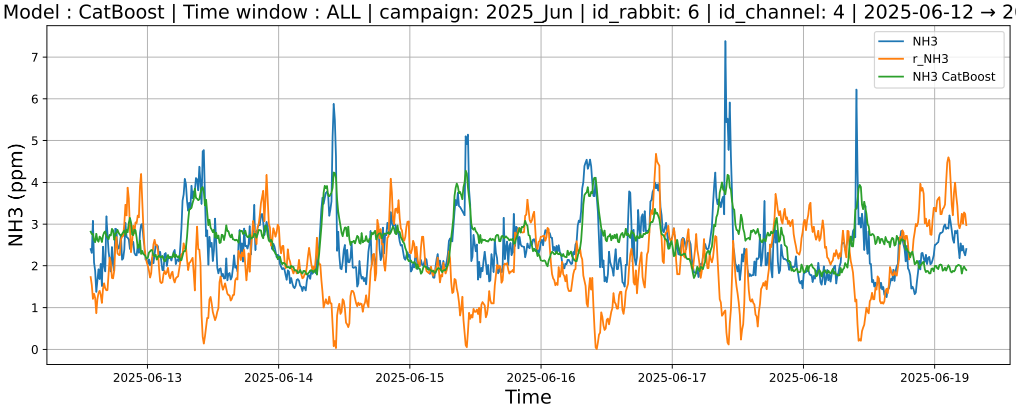
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19

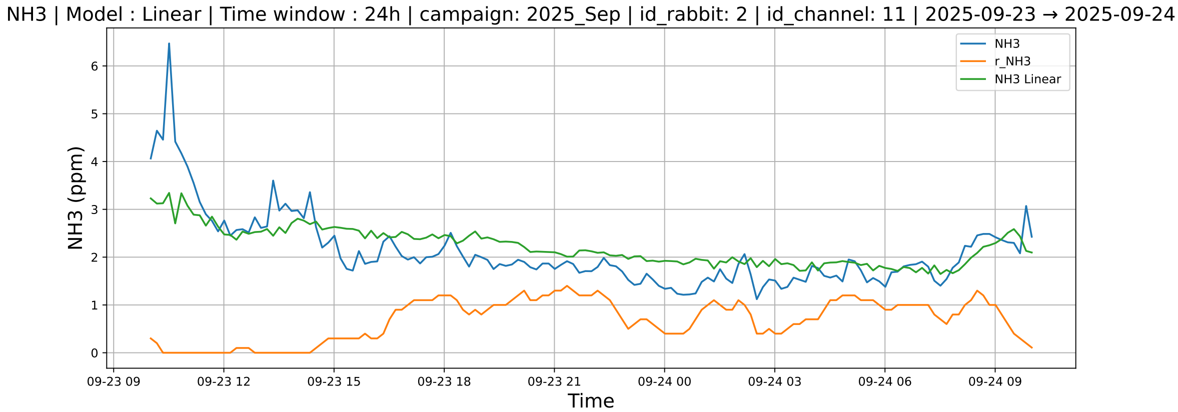


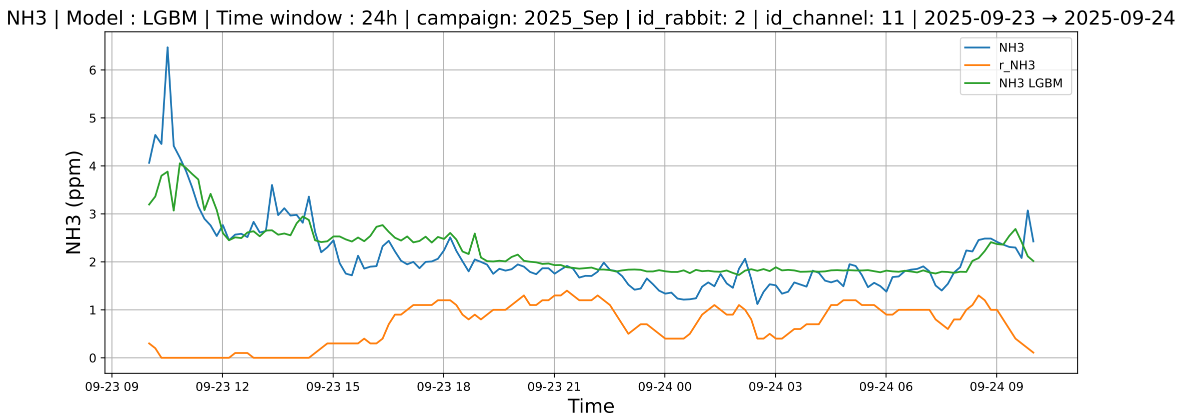
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19

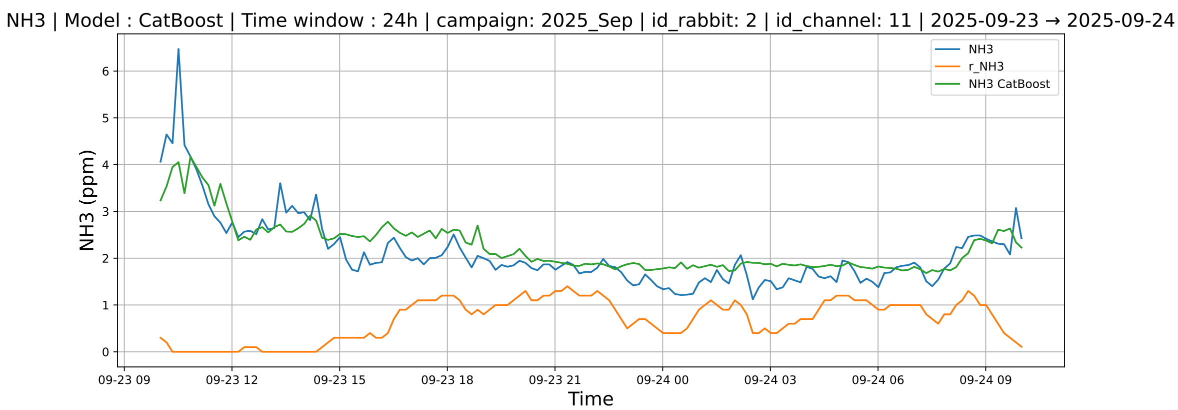


4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

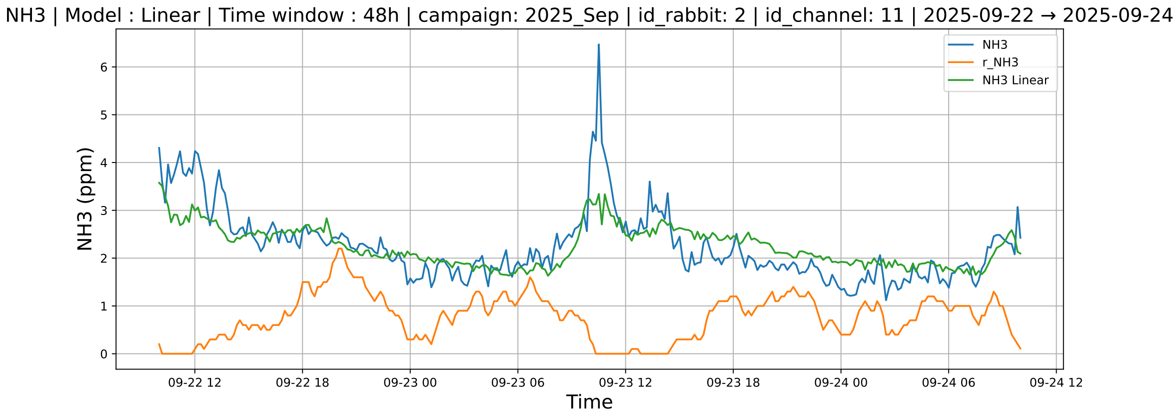
4.1.5.1 Time window : 24

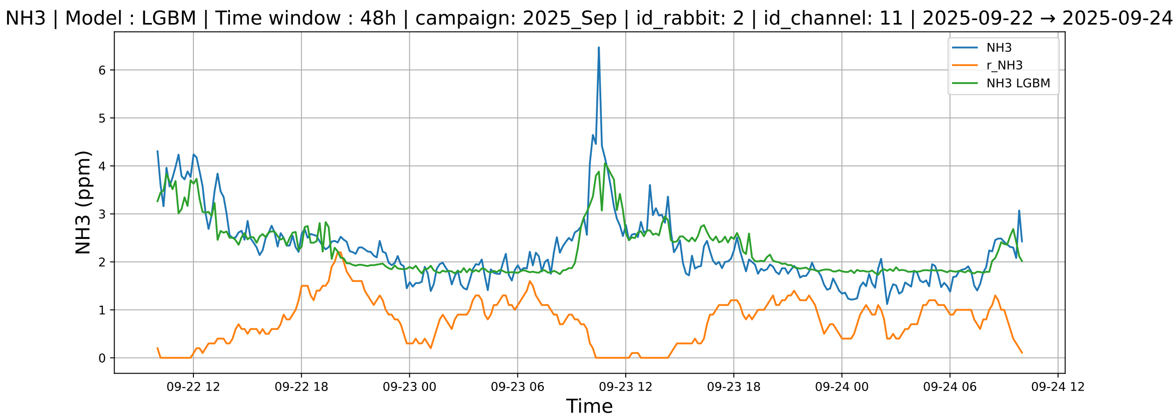


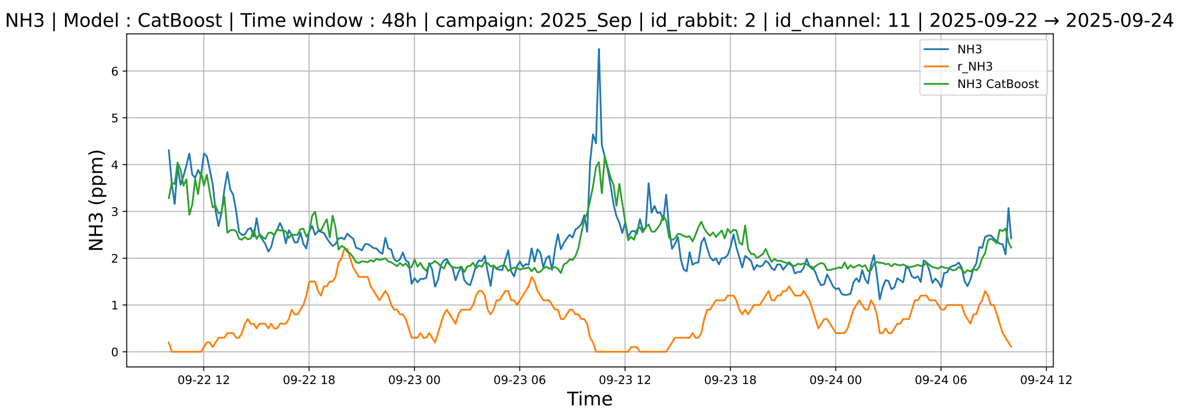




4.1.5.2 Time window : 48

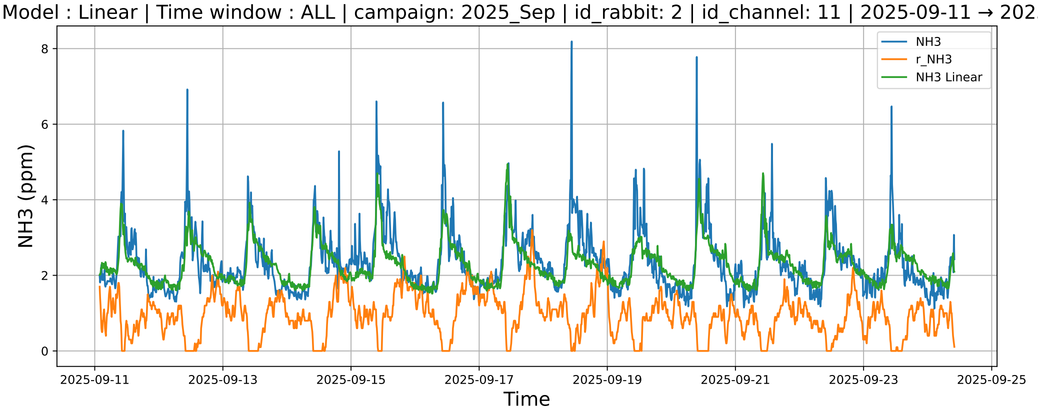




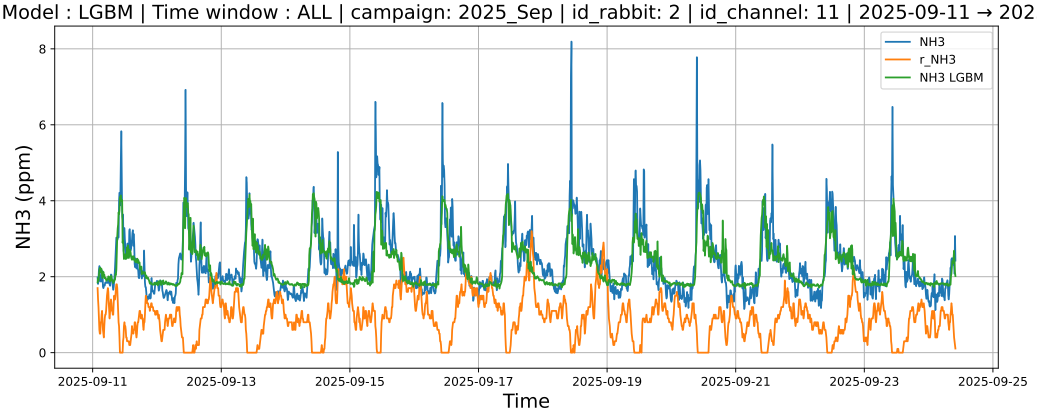


4.1.5.3 Time window : ALL

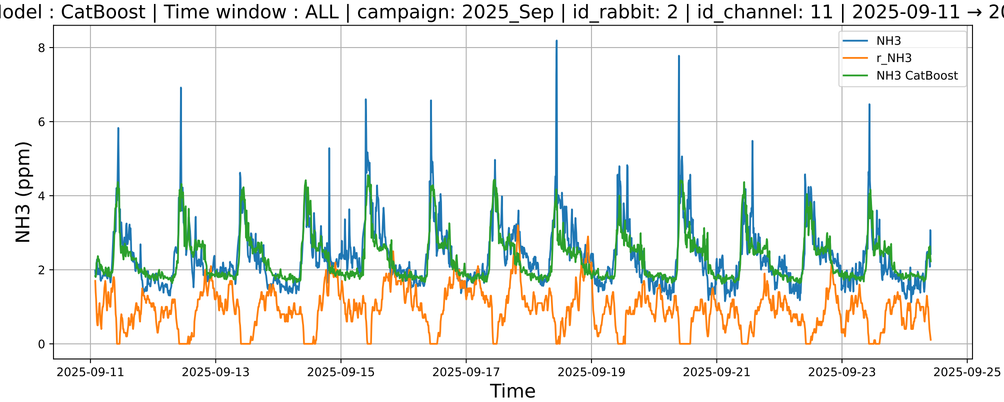
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



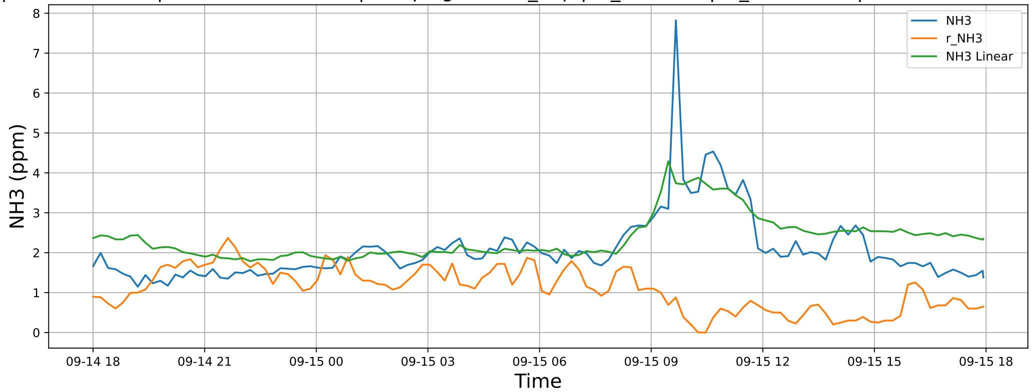
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



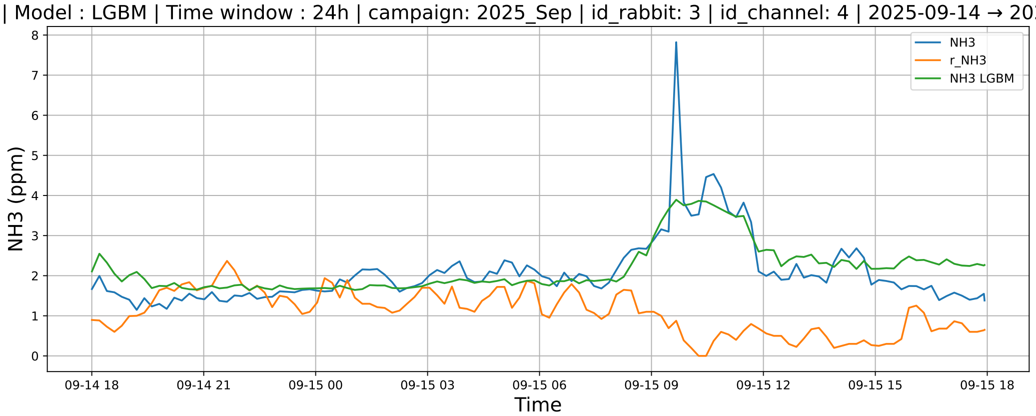
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

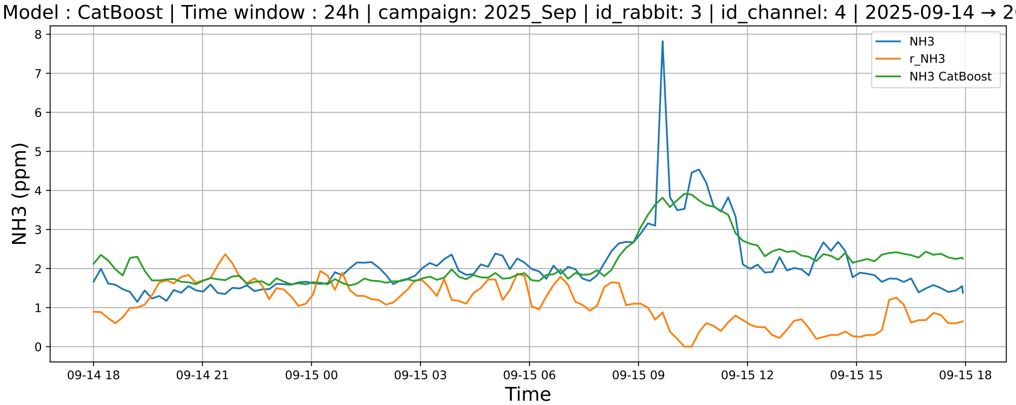
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



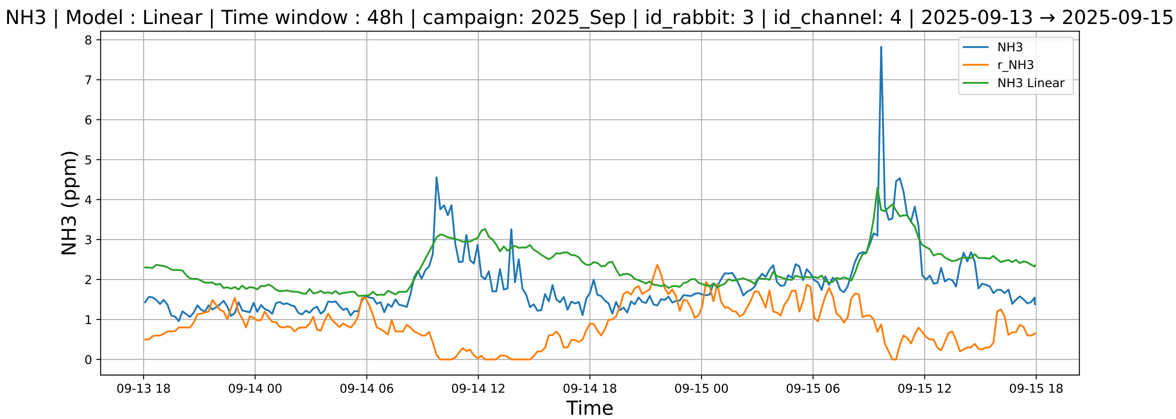
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

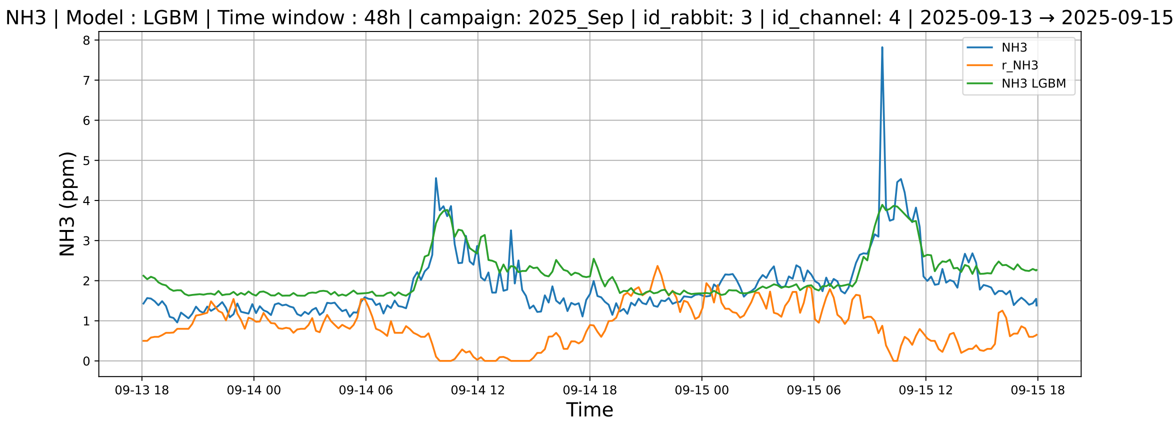


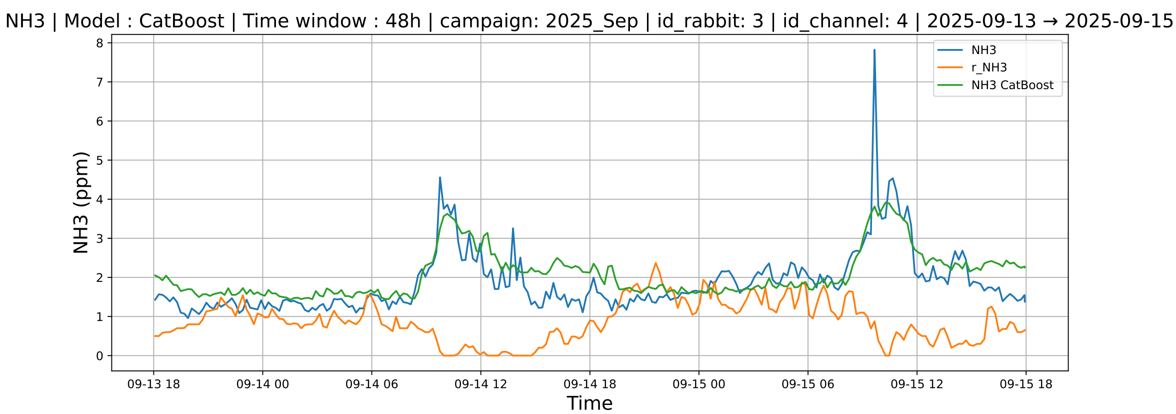
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



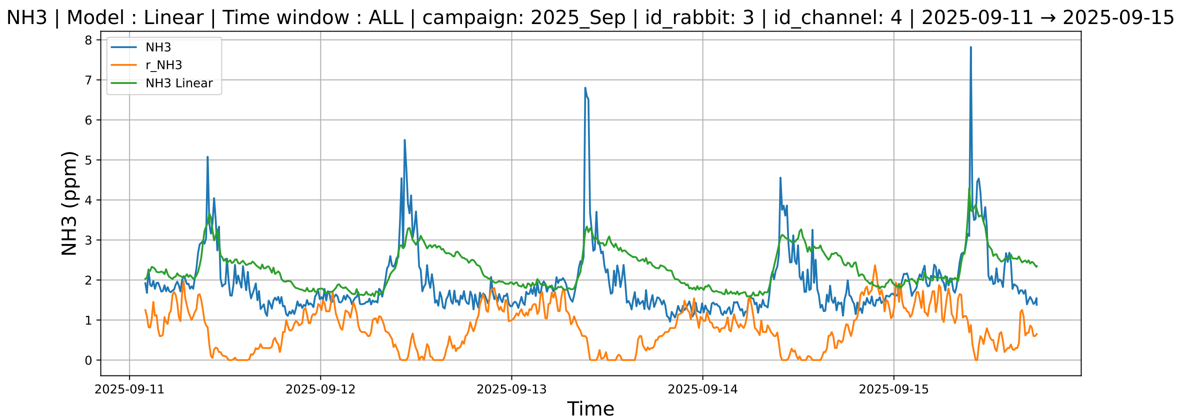
4.1.6.2 Time window : 48

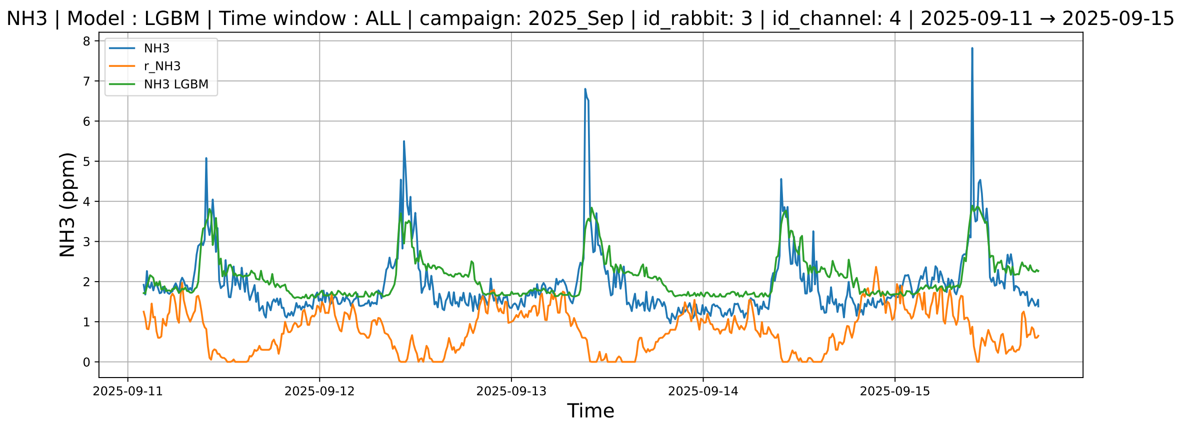


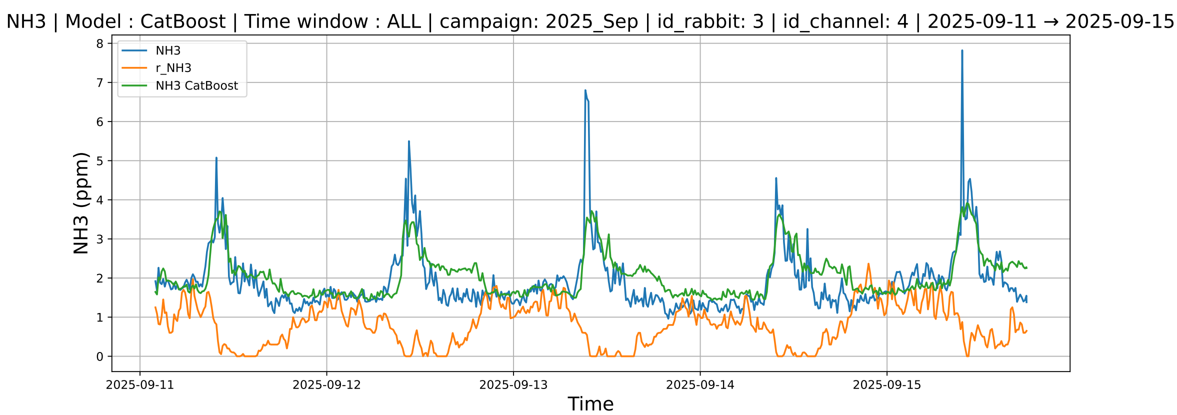




4.1.6.3 Time window : ALL

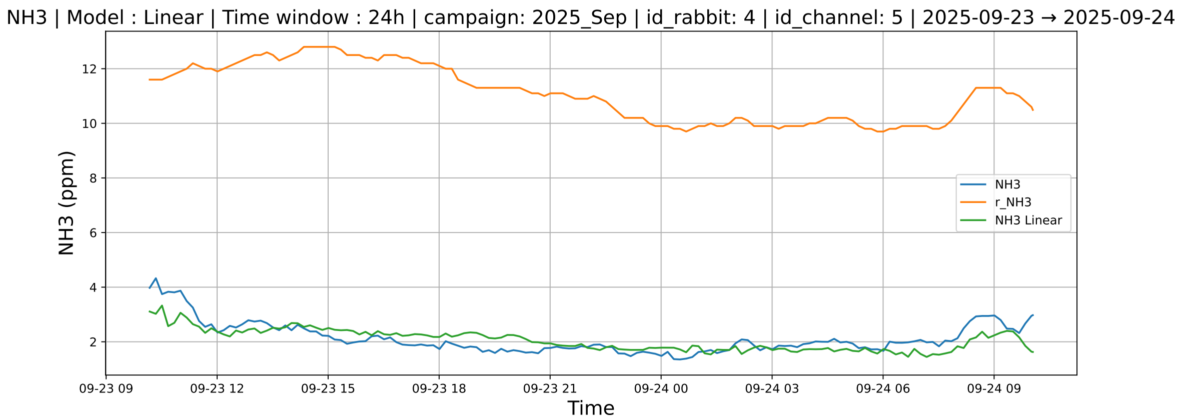


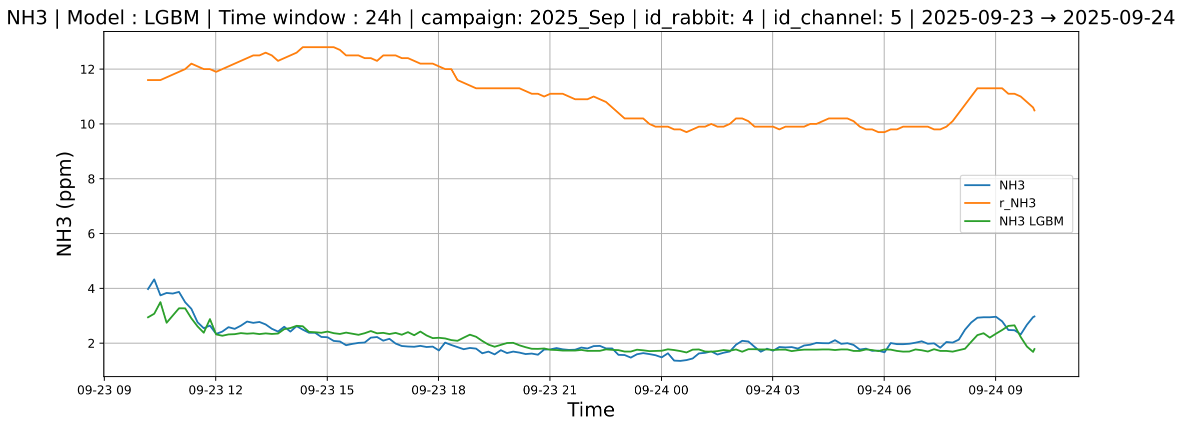


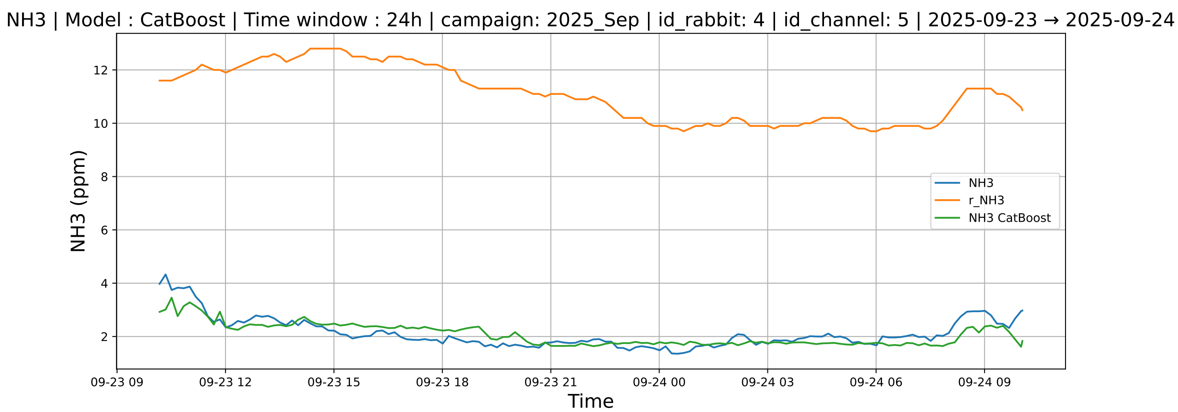


4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

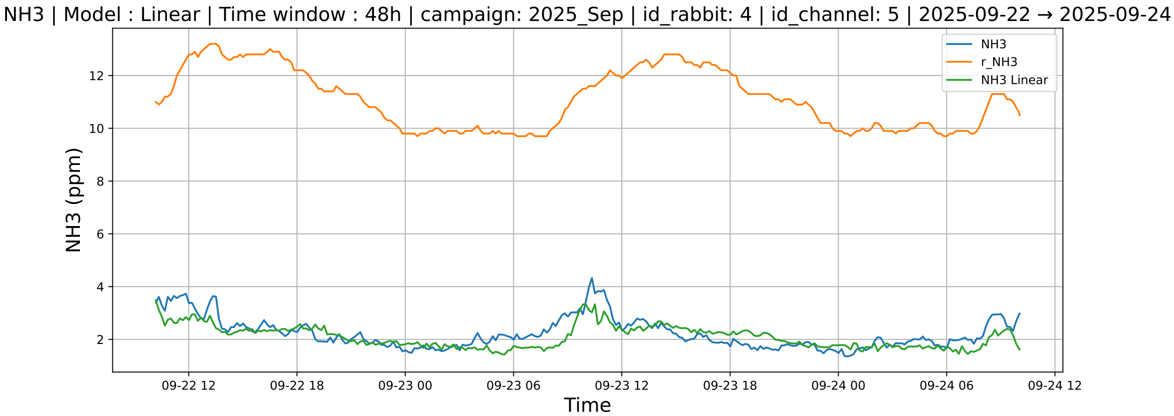
4.1.7.1 Time window : 24

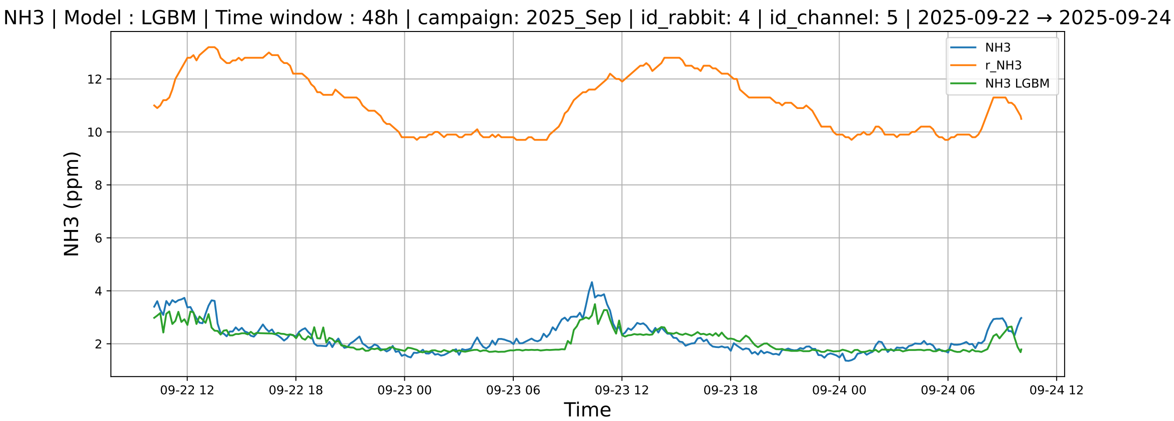


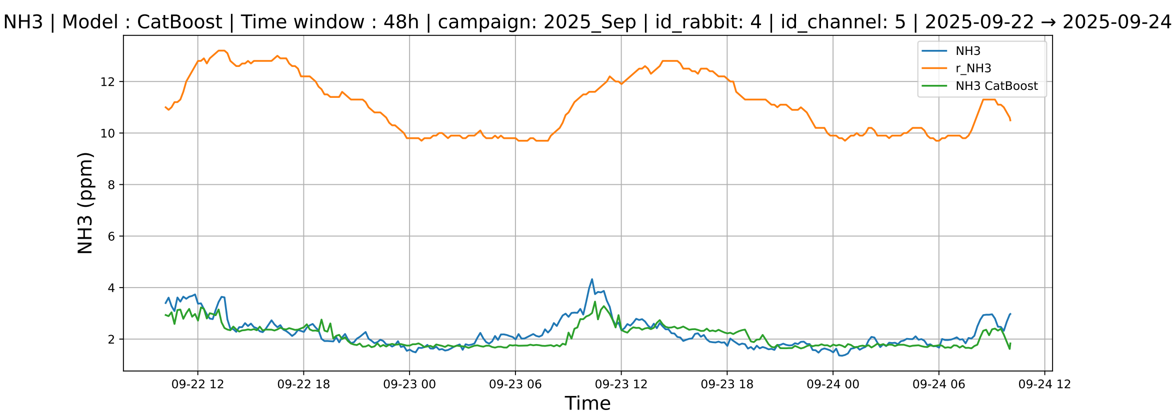




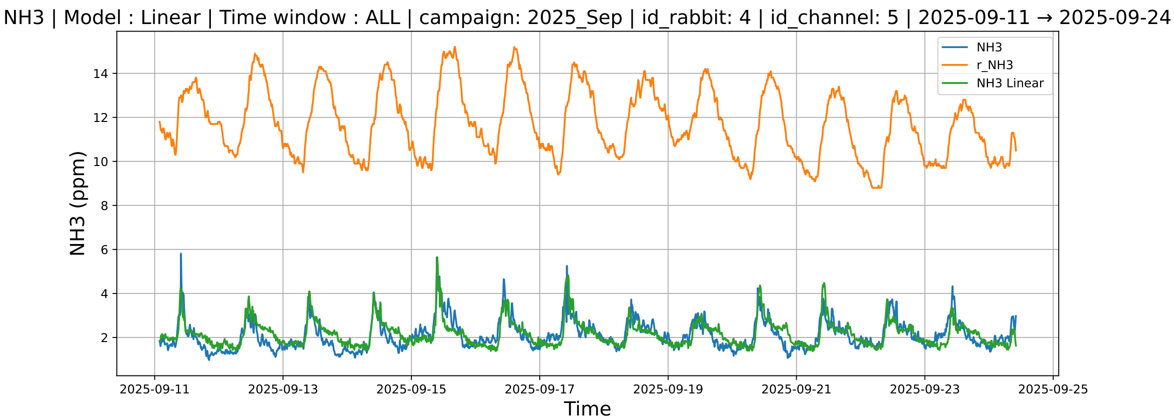
4.1.7.2 Time window : 48

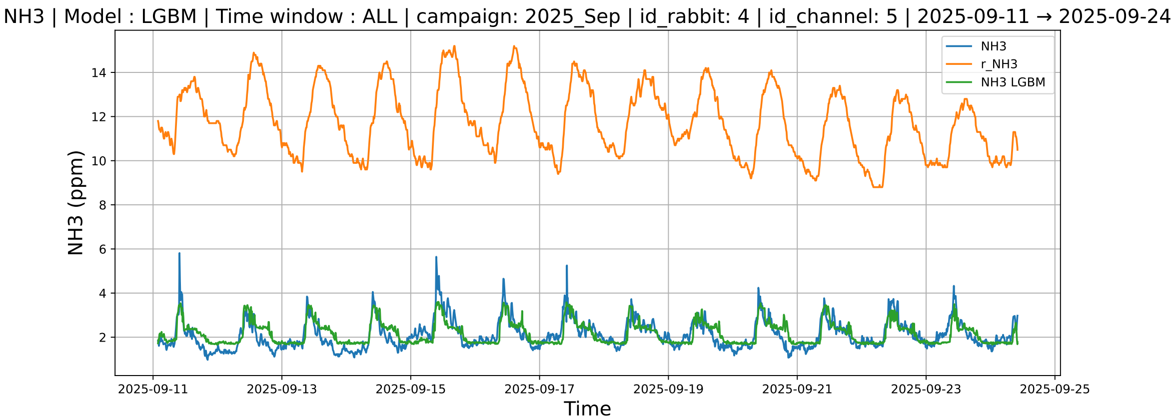


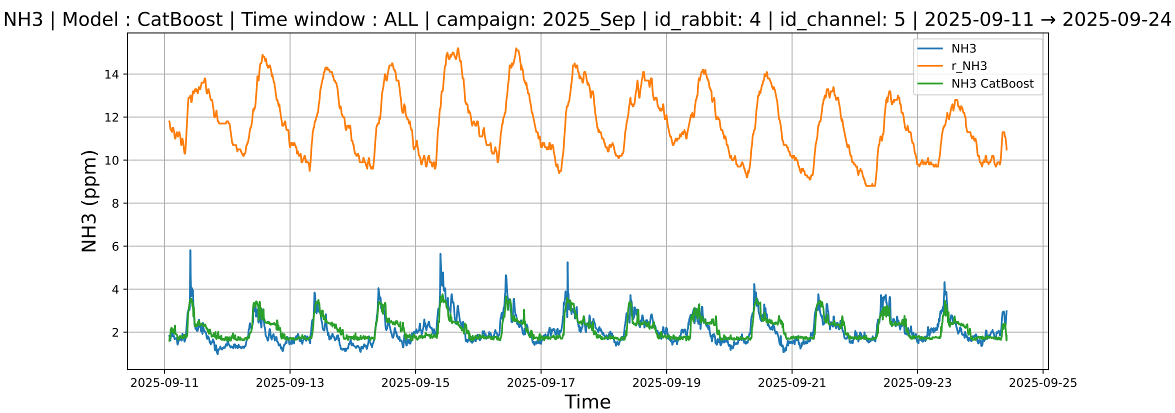




4.1.7.3 Time window : ALL



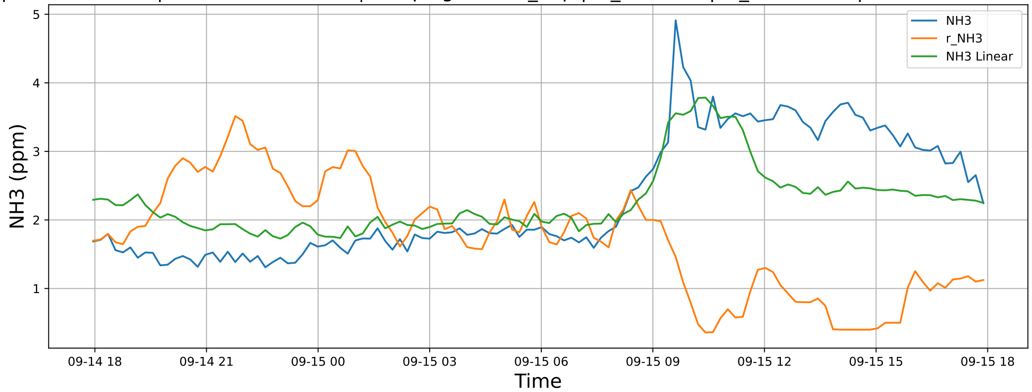




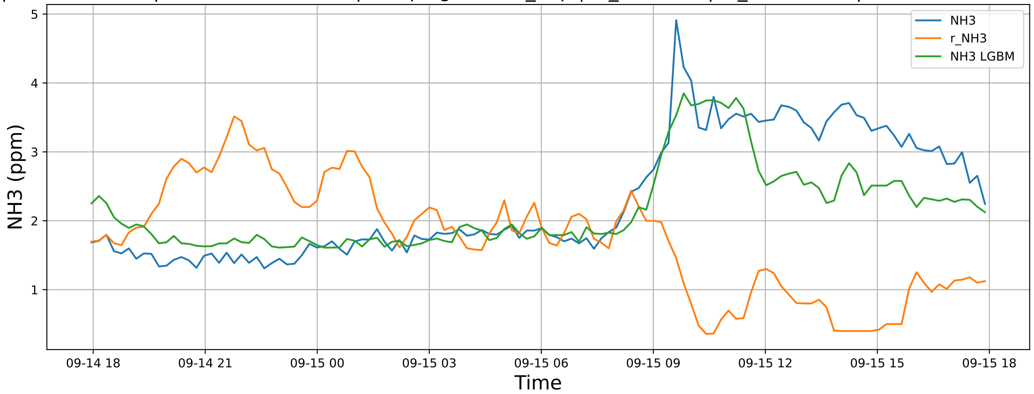
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

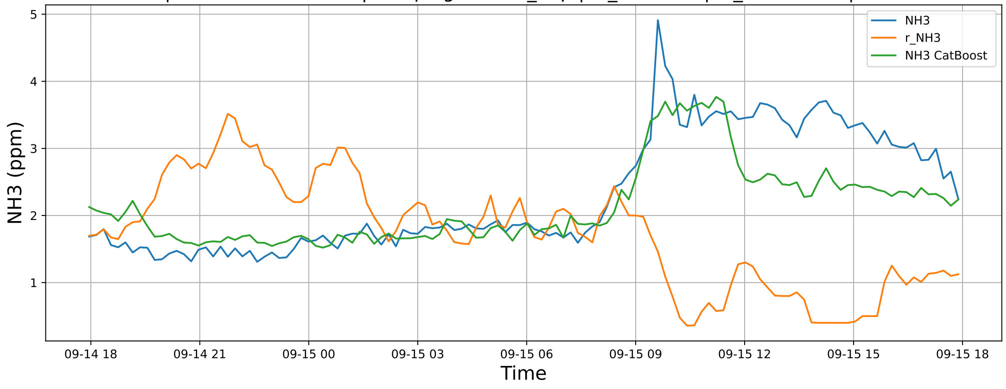
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

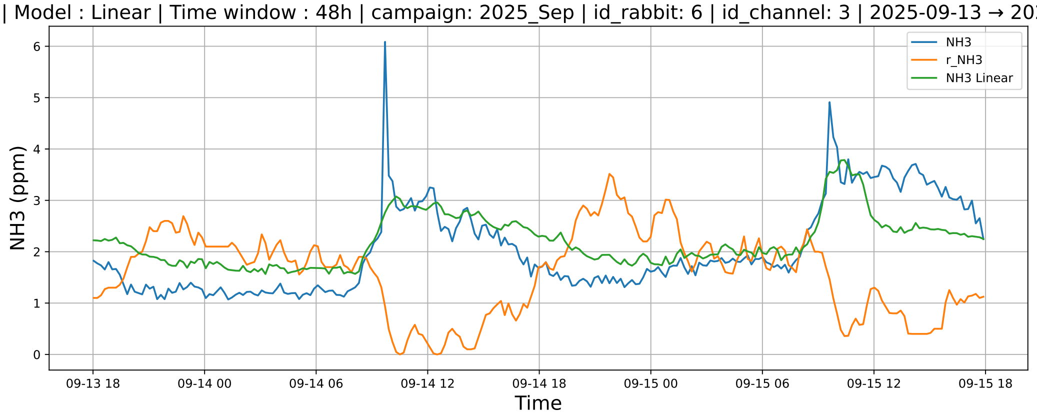


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

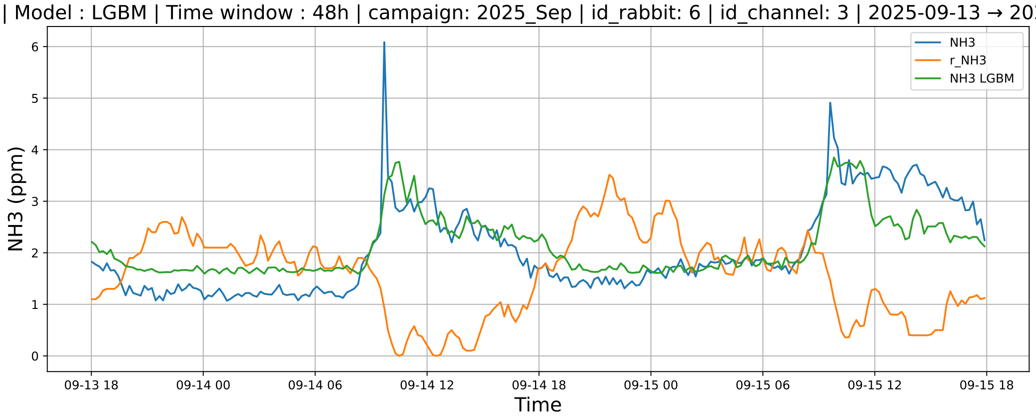


4.1.8.2 Time window : 48

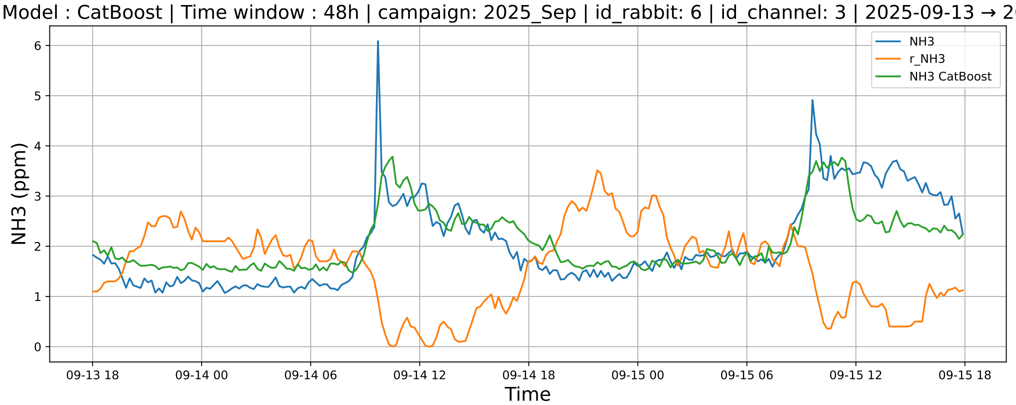
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

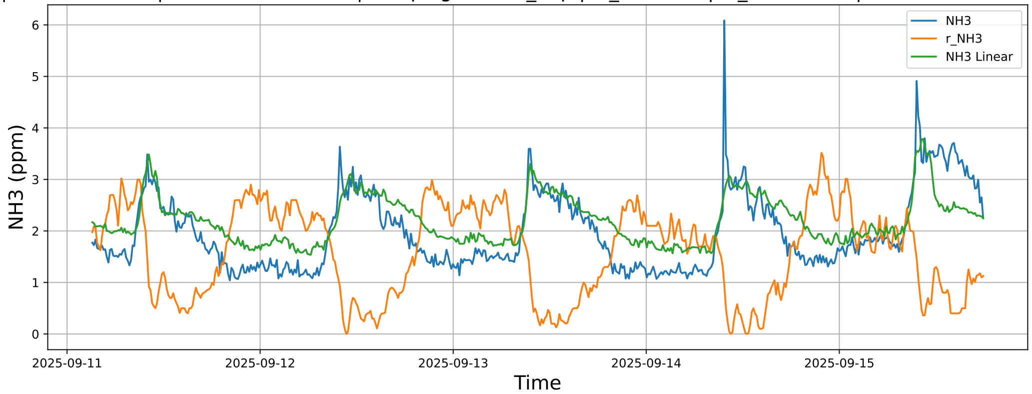


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

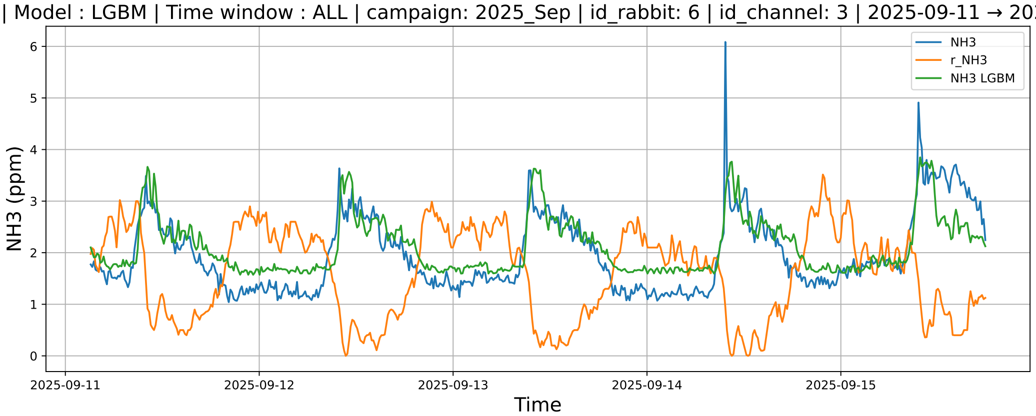


4.1.8.3 Time window : ALL

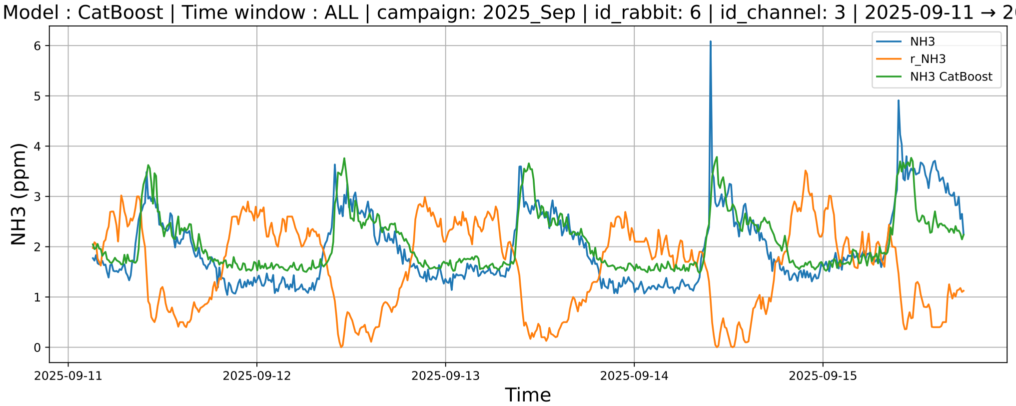
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



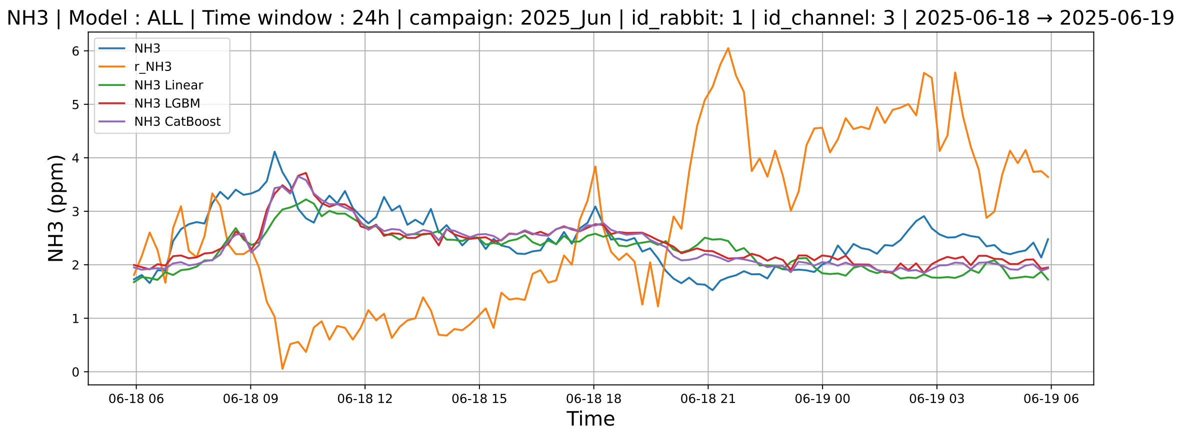
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



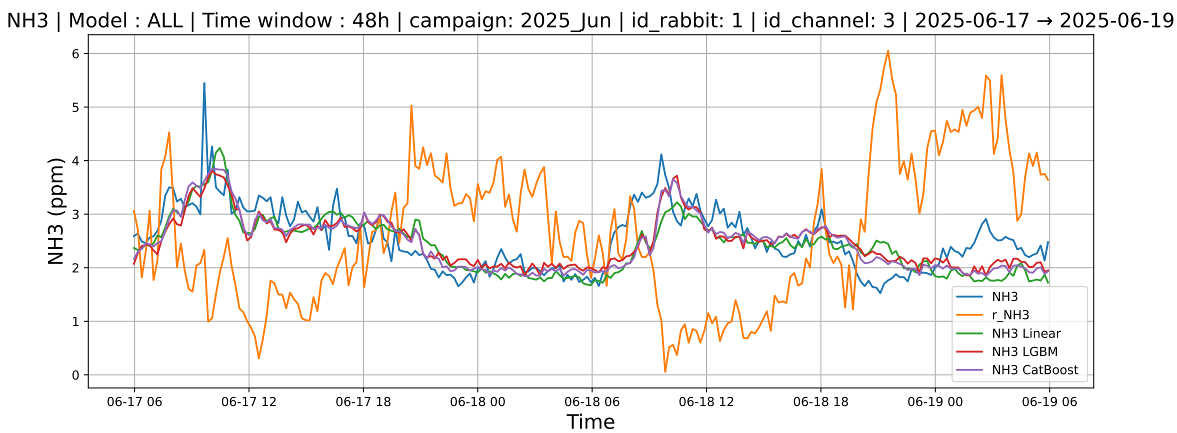
4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

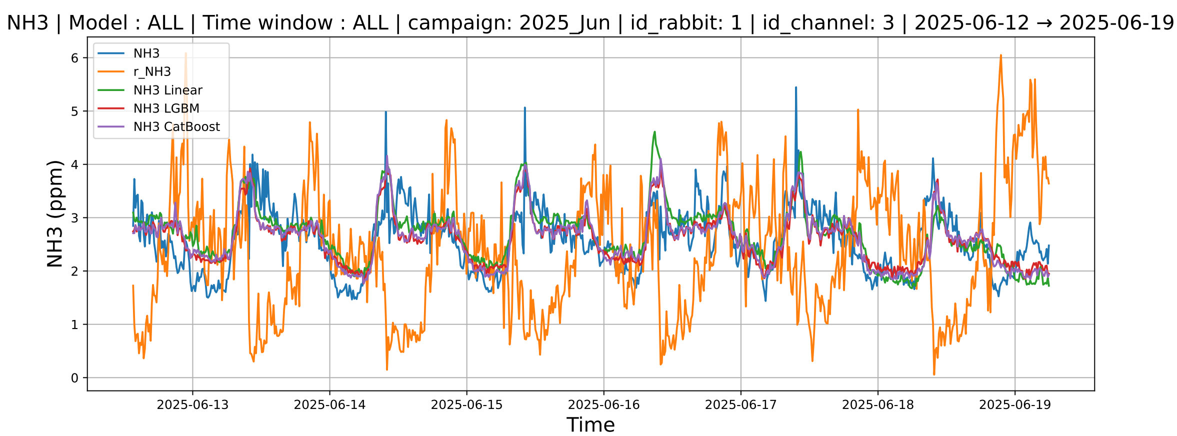
4.2.1.1 Time window : 24



4.2.1.2 Time window : 48

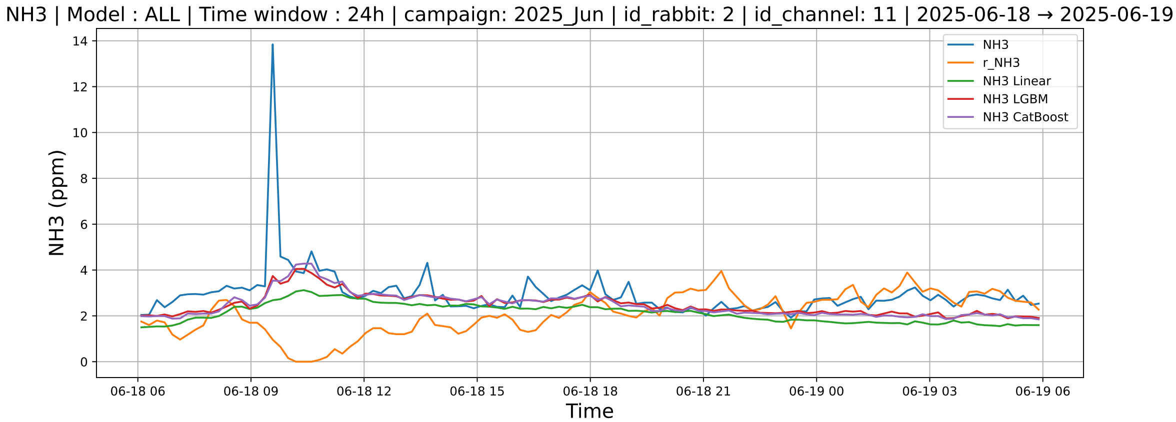


4.2.1.3 Time window : ALL

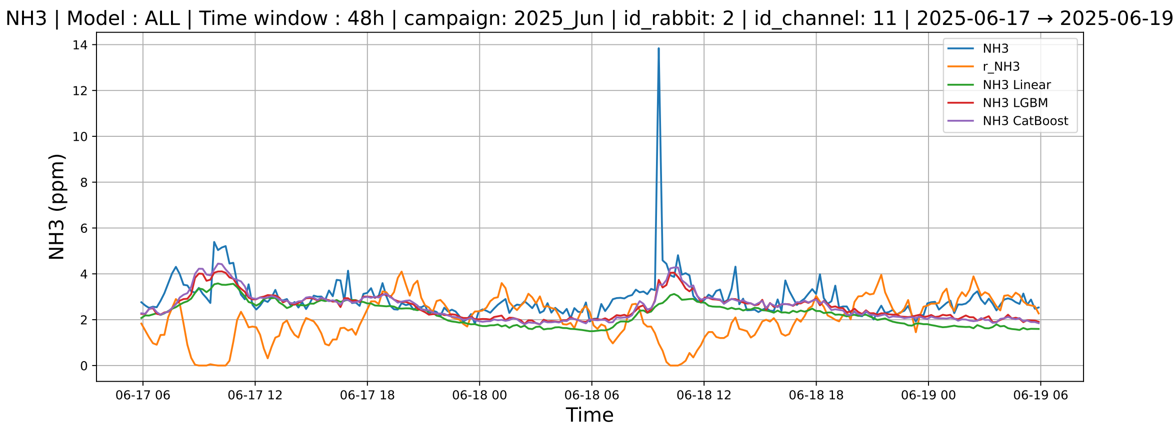


4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

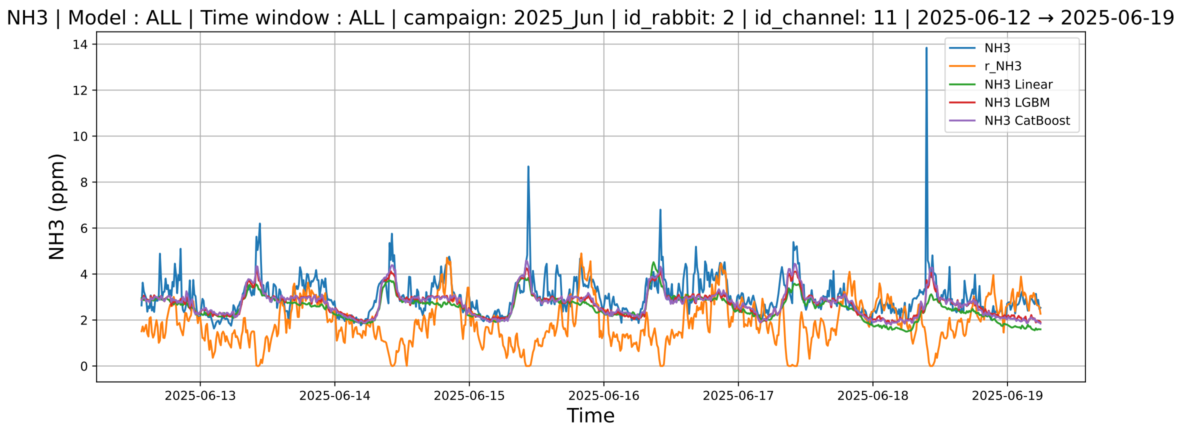
4.2.2.1 Time window : 24



4.2.2.2 Time window : 48

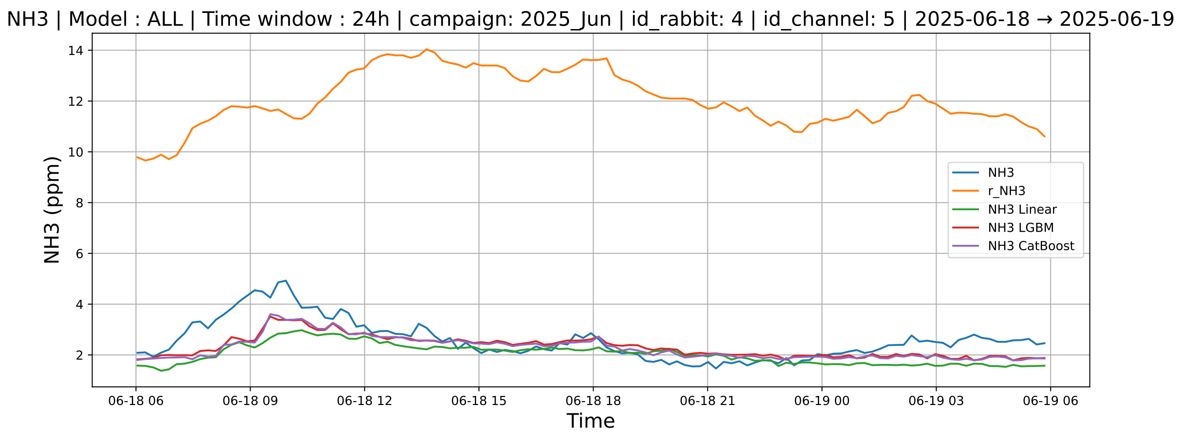


4.2.2.3 Time window : ALL

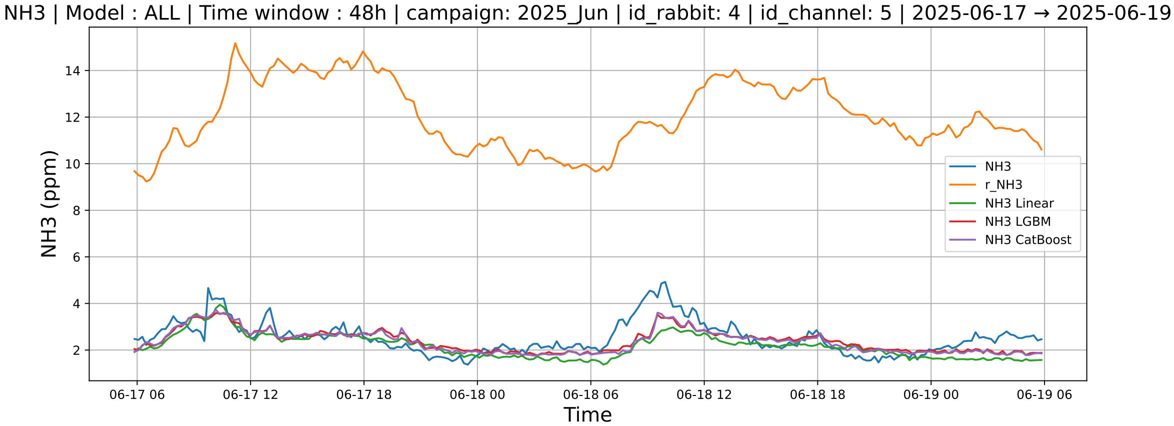


4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

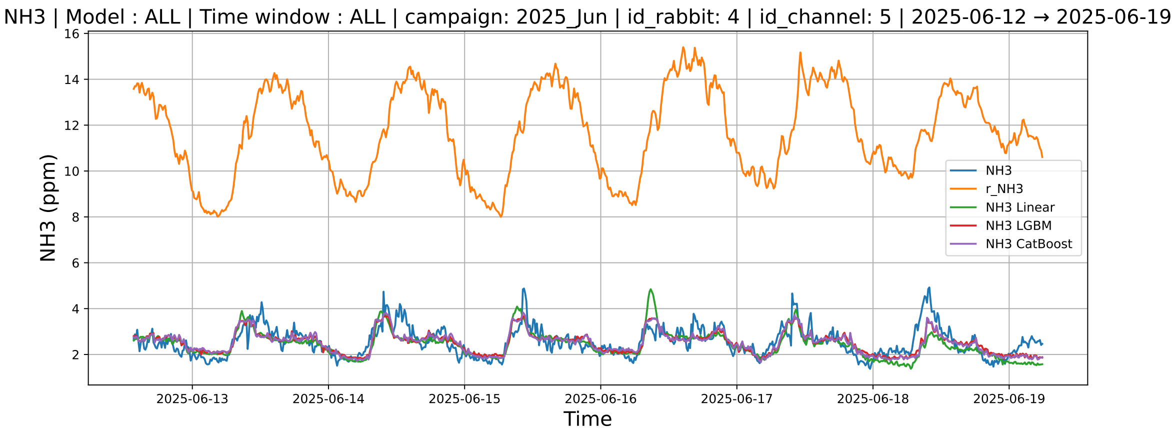
4.2.3.1 Time window : 24



4.2.3.2 Time window : 48

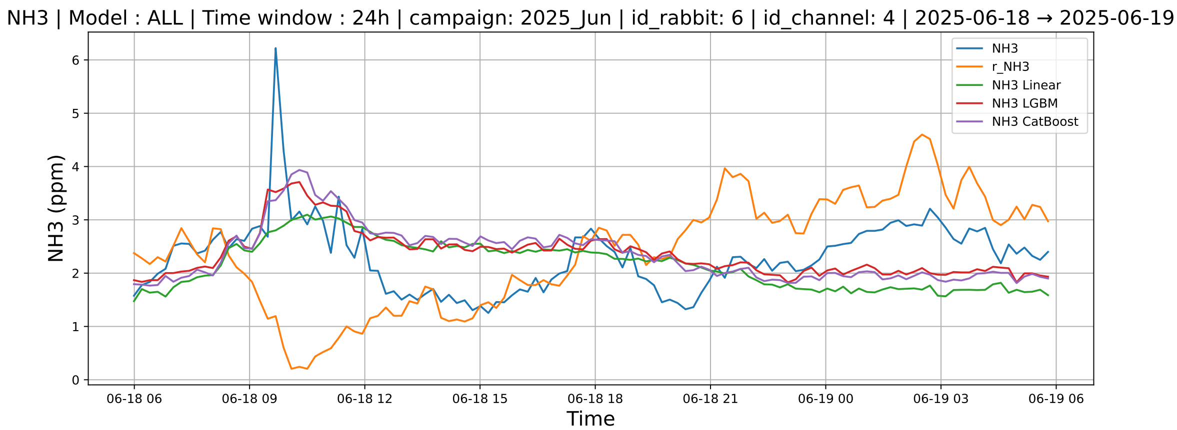


4.2.3.3 Time window : ALL

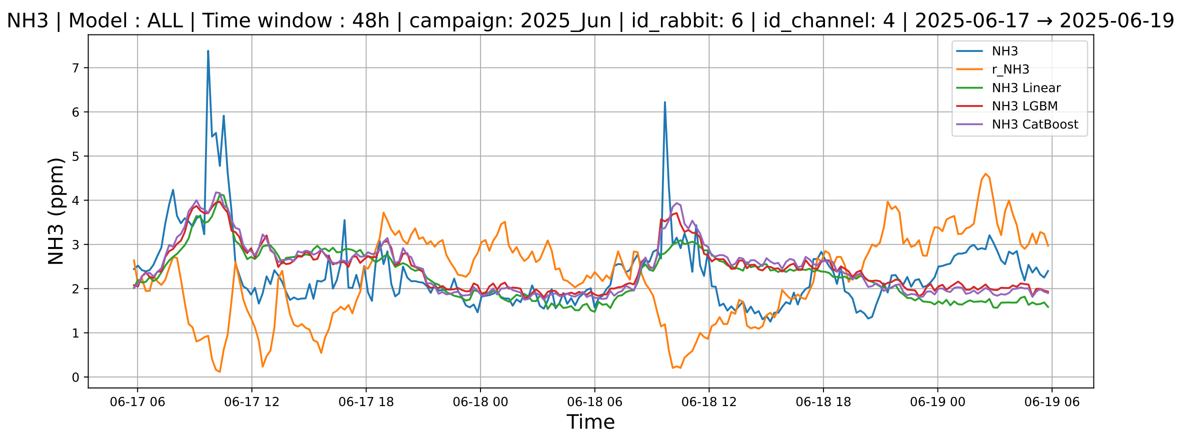


4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

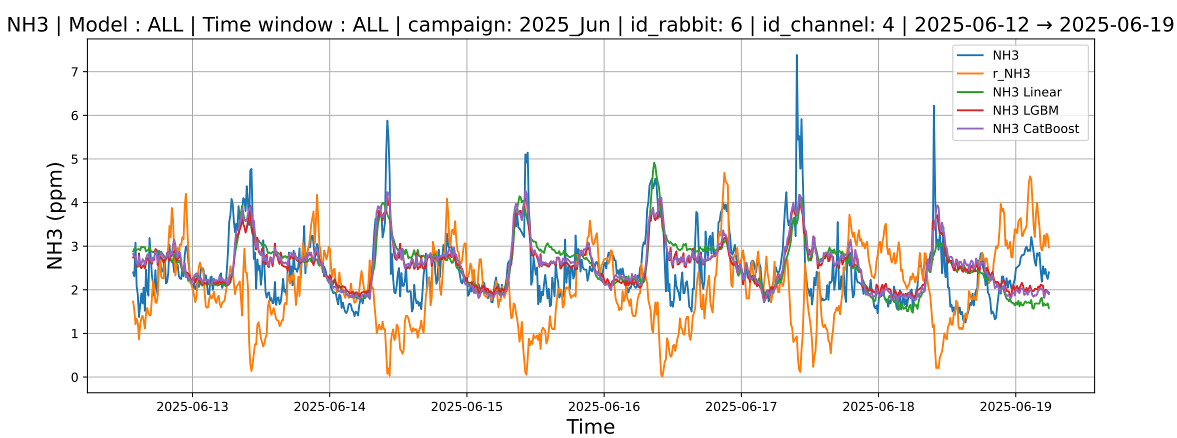
4.2.4.1 Time window : 24



4.2.4.2 Time window : 48



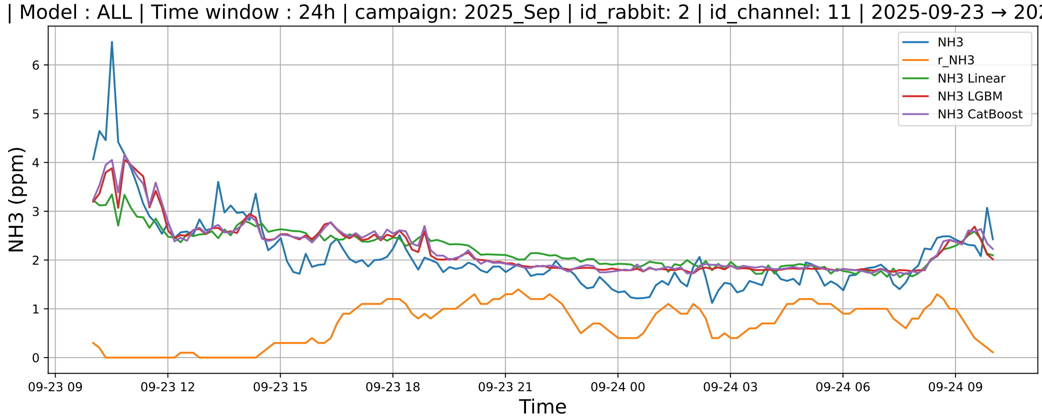
4.2.4.3 Time window : ALL



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

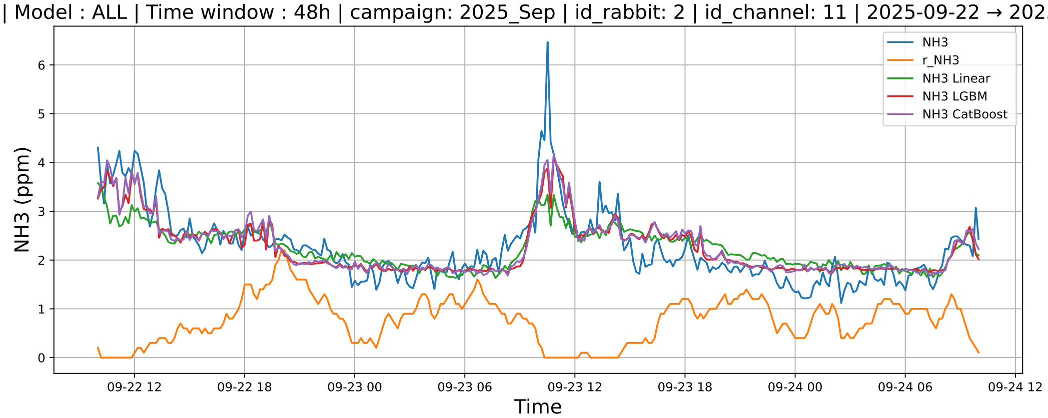
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

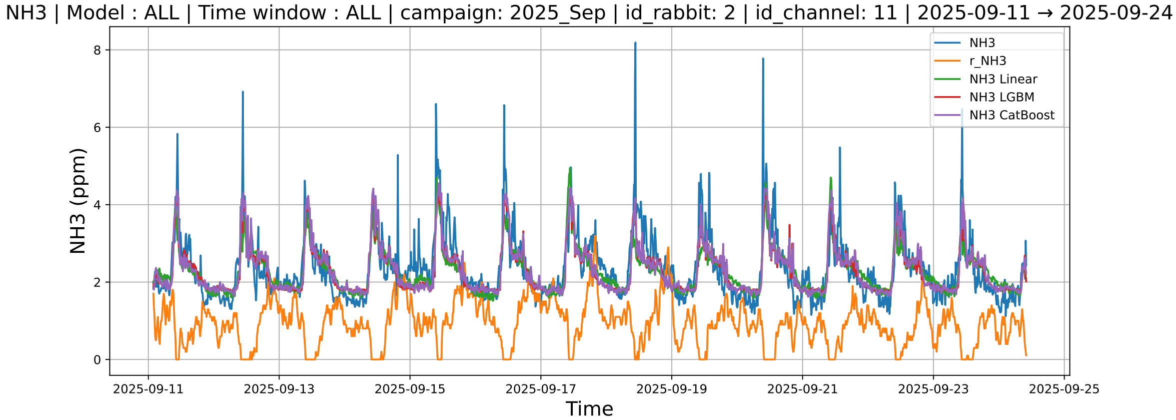


4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

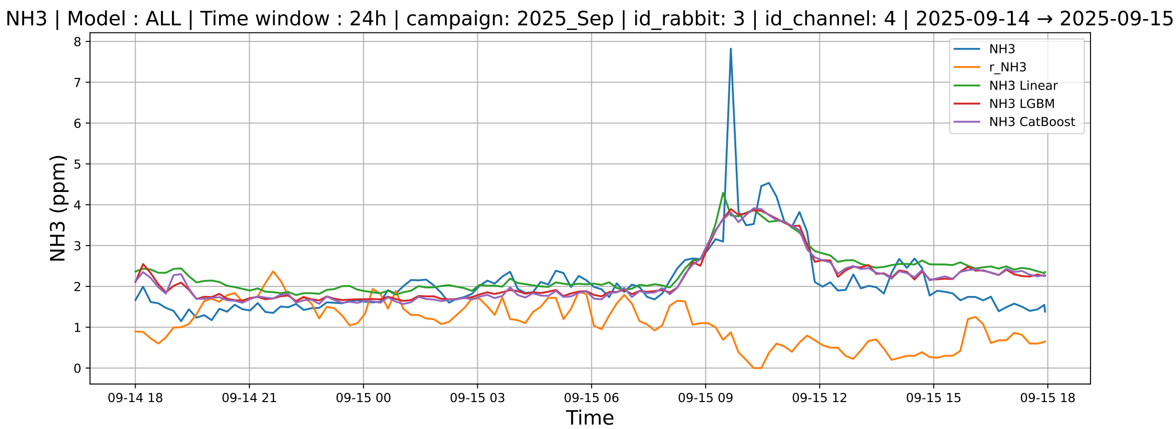


4.2.5.3 Time window : ALL



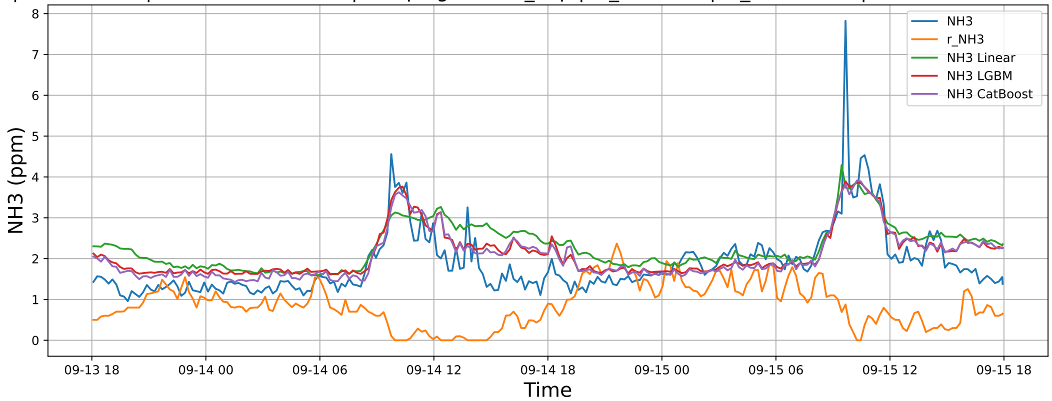
4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.2.6.1 Time window : 24



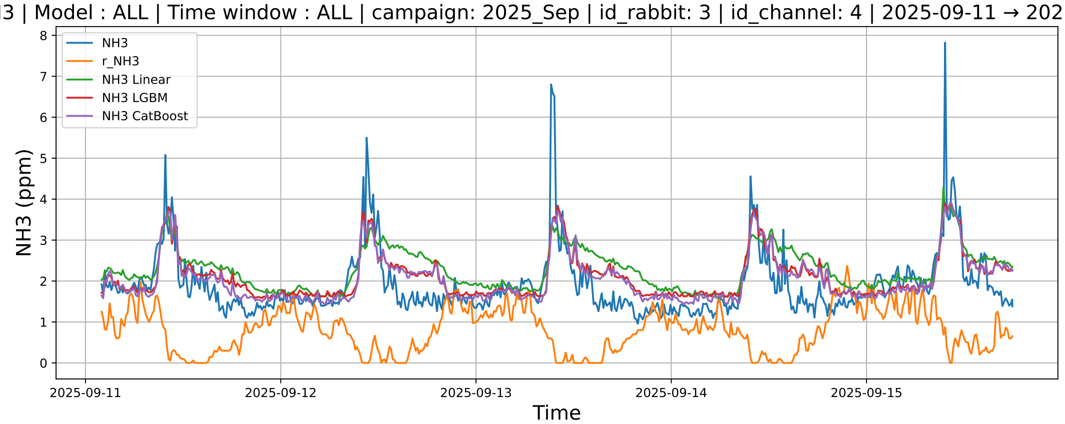
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



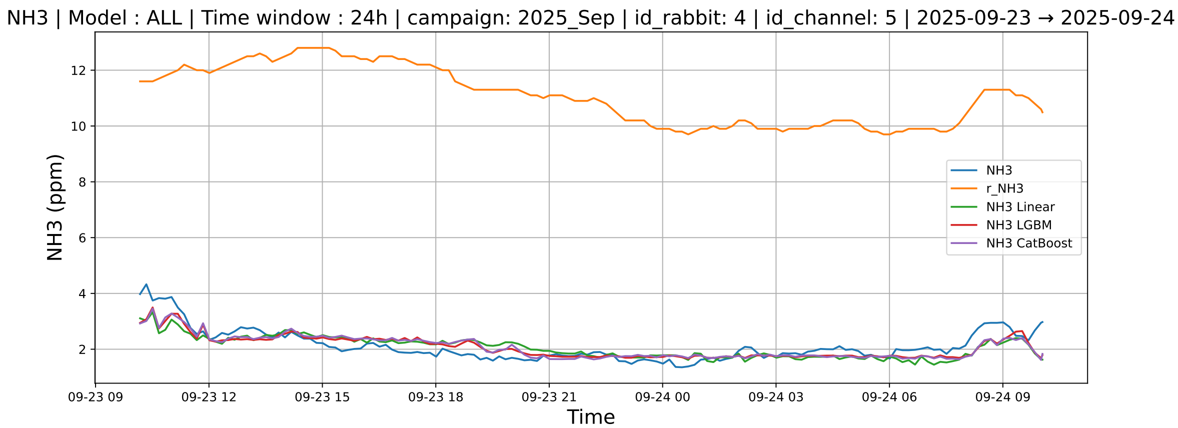
4.2.6.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

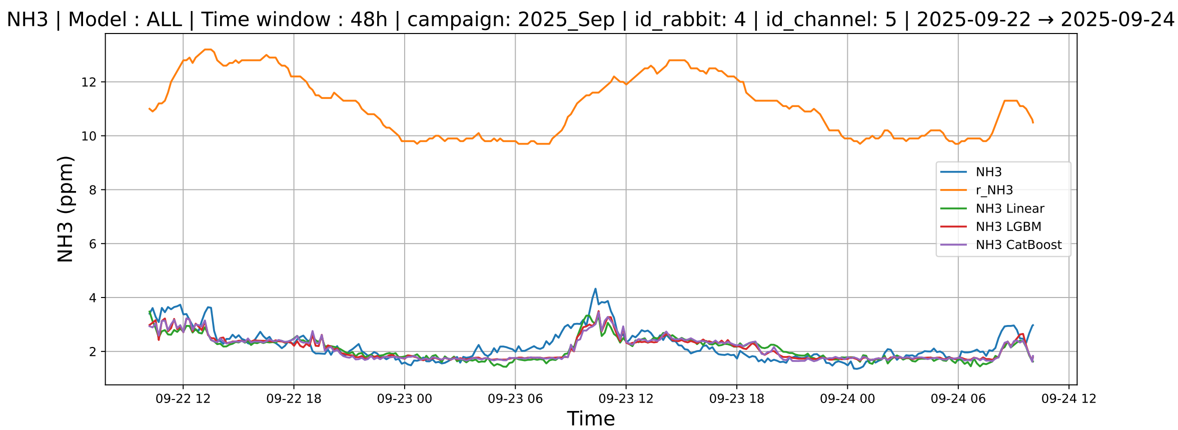


4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

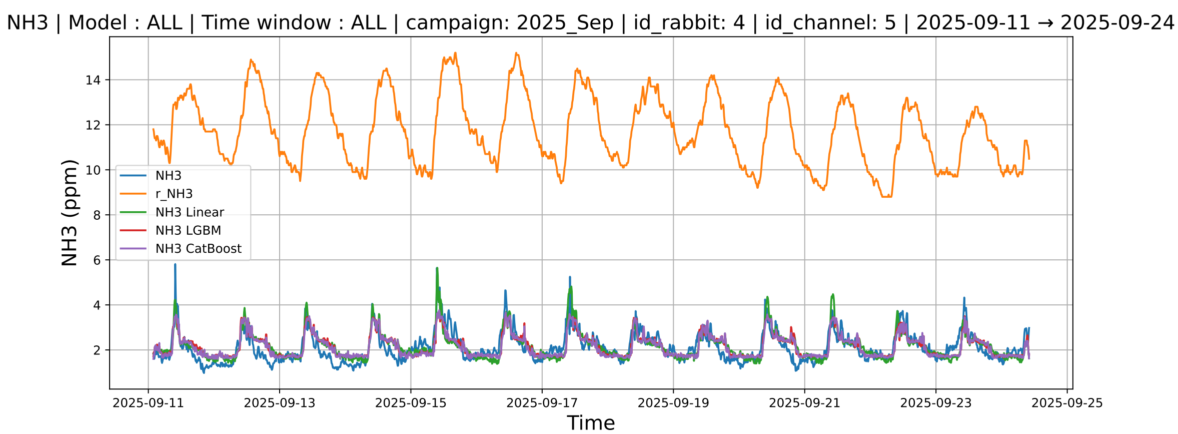
4.2.7.1 Time window : 24



4.2.7.2 Time window : 48

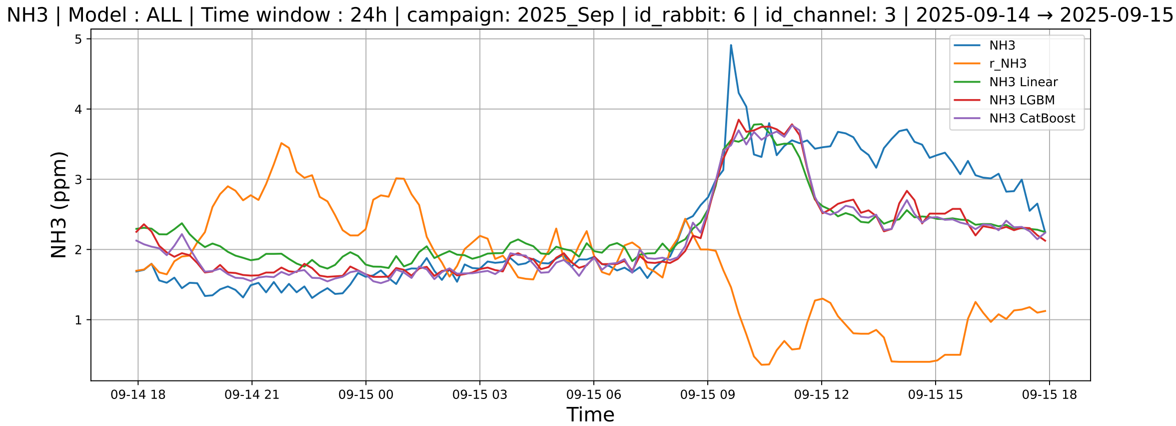


4.2.7.3 Time window : ALL

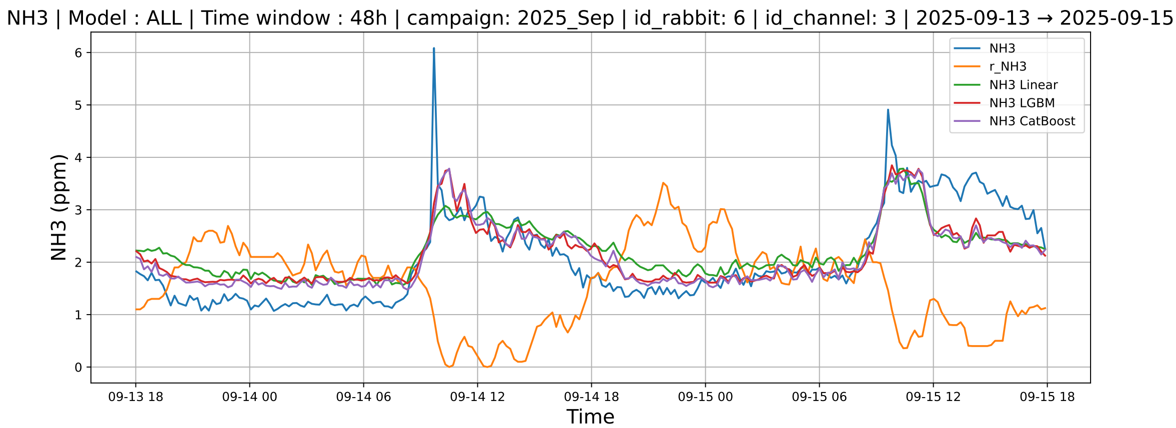


4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

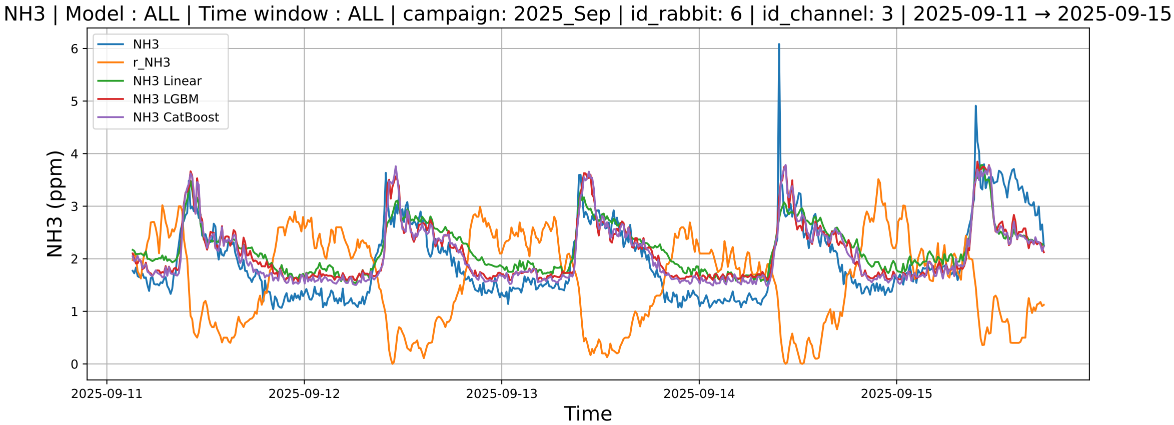
4.2.8.1 Time window : 24



4.2.8.2 Time window : 48



4.2.8.3 Time window : ALL


NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15
NH3
r_NH3
NH3 Linear
NH3 LGBM
NH3 CatBoost
NH3 (ppm)
Time
2025-09-11
2025-09-12
2025-09-13
2025-09-14
2025-09-15
0
1
2
3
4
5
6